

Online Resource 1

Supplementary material for “Quantum-Inspired Cognition and Decision-Making for Autonomous Agents, from Robotics to AI/ML Systems: A Systematic Review”

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This resource contains the data-extraction table of the 67 application studies (Table S1), the PRISMA 2020 checklist (Table S2), the flow of records (Table S3 and Figure S1), the quality-appraisal criteria and judgements (Tables S4 and S5), the search strategy and the record of the targeted search (Table S6), the studies excluded under criterion E5 (Table S7), the classification of the 71 theory and evidence studies (Table S8) and a summary of every graded study (Section S9). The glossary of symbols is in the main article (Table 3). Section numbers refer to the main article. Tables S1 and S5 are also provided as CSV files (Online Resources 3 and 4). Evidence levels: E1 conceptual; E2 formal model; E3 human data; E4 simulation or benchmark; E5 physical system; E6 field deployment.

Table S1: Data-extraction table of the application studies ($n = 67$): domain, model type, evidence level, task, key result, main limitation and basis of extraction (full text or abstract).
[†] also discussed in the companion survey.

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Moreira and Wichert (2014)	Decision models and networks	Quantum-like	E2	Probabilistic inference in Bayesian networks under uncertainty	Large departures from classical inference under high uncertainty; collapse to classical under low uncertainty	No human data; free interference phases	Abstract
Moreira and Wichert (2016)	Decision models and networks	Quantum-like	E3	Sure-thing principle violations (disjunction effect)	Approximates observed violations with automatically set quantum parameters	Heuristic lacks strong psychological justification; aggregate data from few paradigms	Abstract
Moreira (2017)	Decision models and networks	Quantum-like	E3	Modelling paradoxical human decisions with graphical models	Consolidates quantum-like graphical models reproducing sure-thing violations	Few paradigms; aggregate data; heuristic parameter setting	Abstract
Moreira et al. (2020b)	Decision models and networks	Quantum-like	E3	Decision-making with utilities; paradoxical choices	Predicts observed choice distributions; identifies perceived-optimal but irrational decisions; tracks belief evolution	Interference parameters still required; classic laboratory paradigms only	Abstract
Wichert et al. (2020)	Decision models and networks	Quantum-like	E3	Predicting sure-thing principle violations	Reported accurate prediction of paradoxical decisions without fitted phases	Two paradigms; aggregate data; maximum-entropy assumption	Abstract
She et al. (2021)	Decision models and networks	Quantum-like	E4	Multi-attribute decision making with interacting attributes	Reported better predictive capability than classical Bayesian network approaches	Single application; cognitive interpretation of interference unclear	Abstract
Meghdadi et al. (2022)	Decision models and networks	Quantum-like	E3	Biased human selection in decision games	RMSE 3.4 on Prisoner’s Dilemma, ranked first among compared models	Additional loosely constrained parameters; few aggregate datasets	Abstract
Gao et al. (2025)	Decision models and networks	Quantum-like	E3	Sure-thing principle violations; information retrieval	Lowest average and standard deviation of fitting errors among compared models	Existing aggregate datasets; added model complexity	Abstract
Widdows and Peters (2003)	ML, NLP and IR	Quantum-inspired	E4	Word-sense disambiguation / retrieval with logical connectives on word vectors	Negation removes unwanted senses (e.g. legal ‘suit’); operators equal quantum-logic connectives; no quantitative scores in this paper	Qualitative evaluation only; no human data; non-distributivity not exploited empirically	Abstract
Sordoni et al. (2013)	ML, NLP and IR	Quantum-inspired	E4	Ad hoc document retrieval with term dependencies	Reported competitive or improved retrieval over MRF-style dependency models (numbers not verified here)	Vocabulary-sized density matrices are costly; dependencies are hand-selected; no user/cognitive data	Abstract
Lorenz et al. (2023)	ML, NLP and IR	Quantum circuit (QPU)	E5	Sentence and noun-phrase binary classification (MC, RP)	Hardware test error 16.7% (MC) and 32.3% (RP), F-score 0.85/0.75; DisCoCat best on RP	Very small datasets, single hardware run, post-selection overhead, long queue times; no quantum advantage	Full text
Svozil (2025)	ML, NLP and IR	Quantum-inspired	E2	Contextual word embeddings / polysemy	Framework proposal; no quantitative results	No learning procedure, corpus or evaluation; practicality untested	Full text
Stoudenmire and Schwab (2016)	Deep learning and LLMs	Quantum-inspired	E4	Image classification (supervised learning)	Test-set classification error below 1% on MNIST	Single simple benchmark; 1-D ordering of 2-D inputs; no cognitive or agent evaluation	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Zhang et al. (2018)	Deep learning and LLMs	Quantum-inspired	E4	Question answering / answer selection (text matching)	Reported effective against neural and quantum LM baselines (numbers not verified)	Interpretability rests on analogy; text-matching only; no cognitive data	Abstract
Li et al. (2019)	Deep learning and LLMs	Quantum-inspired	E4	Question answering (semantic matching)	Performance comparable to strong CNN and RNN baselines	No gain over baselines; interpretability not user-tested	Abstract
Li et al. (2021a)	Deep learning and LLMs	Quantum-inspired	E4	Multimodal video sentiment analysis (fusion)	Reported competitive with state-of-the-art fusion plus interpretability (numbers not verified)	Task-specific; interpretability argued not measured; no cognitive data	Abstract
Li et al. (2021b)	Deep learning and LLMs	Quantum-inspired	E4	Emotion recognition in conversation (multimodal)	Performance comparable to state of the art	No gain over state of the art; benchmark-only; no human-judgement validation	Abstract
Gkoumas et al. (2021)	Deep learning and LLMs	Quantum-like	E4	Multimodal decision fusion for video sentiment analysis	Substantially outperformed decision- and content-level fusion; handled cases where all unimodal classifiers failed	Incompatibility assumed not tested; benchmark labels, not human decision data	Abstract
Uprety et al. (2024)	Deep learning and LLMs	Quantum-like	E4	Order effects in similarity judgements by LLMs	LLMs show human-like order-effect bias in some settings but not others	Preliminary; prompt-dependent; no QP model fitted in abstract	Abstract
Zhang et al. (2025)	Deep learning and LLMs	Quantum-inspired	E4	Semantic matching / QA	Reported improved/competitive matching (not verified)	Details unverified; density-matrix cost; interpretability not user-tested	Abstract
Maksimovic and Maksymov (2025)	Deep learning and LLMs	Quantum-inspired	E4	Image classification with human-like confidence/uncertainty	Evidence of human-like decision-making; up to 50x faster training than classical counterpart	Human-likeness assessed via simulations, not participant data; datasets unverified	Abstract
Lo et al. (2025)	Deep learning and LLMs	Quantum-like	E4	Contextuality of LLM-derived probabilities in anaphora (co-reference) schema	77,118 (0.148%) sheaf-contextual, 36,938,948 (71.1%) Cbd-contextual; Euclidean distance best predictor; rates rise to 0.50%/81.83% for similar nouns	Single masked LM; synthetic schema; no downstream-task benefit shown	Full text
Aerts et al. (2026)	Deep learning and LLMs	Quantum-like	E4	Bell-inequality and quantum-statistics tests on LLM concept combination and text	Significant Bell-inequality violation; Bose–Einstein statistics in word distributions, mirroring human data	Prompt/version dependence; signalling not controlled in abstract; interpretive leap to convergent cognition	Abstract
Imannezhad et al. (2026)	Deep learning and LLMs	Quantum-like	E4	Probability judgement fallacies (conjunction, disjunction, complementarity) in GPT-5 vs humans	GPT-5 conjunction fallacies ~18.8% vs 58.5% humans; near-perfect complementarity; pattern closer to early QP models than noisy variants	Single proprietary LLM; personas as proxies; numbers to verify	Abstract
Kang (2026)	Deep learning and LLMs	Quantum-like	E4	Auditing question-order effects in LLM sequential judgements	Near-deterministic saturation in 17/18 and 7/8 pairs; label mapping changed QQ verdicts; no residual contextuality certified	Single small model; few items; saturated measurement interface	Abstract
Dong et al. (2008)	Reinforcement learning and agent learning	Quantum-inspired	E4	Reinforcement learning in unknown environments (gridworld navigation)	QRL learned faster than TD(0) and was robust across a much wider range of learning rates (per preprint)	Only classical simulation of the quantum algorithm; small tabular tasks; claimed parallelism speed-up requires quantum hardware	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Dong et al. (2012) [†]	Reinforcement learning and agent learning	Quantum-inspired	E5	Mobile-robot navigation and obstacle avoidance via RL	QiRL converged in about 2000 episodes across start states (TD 1000–4000); robust for learning rates 0.01–0.12 while TD failed for ≥ 0.02 ; real robot reached target room after learning	Tabular representation; single TD baseline; real-robot results qualitative	Full text
Briegel and De las Cuevas (2012)	Reinforcement learning and agent learning	Quantum-inspired	E4	Learning agent design with internal simulation (projective simulation)	Learning curves show a trade-off between forgetting rate and achievable blocking efficiency; larger reflection time speeds learning	Toy environment only; classical simulation; no benchmark comparison with standard RL; quantum version only sketched	Full text
Melnikov et al. (2018)	Reinforcement learning and agent learning	Quantum-inspired	E4	Navigation benchmarks for RL agents (grid world, mountain car)	PS performance similar to Q-learning and SARSA; parameter tuning 1–2 orders of magnitude cheaper	Small tabular tasks; classical only; no deep RL comparison	Abstract
Wei et al. (2022)	Reinforcement learning and agent learning	Quantum-inspired	E4	Experience replay sampling in deep reinforcement learning	Outperforms DRL-PER and DCRL on most games with improved training efficiency (per abstract)	Classical heuristic only; evaluation on game benchmarks; exact gains not verified here	Abstract
Lukac and Perkowski (2007)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Generating emotional variation in humanoid robot behaviour	Architecture and examples of emotion-modulated behaviour; no quantitative results reported	No evaluation or comparison with classical stochastic models; partial testing; simulated quantum layer	Abstract
Aerts et al. (2011b)	Robots and embodied agents	Quantum-like	E1	Conceptual bridge from quantum cognition to AI and robotics	Proposes a two-layer (classical logical plus quantum conceptual) account of thought and quantum-like structure as a route for AI/robotics	No formal model for robots, no implementation or evaluation; speculative claims about computational power of the mind	Full text
Hangl et al. (2016)	Robots and embodied agents	Quantum-inspired	E5	Hierarchical manipulation skill learning by autonomous play	Robot learned sensing and preparation strategies (e.g., flipping to open box); about 80 roll-outs to reach 90% success in simulation	Small hand-designed state and skill sets; classical learning; limited task range	Abstract
Yan et al. (2019)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Representing and transforming robot emotional states	Shows emotion transitions (e.g., anger to relaxation via NOT gates) as unitary operations; argues reduced resource needs	No empirical validation, implementation or classical comparison; benefits require quantum hardware	Abstract
Moreira et al. (2020a)	Robots and embodied agents	Quantum-like	E1	Position on cognitive architectures for decision-making	Proposes quantum-like architecture as a unifying alternative to resource-rational analysis	No formal model specified in detail, no data or implementation; comparative claims not demonstrated	Abstract
Lanza et al. (2020)	Robots and embodied agents	Quantum-like	E2	Robot perception of mutually exclusive percepts	Designed behaviour reproduced in simulation; real quantum hardware showed notable errors	Binary toy scenario; no classical baseline; hardware errors; no demonstrated perceptual benefit	Full text
Hangl et al. (2020)	Robots and embodied agents	Quantum-inspired	E5	Autonomous acquisition of robotic manipulation skills through play	Behaviour composition solved previously unsolvable tasks (e.g., tower disassembly); active learning improved learning speed by about 30% in simulation	Classical PS only; hand-designed skills; supervision needed; limited scalability evidence	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Lanza et al. (2021)	Robots and embodied agents	Quantum-like	E2	Multisensory integration and belief representation in robot perception	Measurement probabilities track similarity of inputs to target states; query operators re-map basis meanings consistently	Simulation only; no temporal integration; single query per prepared state; no classical fusion baseline	Full text
Yan et al. (2021a)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Emotion generation and transition for companion robots	Circuit schemes for generating and transitioning among Plutchik emotions (details not verified)	Apparently formal design only; no verified evaluation or classical comparison	Abstract
Ho and Hoorn (2022)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Modelling robot affect and user-directed attitude in HRI	Complete circuit framework for affective appraisal, involvement–distance and use intentions	No empirical validation or classical comparison; implementation status unclear	Full text
Song et al. (2022)	Robots and embodied agents	Quantum-like	E4	Pedestrian crossing-intention estimation and trajectory prediction for autonomous vehicles	QLB reported to capture boundedly rational crossings and edge events better than classical probability and Social-LSTM	Narrow scenario; interference parameters not clearly learned; quantitative details not verified here	Abstract
Widdows et al. (2023) [†]	Robots and embodied agents	Quantum circuit (QPU)	E5	Implementing quantum cognition models (order effects, disjunction/interference effects) as quantum circuits	Prisoner's Dilemma probabilities (82%, 73%, 64%) reproduced within 0.37%; order-effect outputs within 2–3% of expected values	No new human data or fitting; tiny circuits with no quantum advantage; hardware noise affects order-effect circuits	Full text
Mannone et al. (2024)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Modelling robot 'thinking' and creative behaviour (improvised dance)	Illustrative pose sequences, including feedback-driven unpredictable (creative-like) behaviour and 'overload' responses	Toy model; hand-set states; no behavioural evaluation or classical baseline	Abstract
Daglarli (2025) [†]	Robots and embodied agents	Quantum circuit (design or simulation)	E4	Autonomous behaviour of a smart agent (exploration, survival) in a game simulation	Reported 2× faster convergence to 80% task success and ~15% higher cumulative reward than classical deep RL baseline; gains smaller in reactive, high-uncertainty tasks	Quantum circuits classically simulated; largely qualitative reporting; unclear baselines and statistics; weak justification for algorithm choice	Full text
De Carolis et al. (2025) [†]	Robots and embodied agents	Quantum circuit (design or simulation)	E1	Review and project outline for quantum-enhanced social robotics	Identifies Grover search, quantum walks, PQCs and quantum-inspired emotion models as main tools; recommends hybrid classical–quantum designs given hardware limits	No implementation or evaluation; conflates quantum hardware, quantum-like and quantum-inspired work; limited critical appraisal	Full text
Chella et al. (2026)	Robots and embodied agents	Quantum circuit (design or simulation)	E4	Reactive action selection from robot sensor data (wall following)	Two-sensor configurations best; circuit accuracy about 0.91 (weighted F1 about 0.90); errors near region boundaries	Offline only; no closed-loop robot control; no quantum advantage claimed; hand-defined rule partitions	Abstract
Yukalov et al. (2018)	Multi-agent systems and teams	Quantum-like	E2	Multistep collective decision-making; dynamic disjunction effect	Predicts initial probabilistic deviation, convergence of prospect probabilities to utility factors, and consensus after repeated information exchange; reported to match empirical observations	Stylised interaction rules; validated only against previously published aggregate data	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Ried et al. (2019)	Multi-agent systems and teams	Quantum-inspired	E4	Collective motion / swarm alignment (marching locusts)	Reproduces density-dependent regimes (disorder, switching alignment, stable alignment); responses converge in $\sim 10^2$ steps; drift/diffusion qualitatively match experiments	1-D toy world with two actions; qualitative match only; PS is classical here	Abstract
López-Incera et al. (2020)	Multi-agent systems and teams	Quantum-inspired	E4	Swarm foraging; emergence of collective motion	Distant food: strong alignment (order parameter $\approx 0.8+$), Lévy-like trajectories; near food: weak alignment (≈ 0.2), Brownian trajectories	1-D world, two actions, simple reward; PS run classically	Abstract
Lawless (2020)	Multi-agent systems and teams	Quantum-like	E1	Design and control of autonomous human-machine teams	Argues that interdependence behaves quantum-like and that machines in A-HMTs must communicate causally; no quantitative model fit	Conceptual; analogies not formalised or tested quantitatively; case-study interpretation is post hoc	Abstract
Yan et al. (2021b) [†]	Multi-agent systems and teams	Quantum circuit (design or simulation)	E2	Representing, transforming and fusing emotional states of social/multi-robot systems	Illustrative probability distributions of fused emotions before and after controlled rotations; qubit count grows logarithmically with robot number but preparation cost is $O(2^{4n})$	Conceptual; simulation-only toy examples; no hardware execution, no empirical grounding of emotion model; exponential preparation cost and repeated measurement needed for read-out	Full text
Yukalov et al. (2022)	Multi-agent systems and teams	Quantum-like	E2	Collective decision dynamics in agent networks	Short-memory agents show prolonged probability oscillations; collective information field damps oscillations and forces convergence; reproduces dynamic disjunction effect	Idealised network and communication; no empirical calibration reported in abstract	Abstract
Yukalov (2023)	Multi-agent systems and teams	Quantum-like	E2	Affective (emotion-aware) decision-making for AI	Argues affective AI based on cognition-emotion duality avoids classical behavioural paradoxes	No implementation or empirical test; architecture unspecified	Abstract
Lawless et al. (2023) [†]	Multi-agent systems and teams	Quantum-like	E1	Structure and performance of human-machine teams in open systems	Argues team performance is non-decomposable; proposes trade-off between structural and productive entropy production	Mainly review/position; analogical formalism; no new controlled data	Full text
Essalmi et al. (2025)	Multi-agent systems and teams	Quantum-inspired	E4	Interaction-aware decision-making for automated driving (merging, roundabouts)	QG-G4 reported lower collision and higher success rates than baselines; both quantum models give higher expected payoffs than classical games under some parameters	Simulation only; 2×2 games; parameter-dependent; collision results reported as not reproducible by an independent review	Abstract
Kong et al. (2026)	Multi-agent systems and teams	Quantum-inspired	E4	Group navigation and queue formation in swarms	Polarisation ≈ 0.99 within 100 steps; peak elongation ratio 8.31; optimum at moderate granularity ($m = 15$); polarisation > 0.97 under noise up to 30°	Simulation only; PS classical; no physical robots	Abstract

Study	Domain	Model type	Level	Task or phenomenon	Key result	Main limitation	Basis
Ashtiani and Abdollahi Azgomi (2019)	Human-AI trust	Quantum-like	E4	Computational trust evolution in multi-agent/decision-support systems	Lower MAE and RMSE than BLADE, TidalTrust, max-min, max-mean and Bayesian methods; malicious recommender amplitude ≈ 0.0009 after 10 iterations	Small synthetic scenarios; no human data; no quantum speed-up	Abstract
Pushparaj et al. (2021)	Human-AI trust	Quantum-inspired	E3	Human-automation trust in air traffic control	Acceptance rate under ambiguous conditions 40% per participant; model claimed more robust than questionnaire-based models	Very small sample; no reported quantitative model comparison; limited generalisability	Abstract
Humr et al. (2023)	Human-AI trust	Quantum-like	E2	Temporal evolution of human trust in AI-supported decisions	Proposes approach; no quantitative results in abstract	Short proceedings paper; validation not reported in abstract	Abstract
Roeder et al. (2023)	Human-AI trust	Quantum-like	E3	Trust/reliability calibration in human-AI interaction	Quantum model predicted evolving reliability ratings better than Markov model; clear ERP when expectations were perturbed	Exploratory; N = 7; scripted AI behaviour	Abstract
Humr et al. (2025) [†]	Human-AI trust	Quantum-like	E2	Human-in-the-loop decision-making with AI support	Illustrative simulations of oscillating confidence; cites ~15-point question-order effect captured by QP models	No new empirical data; proposals not implemented in an AI system	Full text
van der Meer et al. (2025)	Human-AI trust	Quantum-like	E3	Dynamics of human trust/reliability ratings in human-AI interaction	Interaction-dependent Hamiltonians reported as a promising means to model trust evolution; no numbers in abstract	Preprint; quantitative evidence not given in abstract; small prior dataset	Full text
Humr and Canan (2026)	Human-AI trust	Quantum-like	E1	AI-supported (military) decision-making; human-AI alignment	Three GenAI applications failed to identify conjunction fallacies; QCAL proposed with research agenda	QCAL not implemented or evaluated; GenAI probe small and qualitative	Abstract
Kak (2018)	Decision models and networks	Quantum-like	E2	Order effects in memory and belief queries of intelligent agents	Argues that inaccessible agent states may be best modelled as quantum states; proposes tests on large medical-treatment databases	No data, implementation or quantitative comparison	Abstract
Hoorn and Ho (2019)	Robots and embodied agents	Quantum circuit (design or simulation)	E2	Concurrent (mixed) emotions and reflection in robot affect	Circuit examples of robot behaviour under superposed affective states	No simulation results, robot or user study; physical claims about neural quantum dynamics are speculative	Abstract
Essalmi et al. (2026) [†]	Multi-agent systems and teams	Quantum-inspired	E4	Interaction-aware decision-making for automated driving with several players	Higher success rates and fewer collisions than classical approaches, especially in highly interactive scenarios	Simulation only; no human drivers or vehicle tests	Abstract
Nebli et al. (2026)	Deep learning and LLMs	Quantum-inspired	E2	Sequence (language) modelling with interference between token paths	Born-rule decoding that keeps phase information is provably more expressive on the constructed task family	No experiments; relevance of interference to natural language untested	Abstract

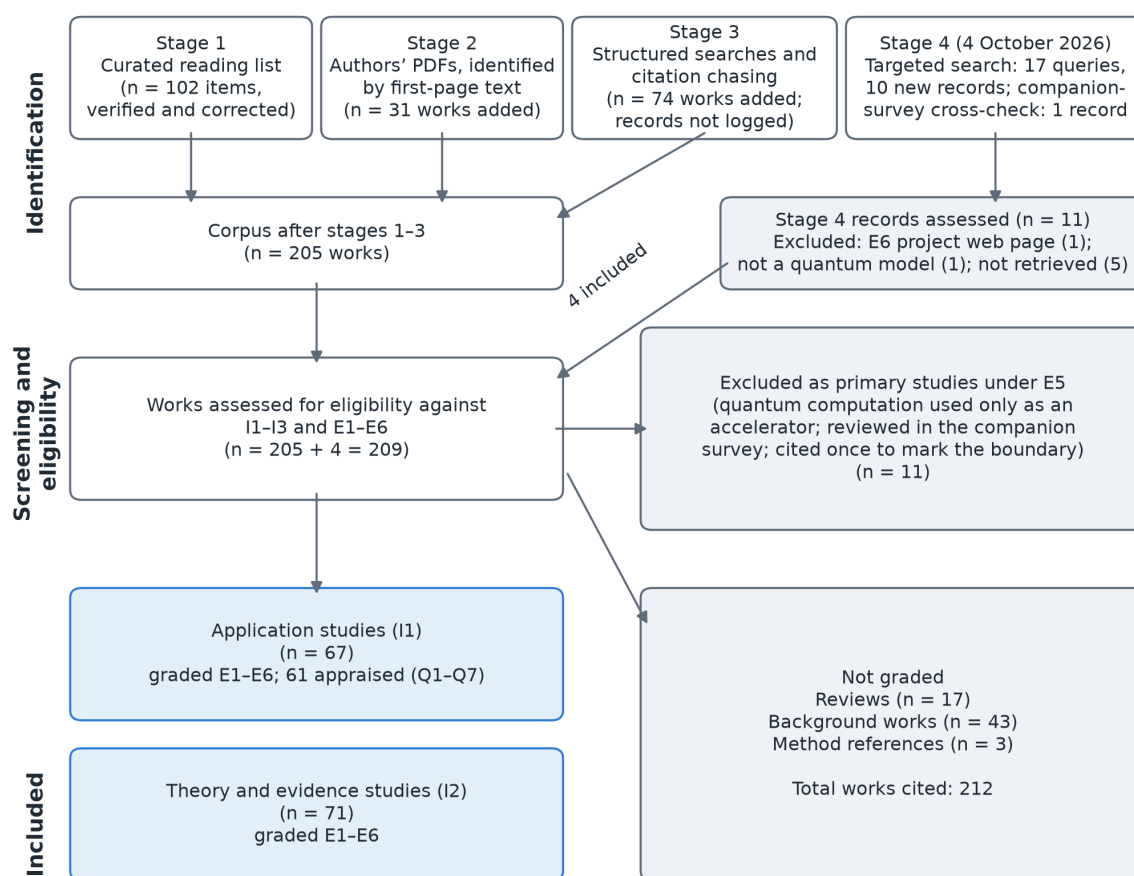
Table S2: PRISMA 2020 checklist: location of each item in the main article and this resource.

Item	Topic	Location
1	Title	Title page (identified as a systematic review)
2	Abstract	Abstract (unstructured, as required by the journal)
3	Rationale	Sections 1.1–1.4
4	Objectives	Section 1.5.1 (RQ1–RQ5)
5	Eligibility criteria	Section 1.5.3 (I1–I3, E1–E6)
6	Information sources	Section 1.5.2; Table S6 (sources and dates)
7	Search strategy	Section 1.5.2; Table S6 (concept blocks of stage 3; exact queries of stage 4)
8	Selection process	Section 1.5.4
9	Data collection process	Section 1.5.5
10a	Data items (outcomes)	Section 1.5.5; Table S1; Online Resource 3
10b	Data items (other variables)	Sections 1.5.5 and 1.5.6 (model type, domain, publication status, basis)
11	Study risk of bias assessment	Section 1.5.5 (criteria Q1–Q7); Tables S4 and S5
12	Effect measures	Not applicable: heterogeneous tasks and metrics; no pooled effect (Section 1.5.6)
13a–f	Synthesis methods	Narrative synthesis by domain and evidence level (Sections 1.5.6 and 13.3); meta-analysis, heterogeneity and sensitivity analyses not applicable
14	Reporting bias assessment	Section 1.5.6 (threats to validity: publication bias)
15	Certainty assessment	Section 1.5.6 (evidence levels E1–E6, Table 1)
16a	Study selection	Sections 1.5.2–1.5.4; Figure S1; Table S3
16b	Excluded studies	Section 1.5.3; Tables S6 and S7
17	Study characteristics	Table S1; Online Resource 3
18	Risk of bias in studies	Section 13.3; Table S5; Online Resource 4
19	Results of individual studies	Table S1; Section S9; Sections 5–12
20a–d	Results of syntheses	Sections 5–12; Section 13.3 and Figure 7
21	Reporting biases	Section 1.5.6
22	Certainty of evidence	Section 13.3; Table 1
23a–d	Discussion	Sections 13–15, 17, 18 and 24; limitations of the review in Section 1.5.6
24a	Registration	Section 1.5 (retrospective OSF registration)
24b	Protocol	Section 1.5 (OSF registration)
24c	Amendments	Sections 1.5.2 and 1.5.3: criterion E5 and the stage 4 search were added after the first corpus was assembled
25	Support	Declarations (no external funding)
26	Competing interests	Declarations
27	Availability of data, code and other materials	Declarations; Online Resources 2–4

Table S3: Flow of records. Counts for stages 1–3 are works retained after verification; the number of records each search returned and the number excluded at each screening step were not logged and cannot be reported.

Stage	Count
Stage 1: curated reading list, checked against publisher, arXiv or repository records	102 items listed; about 20 corrected, 3 unlocatable items replaced by 2 verified papers, 2 non-existent titles replaced, 2 dropped
Stage 2: authors' PDFs identified by first-page text	31 works added
Stage 3: structured searches and citation chasing (two streams)	74 works added; records not logged
Corpus after stages 1–3	205 works

Stage 4: targeted search (4 October 2026) and companion-survey cross-check	17 queries; 10 new records + 1 record; 4 included, 2 excluded, 5 not retrieved (Table S6)
Works assessed against I1–I3 and E1–E6	209
Excluded as primary studies under E5 (cited once; reviewed in the companion survey)	11 (Table S7)
Application studies (I1), graded	67
of which appraised against Q1–Q7 (levels E2–E6)	61
Theory and evidence studies (I2), graded	71
Reviews and background works (I3), not graded	17 and 43
Methodological references	3
Total works cited	212



Counts for stages 1-3 are works added after verification; the number of records each search returned and excluded at each screening step was not logged (Online Resource 1, Table S3).

Figure S1: Flow of records, following the PRISMA 2020 flow diagram. Stages whose record counts were not logged are marked as such (Table S3).

Table S4: Quality-appraisal criteria for application studies. Each criterion was judged met (M), partly met (P) or not met (N); the judgements were not summed into a score. Q6 cannot be judged from an abstract and is marked “?” when the full text was not available.

ID	Criterion	Met (M)	Partly met (P)	Not met (N)
Q1	Model specification	State space, state preparation, operators (projectors, instruments, unitaries or Hamiltonians) and free parameters stated.	Model only sketched, or parameters not all stated.	Model not specified.
Q2	Classical comparison	A classical Bayesian, Markov, utility or neural model evaluated on the same task and data.	Classical model discussed but not evaluated on equal terms, or only weak baselines.	No classical comparison.
Q3	Empirical grounding	Human behavioural or neural data, or a realistic agent or AI task.	Narrow benchmark, aggregate data from a few classic paradigms, or a single small case.	Toy example or no data.
Q4	Parameter discipline	Parameters predicted or constrained a priori (e.g., QQ test, phase heuristics), or the model validated on held-out data.	Some parameters constrained, others fitted to the data they explain.	Free parameters fitted to the data they explain, or no evaluation.
Q5	Quantitative evaluation	Metrics reported with variability (repetitions, confidence intervals or tests).	Metrics without variability, or partly qualitative.	Qualitative only.
Q6	Reproducibility	Code and data released.	Code or data released, or the model fully specified in the paper so that it can be re-implemented.	Neither released (“available on request” counts as not met).
Q7	Critical discussion	Limitations and classical alternative explanations discussed in depth.	Limitations mentioned briefly.	No discussion of limitations.

Table S5: Quality appraisal of the 61 application studies at levels E2–E6 (the six conceptual E1 studies are not appraised). M = met; P = partly met; N = not met; ? = not assessable without the full text. Basis: source of the judgements.

Study	Basis	Q1	Q2	Q3	Q4	Q5	Q6	Q7
Moreira and Wichert (2014)	Abstract	M	M	N	N	P	?	P
Moreira and Wichert (2016)	Abstract	M	P	P	M	P	?	P
Moreira (2017)	Abstract	M	P	P	P	P	?	P
Moreira et al. (2020b)	Abstract	M	M	P	P	P	?	P
Wichert et al. (2020)	Abstract	M	P	P	M	P	?	P
She et al. (2021)	Abstract	M	M	P	P	P	?	N
Meghdadi et al. (2022)	Abstract	M	M	P	N	M	?	P
Gao et al. (2025)	Abstract	M	M	P	P	M	?	P
Widdows and Peters (2003)	Abstract	M	N	P	N	N	?	P
Sordoni et al. (2013)	Abstract	M	M	M	M	M	?	P
Lorenz et al. (2023)	Full text	M	P	P	M	M	M	M
Svozil (2025)	Full text	P	N	N	N	N	N	P
Stoudenmire and Schwab (2016)	Abstract	M	P	M	M	M	?	P
Zhang et al. (2018)	Abstract	M	M	M	M	M	?	P
Li et al. (2019)	Abstract	M	M	M	M	M	?	P
Li et al. (2021a)	Abstract	M	M	M	M	M	?	P
Li et al. (2021b)	Abstract	M	M	M	M	M	?	P
Gkoumas et al. (2021)	Abstract	M	M	M	M	M	?	P

Study	Basis	Q1	Q2	Q3	Q4	Q5	Q6	Q7
Upreti et al. (2024)	Abstract	P	P	M	N	P	?	P
Zhang et al. (2025)	Abstract	P	M	M	M	M	?	?
Maksimovic and Maksymov (2025)	Abstract	M	M	N	N	P	?	P
Lo et al. (2025)	Full text	M	N	M	M	M	M	M
Aerts et al. (2026)	Abstract	P	N	P	M	P	?	P
Imannezhad et al. (2026)	Abstract	M	M	M	M	M	?	M
Kang (2026)	Abstract	M	M	P	M	M	?	M
Dong et al. (2008)	Abstract	M	M	N	P	P	?	P
Dong et al. (2012)	Full text	M	P	M	P	P	N	P
Briegel and De las Cuevas (2012)	Full text	M	N	N	N	P	N	P
Melnikov et al. (2018)	Abstract	M	M	P	P	M	?	P
Wei et al. (2022)	Abstract	M	M	M	M	M	?	P
Lukac and Perkowski (2007)	Abstract	P	N	N	N	N	?	N
Hangl et al. (2016)	Abstract	M	P	M	P	M	?	P
Yan et al. (2019)	Abstract	M	N	N	N	N	?	N
Lanza et al. (2020)	Full text	M	N	P	N	P	P	P
Hangl et al. (2020)	Abstract	M	P	M	P	M	?	P
Lanza et al. (2021)	Full text	M	N	P	N	P	M	P
Yan et al. (2021a)	Abstract	P	N	N	N	N	?	?
Ho and Hoorn (2022)	Full text	M	P	N	N	N	N	P
Song et al. (2022)	Abstract	P	M	M	N	P	?	P
Widdows et al. (2023)	Full text	M	P	P	N	M	P	M
Mannone et al. (2024)	Abstract	P	N	N	N	N	?	P
Daglarli (2025)	Full text	P	P	P	N	P	N	N
Chella et al. (2026)	Abstract	M	M	P	M	M	?	M
Yukalov et al. (2018)	Abstract	M	P	P	P	P	?	P
Ried et al. (2019)	Abstract	M	P	P	P	P	?	P
López-Incera et al. (2020)	Abstract	M	N	P	P	M	?	P
Yan et al. (2021b)	Full text	M	N	N	N	P	N	P
Yukalov et al. (2022)	Abstract	M	N	N	P	P	?	P
Yukalov (2023)	Abstract	P	N	N	N	N	N	P
Essalmi et al. (2025)	Abstract	M	M	P	N	M	?	P
Kong et al. (2026)	Abstract	M	N	P	P	M	?	P
Ashtiani and Abdollahi Azgomi (2019)	Abstract	M	M	P	N	M	?	P
Pushparaj et al. (2021)	Abstract	P	N	M	N	P	?	P
Humr et al. (2023)	Abstract	P	N	N	N	N	?	P
Roeder et al. (2023)	Abstract	M	M	M	P	M	?	P
Humr et al. (2025)	Full text	M	P	N	N	P	N	P
van der Meer et al. (2025)	Full text	M	N	M	P	M	N	P
Kak (2018)	Abstract	P	P	N	N	N	?	P
Hoorn and Ho (2019)	Abstract	P	N	N	N	N	?	P
Essalmi et al. (2026)	Abstract	M	M	P	N	M	?	P
Nebli et al. (2026)	Abstract	M	N	N	M	N	?	M

Totals (met / partly met / not met / not assessable): Q1 47/14/0/0; Q2 24/16/21/0; Q3 19/24/18/0; Q4 18/16/27/0; Q5 27/22/12/0; Q6 3/2/9/47; Q7 7/48/4/2.

Search strategy

Stages 1–3. Sources: Google Scholar, Semantic Scholar, arXiv, PubMed and publisher sites, with backward and forward citation chasing from the main reviews (Pothos and Bussemeyer, 2022; Huang et al., 2025b; Widdows et al., 2021; Meyer et al., 2022). The two search streams covered (a) applications of quantum cognition to robots, agents, human–machine teams and reinforcement learning, and (b) quantum-cognition theory, critiques, machine learning, large language models and neuroscience published from 2016 to 2026. The exact strings and record counts were not logged. To support replication, the following concept blocks reproduce the scope of the two streams; they are a reconstruction, not a log:

("quantum cognition" OR "quantum-like" OR "quantum probability" OR "quantum decision theory" OR "quantum-inspired" OR "projective simulation")
AND(robot* OR agent* OR "human-robot" OR autonomous OR "multi-agent" OR swarm OR "reinforcement learning" OR "machine learning" OR "language model*" OR trust OR automation)
or, for stream (b), AND ("order effect*" OR "conjunction fallacy" OR "disjunction effect" OR "sure-thing" OR interference OR contextuality OR "Bayesian network*" OR critique)

Stage 4 (4 October 2026). Exact queries, results and decisions are given in Table S6.

Table S6: Targeted search of 4 October 2026: queries and decisions. The full log, including the verification queries, is in the method workbook (targeted_search_record.md).

#	Query	New records and decisions
T1	quantum-like model human-robot interaction decision-making robot 2024 2025	Genoa research-project page (excluded, E6)
T2	quantum cognition large language model order effects 2025 arXiv quantum probability	Kak (2018) and Nebli et al. (2026) (included); arXiv:2510.13894 (not retrieved)
T3	quantum probability model trust automation human-autonomy teaming 2024 2025 study	Flight-test trust case study (excluded, not a quantum model); arXiv:2410.20496 and arXiv:2503.16227 (not retrieved)
T4	quantum-like Bayesian network decision support autonomous driving or robot 2023 2024 2025	No new record
T5	quantum decision theory artificial agents reinforcement learning quantum-like agent Yukalov 2024 2025	No new record
T6	quantum-inspired cognitive model robot emotion social robot 2024 2025 quantum-like	Hoorn and Ho (2019) (included); J. Phys.: Conf. Ser. 2115, 012040 (not retrieved)
V1–V11	Eleven verification queries (titles, authors and abstracts of the records above)	arXiv:2608.14691 seen as a title only (not retrieved)
Other	Cross-check against the companion survey's corpus	Essalmi et al. (2026) (included)

Table S7: Studies excluded as primary studies under criterion E5 (quantum computation used only to accelerate a robot or learning task, without a cognitive or decision model). They are reviewed in the companion survey (Guru and Chauhan, 2026) and cited once in the main article to mark the boundary.

Study	Method
Chen et al. (2020)	Variational quantum circuits as Q-function approximators
Jerbi et al. (2021)	Parametrised quantum policies; learning separations on constructed tasks
Skolik et al. (2022)	Variational deep Q-learning (CartPole)
Acuto et al. (2022)	Variational quantum soft actor-critic for a simulated arm
Sinha et al. (2025)	Quantum critic for self-driving navigation (Nav-Q)
Hohenfeld et al. (2024)	Variational deep Q-learning for robot navigation
Yun et al. (2022)	Quantum multi-agent reinforcement learning
Paparo et al. (2014)	Quantum speed-up of agent deliberation (quantum walks)
Chella et al. (2022)	Grover search for robot motion planning
Mannone et al. (2022)	Quantum-circuit models of swarm interaction rules
Khoshnoud et al. (2020)	Quantum communication for cooperating robots

Table S8: Theory and evidence studies ($n = 71$), graded on the evidence scale: phenomenon, key result and basis of extraction.

Study	Level	Phenomenon	Key result	Basis
Khrennikov (1999)	E2	Formal description of cognitive and social information processes	No empirical results; provides a p-adic formalism for information dynamics	Full text
Aerts et al. (2000)	E2	Logical structure of disjunction; concept combination	Quantum disjunction shown non-equivalent to classical 'or' owing to EPR-like correlations	Abstract
Busemeyer et al. (2006)	E2	Choice and response-time dynamics in decision-making	Quantum dynamics produce interference effects not possible under Markov models; predictions contrasted	Abstract
Pothos and Busemeyer (2009)	E3	Sure-thing principle violations in prisoner's dilemma and two-stage gambles	Interference reproduces violations of the law of total probability that classical models cannot	Abstract
Busemeyer et al. (2009)	E3	Choosing between Markov and quantum dynamic models of decision	Provides parameter-free tests discriminating Markov vs quantum models; evidence reviewed on both properties	Abstract
Aerts (2009)	E3	Concept combination; membership weights for conjunctions and disjunctions	Data shown incompatible with classical models; quantum model reproduces over- and underextension	Full text
Khrennikov (2010)	E2	Foundations of quantum-like modelling in psychology, economics and finance	Systematic quantum-like framework linking contextuality and interference to QP representations	Abstract
Trueblood and Busemeyer (2011)	E3	Order effects in sequential evidence evaluation (inference)	Quantum model reproduced the pattern of order effects better than the anchoring-and-adjustment model (per Bruza 2015)	Abstract
Busemeyer et al. (2011)	E3	Probability judgement fallacies (conjunction, disjunction, inverse, conditional)	Quantum model explains more of 15 major findings than three rival theories	Full text
Aerts and Sozzo (2011)	E3	Concept combination; entanglement between concepts	CHSH value $2.4197 > 2$, interpreted as violation of Bell's inequality	Full text
Aerts et al. (2011c)	E2	Concept combination, decision paradoxes and information retrieval	Argues quantum structure is a structural model for human and artificial thought; proposes SCoP-based knowledge organisation	Full text
Aerts et al. (2011a)	E3	Sure-thing principle violations; disjunction effect (Hawaii problem)	Quantum model reproduces 54%/57% vs 32% choice pattern as interference of conceptual landscape	Full text
Yukalov and Sornette (2011)	E3	Decision under risk and uncertainty; classical paradoxes	Explains disjunction effect, conjunction fallacy, Allais and Ellsberg-type paradoxes	Abstract
Busemeyer and Bruza (2012)	E3	Order effects, judgement errors, concepts, memory, decision dynamics	Unified QP account of multiple anomalies; comparisons with classical/Markov models	Full text
Pothos and Busemeyer (2013)	E3	General case for QP cognitive modelling	Argues many judgement and decision anomalies have natural QP accounts	Full text
Wang et al. (2013)	E1	Programmatic case for quantum models of cognition	Frames QP as a general cognitive modelling framework and lists open questions	Abstract
Wang and Busemeyer (2013)	E3	Question order effects in attitude surveys	QQ equality supported in the data examined	Abstract
Aerts et al. (2013)	E3	Concept combination and conceptual dynamics	Quantum models reproduce guppy-type effects, over/underextension and entanglement	Abstract
Wang et al. (2014)	E3	Question order effects in national surveys	QQ equality statistically supported; anchoring-adjustment models fail to reproduce patterns	Abstract
Khrennikov et al. (2014)	E2	Consistency of quantum measurement models with order effects and response replicability	Projective quantum models cannot jointly reproduce order effects and response replicability	Abstract
Bruza et al. (2015)	E3	Review of quantum vs Bayesian models of judgement and decision	QP accounts for order effects, QQ equality (70 surveys), total-probability violations; classical and QP coincide for compatible events	Full text
Busemeyer and Wang (2015)	E1	Introduction to quantum cognition for psychologists	Summarises QP explanations of order effects, fallacies and sure-thing violations	Abstract
Kvam et al. (2015)	E3	Choice-confidence interference in perceptual evidence accumulation	Intermediate choice changed later confidence (interference); quantum walk fitted better than Markov	Abstract
Yearsley and Busemeyer (2016)	E2	Tutorial on constructing quantum cognitive models	Step-by-step construction of order-effect models and derivation of the QQ equality	Abstract
Wang and Busemeyer (2016)	E3	Interference of categorisation on decision-making	Interference (violation of total probability) in some conditions; BAE model predicted the pattern	Abstract
Yearsley and Pothos (2016)	E3	Effect of repeated judgements on belief change (quantum Zeno effect)	Frequent judgements associated with slower belief change, consistent with a Zeno effect	Abstract

Study	Level	Phenomenon	Key result	Basis
Yukalov and Sornette (2016)	E2	Foundations of quantum probability for measurement and decision-making	Predicted 0.35/0.65 vs observed 0.37/0.63 cooperation/defection; conditions for classical reduction	Full text
Dzhafarov and Kujala (2016)	E2	Defining and testing contextuality in systems of random variables	Criterion distinguishing true contextuality from direct context influences on marginals	Abstract
Yearsley (2017)	E2	Tutorial on advanced quantum-cognition tools	Framework for incorporating noise and imperfect measurement in QP models	Abstract
Pothos et al. (2017)	E2	Normative (rational) status of quantum probabilistic judgement	Argues QP reasoning can be principled rather than simply erroneous under incompatible perspectives	Abstract
Trueblood et al. (2017)	E3	Probabilistic inference judgements (order effects, inverse fallacy, Markov violations, anti-discounting)	Accounts for five inference phenomena; representations become more classical with task familiarity	Abstract
Broekaert et al. (2017)	E2	Time evolution of mental states in decision-making	Clarifies relations and explanatory value of increasingly complex dynamics	Abstract
Aerts et al. (2017)	E4	Conjunction fallacy	Emergence effects, not order effects, identified as the main mechanism	Abstract
Aerts and Sassoli de Bianchi (2017)	E3	Question order effects and response replicability in sequential survey judgements	Exact fits to both data sets; argues the data are irreducibly non-Hilbertian, that the QQ-equality can hold for non-Hilbertian models, and that order effects plus replicability are jointly modelled	Full text
Bussemeyer and Wang (2018)	E3	Context effects across subsets of judged variables	HSM fits compare favourably with Bayesian networks, demonstrating viability	Abstract
Cervantes and Dzhafarov (2018)	E3	Contextuality in paired human choices	Marginal selectivity violated, yet true contextuality detected beyond direct influences	Abstract
Bussemeyer et al. (2019)	E3	Belief change during evidence monitoring; sequential probability ratings	Small interference effects (low coherence; 3/11 participants); summed G^2 diff favoured quantum (350, 306, 260, 40), but versus Markov with drift variability quantum wins only at low coherence	Full text
Basieva et al. (2019)	E3	True contextuality versus direct influences in human decisions	Prior claimed demonstrations explained by direct influences; new experiments show true contextuality	Abstract
Bussemeyer et al. (2020)	E2	Dynamics of choice, confidence and decision time	Unifies random-walk and quantum-walk dynamics in one process	Abstract
Ozawa and Khrennikov (2020)	E2	Question order effect combined with response replicability	Shows both effects can be jointly modelled, overturning the no-go for this class	Abstract
Yukalov (2020)	E2	Decision making over time; dynamic inconsistency	Argues dynamic inconsistencies are resolved in QDT; quantum-classical correspondence	Abstract
Kvam et al. (2021)	E3	Constructed preference and effect of prior choice on preference strength	Oscillating preference strength and choice-induced shifts captured by the open-system model	Abstract
Ozawa and Khrennikov (2021)	E3	Joint modelling of question order effect, response replicability and QQ-equality	Instruments satisfying QOE, RRE and QQE exist; model closely reproduces Clinton–Gore statistics with an order-independent prior state	Full text
Epping et al. (2023)	E3	Preferential choice and response time	Pure Markov model favoured for most participants; a substantial number required the open system model	Abstract
Khrennikov (2023)	E2	Quantum-like modelling of biological and cognitive information processing	Argues open-system formalism is the natural basis for quantum-like biology and cognition	Abstract
Khrennikov (2024)	E2	Foundational framework for contextual probability update covering classical and quantum measurements	Kolmogorov, von Neumann and quantum-instrument measurements framed within CMM; contextual definitions of order effect, response replicability, interference and Bell-type violations	Full text
Waddup et al. (2023)	E3	Constructive memory; sequential measurement effects	Violation of a temporal Bell inequality observed in one case	Abstract
Veloz and Sobetska (2024)	E3	Conjunction fallacy: coverage of possible probability-judgement patterns in the literature	Data cluster in a few sub-volumes; single fallacy well explored, no-fallacy and double-fallacy regions under-explored	Full text
Huang et al. (2025a)	E3	Probability judgement fallacies	Model compared favourably with the Bayesian Sampler; systematic overestimation of probabilities discovered	Abstract
Huang et al. (2025c)	E2	Recognition heuristic and hindsight bias in paired-comparison judgements	Deterministic heuristic recovered under measurement; probabilistic choice and hindsight bias emerge from superposition and projective update	Full text
Fuyama et al. (2025)	E2	Characterising measurement classes for question order, response replicability and QQ-equality	$SR\bar{P}$ measurements reproduce QOE with RRE and can satisfy or violate the QQ-equality; non-classicality can arise from non-commuting state-update maps with commuting observables	Full text

Study	Level	Phenomenon	Key result	Basis
Asano and Khrennikov (2026)	E2	Temporal dynamics of deliberation and decision, including strategic games	Framework linking GKSL steady states to decisions, stabilisation of non-Nash outcomes, and predicted beat envelopes in conviction	Full text
Bruza et al. (2009)	E2	Word association and cued recall in the mental lexicon	Initial quantum models outlined and experimental tests proposed	Abstract
Blutner et al. (2013)	E2	Borderline vagueness and borderline contradictions	Classical model falls short; quantum interference explains borderline contradictions	Abstract
Pothos et al. (2013)	E2	Similarity judgement (asymmetry, triangle inequality, diagnosticity)	Asymmetry as order effect; triangle-inequality and diagnosticity effects accommodated	Abstract
Trueblood and Hemmer (2017)	E3	Episodic recognition memory: over-distribution and subadditivity	GQEM accounts for over-distribution and subadditivity	Abstract
Busemeyer et al. (2025)	E2	Effect of episodic memory of prior judgements on later judgements (AA, AB/BA, ABA paradigms)	All three accounts cover the combined findings but differ in mechanism (belief-type updates vs non-commuting projections) and in allowing changed answers on repeated questions	Full text
Costello and Watts (2014)	E3	Biases in probability judgement	Near-normative judgements for noise-cancelling expressions; four biases explained by noise	Abstract
Boyer-Kassem et al. (2016)	E3	Conjunction fallacy	Results at odds with quantum-like model predictions	Abstract
Dzhafarov et al. (2016)	E3	Contextuality in behavioural and social data	No evidence of true contextuality in the reviewed datasets	Abstract
Moreira and Wichert (2017)	E3	Question order effects in opinion polls	Both models reproduce the order effects; interference terms play no role, so the classical projection model suffices	Full text
Busemeyer and Wang (2017)	E3	Sequential opinion measurement (ABA design)	Moderate changes in repeated A judgements; correlation pattern consistent with quantum predictions	Abstract
Kellen et al. (2018)	E2	Mirrored question order effects (QQ equality)	Classical models can reproduce mirrored order effects	Abstract
Costello and Watts (2018)	E3	Invariants in probabilistic reasoning (QQ, addition law, Bayes rule identities)	Identities hold despite biases; consistent with noisy classical reasoning	Abstract
Martínez-Martínez and Sánchez-Burillo (2016)	E2	Choice under uncertainty; Prisoner's Dilemma (sure-thing principle violation)	Stationary distribution of the QSW yields choice probabilities that can violate the law of total probability / sure-thing principle; unique stationary state proved (Spohn's theorem)	Abstract
Waddup et al. (2021)	E3	Contextuality in two-person strategic interaction	Evidence of sensitivity to context; quantum model fitted better than a matched classical model	Abstract
Busemeyer et al. (2017)	E2	Neural implementation of quantum cognition operations	Proposes classical neural realisations of quantum-cognition operations (no quantitative results)	Abstract
Khrennikov et al. (2018)	E2	Neural basis of quantum-like decision statistics	Decision state is a steady state reached with exponentially fast convergence (analytical)	Abstract
Khrennikov et al. (2025b)	E2	Neural grounding of quantum-like cognition	Formal coupling proposed; no quantitative results known	Abstract
Khrennikov et al. (2025a)	E2	Neural oscillations as a substrate for quantum-like cognition and AI	Conceptual/formal bridge proposed; no quantitative results known	Abstract
Khrennikov and Yamada (2025)	E2	Neural grounding of quantum-like entanglement in cognition ('mental entanglement')	Formal construction showing how classical neuronal networks can yield QL entangled states and observables; mapping to EEG/MEG functional connectivity measures	Full text

S9 Study summaries

One summary per graded study: the 67 application studies, grouped by domain as in Table S1, and the 71 theory and evidence studies. The level, the model type and the basis of each summary (full text or abstract) are given in brackets; summaries based on an abstract should be read with that limitation. The same text, with summaries of the reviews and background works, is in the data package (study_summaries.csv).

S9.1 Application studies

Decision models and networks

Moreira and Wichert (2014) (E2; Quantum-like; abstract). Moreira and Wichert (2014) argued that classical probabilistic graphical models cannot fully simulate human decision making, because people rely on limited data and heuristics that lead to violations of the law of total probability. Drawing on findings that quantum probability can describe such decisions, with thoughts treated as superposed waves that interfere before a choice, they proposed a quantum-like Bayesian network for computer science, which they describe as

the first such probabilistic system in that field. In the network, classical probabilities are replaced by quantum amplitudes, so that inference over unobserved variables includes interference terms. The authors tested the approach on two well-known Bayesian networks from the literature. The results showed that the quantum-like network can drastically change probabilistic inferences when uncertainty is very high, that is, when no evidence is observed, whereas under low uncertainty it collapses to its classical counterpart. All computation runs on classical hardware. A limitation is that the evaluation uses textbook networks rather than human data, and the interference phases are free parameters without a principled setting, a problem later addressed by similarity and balance heuristics (Moreira and Wichert, 2016; Wichert et al., 2020). For autonomous agents, the paper establishes the basic mechanism by which a graphical belief model can depart from classical inference precisely when evidence is scarce.

Moreira and Wichert (2016) (E3; Quantum-like; abstract). Moreira and Wichert (2016) proposed a quantum-like Bayesian network (QLBN) to accommodate paradoxical findings that violate Savage’s sure-thing principle, such as the disjunction effect. The network keeps the directed-graph structure of a classical Bayesian network but replaces classical probabilities with quantum probability amplitudes, so that marginalising over an unobserved node produces interference terms. Because such networks suffer from exponential growth in the number of quantum phase parameters, the authors introduce a similarity heuristic that sets the phases automatically from vector similarities between the states involved. This, they argue, makes the model general and predictive, whereas earlier quantum models were explanatory only and did not generalise to more complex decision scenarios. The model is evaluated against published experimental results from the Prisoner’s Dilemma game (five studies) and the Two-Stage Gambling game (three studies), where it approximates the observed violations. A limitation is that the heuristic’s psychological justification is weak and predictions depend on how states are embedded as vectors; the evaluation also relies on aggregate data from a small set of classic paradigms. All computation runs on classical hardware. For autonomous agents, the QLBN offers a graph-based inference engine able to reproduce human-like deviations from Bayesian updating, which is useful for modelling human partners’ choices under uncertainty.

Moreira (2017) (E3; Quantum-like; abstract). Moreira (2017) is Catarina Moreira’s doctoral thesis at Instituto Superior Técnico, which brings together her work with Andreas Wichert on quantum probabilistic graphical models for cognition and decision. The thesis develops quantum-like Bayesian networks, in which classical probabilities are replaced by quantum amplitudes so that interference terms arise during inference, and addresses the problem of setting the resulting phase parameters, including through heuristic approaches. It applies these models to paradoxical decision findings, notably violations of the sure-thing principle in the Prisoner’s Dilemma and two-stage gambling paradigms, building on the published papers (Moreira and Wichert, 2014) and (Moreira and Wichert, 2016). All models are quantum-like, implemented on classical computers. Its limitations mirror those of the constituent papers: evaluation on a small set of classic paradigms with aggregate data, and parameter-setting heuristics with limited psychological grounding. For autonomous agents, the thesis is the most complete single reference on quantum-like graphical models, which are the natural bridge between human-like probabilistic reasoning and the Bayesian-network machinery used in robotics and AI decision support.

Moreira et al. (2020) (E3; Quantum-like; abstract). Moreira et al. (2020b) proposed a framework for describing the probabilistic nature of decision making by extending quantum-like Bayesian networks with the notion of maximum expected utility, forming quantum-like influence diagrams. The key idea is that quantum interference effects produced during inference in the quantum-like Bayesian network alter the probabilities used to compute the expected utility of each decision, allowing the model to represent paradoxical, sub-optimal and irrational choices. The authors report that the model can predict the probability distributions found in different experiments from the literature by modelling uncertainty through interference; can identify decisions that decision makers perceive as optimal within their belief space but that are irrational with respect to expected utility theory; and can show how a decision maker’s beliefs evolve within a scenario. The model runs on classical hardware and is evaluated against published experimental data rather than new studies. A limitation is that interference parameters must still be set or fitted and the tested scenarios are classic laboratory paradigms. The authors highlight implications for decision support systems handling human data. For autonomous agents, influence diagrams are directly usable in planning, so this model offers a practical path to embedding human-like utility distortions in agent decision support.

Wichert et al. (2020) (E3; Quantum-like; abstract). Wichert et al. (2020) addressed a weakness of quantum-like Bayesian networks: inference relies on a Bayes normalisation factor to turn interference-affected likelihoods into probabilities, an operation with unclear interpretation that produces extremely skewed intensity waves and makes it hard to predict irrational decisions. The authors propose the law of balance, a formalism in

which the intensity waves produced by quantum interference are balanced so that they cancel during normalisation. On this representation they add the law of maximum uncertainty, which predicts paradoxical choices by selecting the wave amplitudes with the highest entropy, thereby removing free phase parameters. The authors report that the two laws together accurately predict experiments showing paradoxical or irrational decisions in the Prisoner’s Dilemma game and the Two-Stage Gambling game. The approach runs entirely on classical computers. A limitation is that validation is confined to these two classic paradigms and to aggregate probabilities, and the maximum-entropy choice is a modelling assumption rather than an empirically derived one. The paper is also somewhat technical in its notation. For autonomous agents, parameter-free prediction of human deviations from classical probability is attractive, because an agent could anticipate a partner’s choices without fitting phases to each new situation.

She et al. (2021) (E4; Quantum-like; abstract). She et al. (2021) addressed multi-attribute decision making (MADM), where conventional methods assume that attributes are independent even though they often interact. The authors propose a MADM model in which interference effects between attributes are represented with a quantum-like Bayesian network, following the view that quantum probability better reflects human cognition. The magnitude of interference is expressed by an interference value that is calculated using belief entropy, a measure from evidence theory, rather than being fitted freely. The model is applied to an assessment of industrial economic benefit, where the authors report better predictive capability than classical Bayesian network approaches. The method is quantum-like and runs on classical computers. A limitation is that evaluation rests on a single application domain, and it is not clear that the interference captured corresponds to human cognitive interference rather than statistical dependence among attributes. For autonomous systems, the paper illustrates how quantum-like interference terms are being carried from cognitive modelling into engineering decision support, where attributes are correlated and uncertain.

Meghdadi et al. (2022) (E3; Quantum-like; abstract). Meghdadi et al. (2022) proposed a quantum-like cognitive framework for modelling biased human selection behaviour, arguing that choices usually labelled irrational can be treated as systematically biased by emotion and past experience. Their model, a Biased Entangled Quantum-like Bayesian Network (BEQBN), simulates society as a quantum system: decision makers are represented within an entangled quantum-like Bayesian network, and phase parameters together with bias potential functions encode emotional influences and prior experience. The model runs on classical computers and is evaluated on two empirical decision tasks drawn from the literature, including the Prisoner’s Dilemma. The authors report a root-mean-square error of 3.4 on the Prisoner’s Dilemma task, ranking first when compared with classical Bayesian networks and six other quantum-like models, and superior predictive performance across both equal and unequal probability scenarios. A limitation is that the bias functions add parameters whose psychological meaning is only loosely constrained, and the evaluation relies on a small number of aggregate datasets from classic paradigms, raising a risk of over-fitting. The comparison set, though broad, is drawn from the same family of quantum-like models. For multi-agent and human–robot systems, the explicit bias terms offer a route to modelling how affect and history shape a human partner’s choices.

Gao et al. (2025) (E3; Quantum-like; abstract). Gao et al. (2025) proposed a quantum-like evidence networks decision-making model (QLENDM) to address deviations from the sure-thing principle observed in cognitive experiments. Unlike earlier approaches, such as the quantum dynamic Markov model, quantum prospect decision theory and quantum-like Bayesian networks, which operate in probability theory, QLENDM operates within evidence (Dempster–Shafer) theory. It uses quantum-like basic probability assignments, which the authors argue better represent global and local ignorance present in cognitive experiments than a probability distribution. The parameters governing interference are fitted heuristically from the distance between quantum-like basic probability assignments, based on focal element structure. The model is applied to cognitive science datasets and to information retrieval, and the authors report that it attains the minimum average and standard deviation of fitting errors among the compared models, better predicting participants’ behaviour. All computation is classical. A limitation is that comparisons rely on existing aggregate datasets, and the added evidence-theoretic machinery increases model complexity relative to the gain. For autonomous agents that already use evidential reasoning in sensor fusion, the model shows how quantum-like interference can be combined with explicit representations of ignorance.

Kak (2018) (E2; Quantum-like; abstract). Kak (2018) examined the assumptions usually made about the internal decision environment of intelligent agents, in particular how memory and belief states are queried. He discussed classical and non-classical explanations of order effects, in which the response to one query depends on the query asked before it, and asked whether states that are inaccessible to introspection are best modelled as quantum states. A maze network is used to contrast classical and quantum decision scenarios, and the author proposes testing the quantum hypothesis on large databases such as those of medical treatment and

drug efficacy. The paper is conceptual: it offers no data, implementation or quantitative comparison. For autonomous agents, it is an early argument that the order in which an agent queries its own memory or its users is a design variable.

ML, NLP and IR

Widdows and Peters (2003) (E4; Quantum-inspired; abstract). Widdows and Peters (2003) asked whether logical connectives could be defined directly on distributional word vectors, so that the geometric flexibility of vector semantics is combined with the crispness of Boolean querying. They defined negation as orthogonal projection, where 'a NOT b' removes from a the component along b, and disjunction as the linear subspace spanned by several vectors. Extended negation, 'a NOT (b1 OR ... OR bn)', projects a onto the orthogonal complement of that subspace. The authors showed that these operators coincide with the connectives of Birkhoff-von Neumann quantum logic. This was the first explicit use of quantum logic in natural language processing. The operators were tested on word vectors built from roughly 173 million words of New York Times text. Qualitative neighbour lists showed that negation removes unwanted senses of ambiguous words, for example the legal sense of 'suit' or the sporting sense of 'play'. The paper itself gives no quantitative retrieval scores and defers them to companion work. The approach is quantum-inspired: it borrows the lattice of subspaces and runs entirely on classical hardware. Its relevance for autonomous agents is that it supplies a projective, non-distributive semantics for concept negation and exclusion. This is the same projector algebra used by QP decision models.

Sordoni et al. (2013) (E4; Quantum-inspired; abstract). Sordoni et al. (2013) proposed the Quantum Language Model (QLM) for ad hoc retrieval. The aim was to capture dependencies between query terms without the combinatorial blow-up of n-gram or Markov random field (MRF) models. In QLM, single terms and compound term dependencies are encoded as projectors in a Hilbert space whose dimension equals the vocabulary. A query or document is a density matrix, estimated by maximum likelihood from the sequence of projectors it contains. The density matrix is thus a generalised probability distribution over subspaces rather than over sets of terms. Documents are then ranked by a divergence between query and document density matrices, a quantum generalisation of the KL-divergence ranking of classical language models. The model was evaluated on standard TREC ad hoc collections and compared with bag-of-words and MRF-based dependency baselines. It was reported to be competitive with or better than these baselines. Numerical gains are not reproduced here because the full text was not read. The approach is quantum-inspired: it borrows the density-matrix formalism but runs on classical hardware. Its main limitation is the cost of estimating a vocabulary-sized density matrix. The paper matters because it introduced density matrices as learnable state representations, which later neural quantum-inspired models inherited.

Lorenz et al. (2023) (E5; Quantum circuit (QPU); full text). Lorenz et al. (2023) ran compositional natural language processing (NLP) models on real quantum hardware. They exploited the formal correspondence between the DisCoCat model of meaning and quantum theory: a pregroup-grammar parse of a sentence is compiled into a parametrised quantum circuit. Word meanings are trainable circuit parameters, and the output qubit's measurement statistics give the class label. Quantum analogues of a bag-of-words model and a word-sequence model served as syntax-free and word-order-only baselines. There were two binary tasks. The first was meaning classification (MC), with 130 grammar-generated sentences about food or IT. The second was relative-pronoun (RP) classification, with 105 noun phrases from RelPron. Models were trained with SPSSA in noiseless simulation and then on IBM's 5-qubit superconducting device `ibmq_bogota`. For DisCoCat, the simulated train and test errors after 500 iterations were 16.9% and 20.2% for MC, and 9.4% and 27.7% for RP. On hardware, the test errors were 16.7% for MC and 32.3% for RP, with F-scores of 0.85 and 0.75. As expected, syntax-sensitive DisCoCat did best on RP, while simpler models suited MC. The main limitations are the tiny datasets, a single hardware run per task, costly post-selection, and hardware queues of about 12 to 72 hours. The authors explicitly claim no quantum advantage. For autonomous agents, the work is a benchmark for what genuinely quantum language processing can do on current noisy hardware.

Svozil (2025) (E2; Quantum-inspired; full text). Svozil (2025) proposed an alternative to dynamic, transformer-based handling of polysemy. Each word is a single static unit vector in a Hilbert space, and linguistic contexts are maximal observables, that is, orthonormal bases. A word vector gets its specific meaning from the basis, or context, in which it is considered. Because in dimension three or higher one vector can belong to several mutually complementary bases, called intertwining contexts, 'run' can sit in separate contexts with 'exercise', 'business' or 'river'. The construction is illustrated with orthogonality hypergraphs and the twelve-element 'firefly' logic. The author also suggests building self-adjoint operators from contextual bases as a more compact encoding. The paper is a short conceptual note of about five pages. It has no learning algorithm, no corpus and no evaluation, and it leaves learning of the intertwined bases from data to future

work. The idea is quantum-inspired and entirely classical. It matters for autonomous agents because it gives a Kochen–Specker-style account of context-dependent meaning. An agent’s language or concept module could represent complementary interpretations without a separate vector for each sense.

Deep learning and LLMs

Stoudenmire and Schwab (2016) (E4; Quantum-inspired; abstract). Stoudenmire and Schwab (2016) adapted optimisation algorithms for tensor networks, developed in quantum many-body physics, to supervised learning. Each input is mapped by a local feature map into a product state, which is a tensor product of small vectors, one per pixel. The weight tensor of a linear classifier over this exponentially large space is then parameterised as a matrix product state (MPS), also known as a tensor train. Training uses a DMRG-like sweeping algorithm. It updates pairs of adjacent tensors and adapts the bond dimension, the rank of each bond, through singular value decomposition. On the MNIST handwritten-digit dataset the authors report a test-set classification error below 1%. They also discuss the extra structure the tensor-network form gives the model and suggest a possible generative interpretation. This is a quantum-inspired method in the strict sense: it uses the mathematics of quantum states, but everything runs on classical hardware, and no cognitive claim is made. Limitations include evaluation on a single, simple benchmark, and a fixed one-dimensional ordering of the input that ignores two-dimensional image structure. For autonomous agents and robots, it shows that Hilbert-space state representations can be compressed and trained efficiently at scale. This bears on the feasibility of quantum-like models in real agents.

Zhang et al. (2018) (E4; Quantum-inspired; abstract). Zhang et al. (2018) proposed a quantum many-body wave function inspired language model (QMWF-LM) for text matching. A word is a quantum state over basis meanings, and a sentence or text is a many-body wave function formed from the tensor product of its word states. The high-order interactions between words are therefore held in a single, exponentially large coefficient tensor. The authors avoid building this tensor explicitly. Projecting the global representation onto local, product-state representations is argued to be equivalent to tensor decomposition, and this can be realised by a convolutional neural network (CNN). The resulting model thus gives a quantum-many-body reading of CNN-based matching. It was evaluated on question answering and answer selection benchmarks against classical neural and earlier quantum language model baselines, and was reported to be effective. Specific scores are not given here because the paper’s full text was not read. The model is quantum-inspired and runs on classical hardware. The analogy constrains the architecture rather than implying any physical quantum process. Its limitations are that interpretability rests on the analogy and that the evaluation covers only text matching. For autonomous agents, the paper shows how tensor-product state spaces from QP cognition can be learned with standard deep-learning tools.

Li et al. (2019) (E4; Quantum-inspired; abstract). Li et al. (2019) introduced the Complex-valued Network for Matching (CNM), which applies quantum-probability mathematics to language modelling. Words are pure quantum states and sentences are mixed states, so each sentence is a density matrix. All of these are represented in a single complex-valued vector space. Each word embedding has an amplitude and a phase, and a sentence’s density matrix is a weighted mixture of its word states. Matching between a question and a candidate answer is computed through trainable quantum measurements, that is, projections onto learned measurement vectors. The network’s components are constrained so that each has an explicit physical meaning, which the authors present as a route to interpretability. On two standard question-answering benchmarks, CNM performed comparably to strong CNN and RNN baselines. The abstract gives no numbers, and none are quoted here. The paper won the NAACL 2019 best explainable NLP paper award. The model is quantum-inspired: it borrows the density-matrix and measurement formalism but runs entirely on classical hardware. Limitations are that performance is comparable to, not better than, baselines, and that the interpretability claims are not tested with users. For autonomous agents, CNM shows that learned density-matrix states and projective measurements can serve as an interpretable decision layer.

Li et al. (2021) (E4; Quantum-inspired; abstract). Li et al. (2021a) addressed multimodal fusion for video sentiment analysis with a quantum-inspired framework. The goal was to make interactions between text, visual and acoustic modalities explicit and interpretable. Unimodal features are embedded as complex-valued quantum states. Interactions within and across utterances are modelled with tensor products and density matrices, so that non-separable, entangled-like structure stands for inter-modal correlation. Sentiment is read out through learned quantum measurements. The architecture is a complex-valued neural network trained end to end on classical hardware. It was evaluated on standard multimodal video sentiment benchmarks and compared with state-of-the-art fusion networks. The authors report competitive effectiveness together with greater transparency about how modalities combine. Specific benchmark scores are not given because the full text was not read. The model is quantum-inspired rather than quantum-like: it does not fit human judgement

data, and 'entanglement' refers to a mathematical structure, not a physical resource. Its limitations are that it is specific to sentiment tasks and that interpretability is argued rather than measured. For autonomous agents, the paper offers a template for fusing heterogeneous sensor streams in a Hilbert space while keeping cross-modal dependencies inspectable.

Li et al. (2021) (E4; Quantum-inspired; abstract). Li et al. (2021b) built on an analogy between quantum measurement and emotion recognition in conversation. They framed the recognition of a speaker's emotion in a dialogue as measuring a quantum system whose state depends on the conversational context. The resulting quantum-inspired neural network has quantum-like layers built from complex-valued operations. These layers capture contextual dependencies across dialogue turns and fuse textual, acoustic and visual inputs within a single Hilbert-space representation. The network's weights are learned with an existing algorithm for training such complex-valued quantum-inspired layers, so the model can be trained in a data-driven, end-to-end way. The authors evaluated it on two standard conversational emotion-recognition benchmarks. The abstract reports performance comparable to state-of-the-art methods, with no numbers, and none are quoted here. The authors also argue that it offers a quantum view for understanding conversational emotion recognition. The model is quantum-inspired and runs on classical hardware. Its limitations are that it matches rather than exceeds existing systems, that it is evaluated only on benchmark corpora, and that it has no validation against human judgements. For autonomous agents and social robots, it shows how measurement-based, context-sensitive state updating can be built into a trainable affect-recognition module.

Gkoumas et al. (2021) (E4; Quantum-like; abstract). Gkoumas et al. (2021) treated video sentiment analysis as a decision-fusion problem subject to cognitive-bias-like effects. Following quantum cognition, they hypothesised that the sentiment judgements from different modalities may be incompatible: their order matters, and they cannot be measured jointly for a single decision. Each utterance is represented as a superposition state in a complex-valued Hilbert space. The unimodal classifiers, for text, vision and audio, are modelled as incompatible observables and implemented as positive-operator-valued measures (POVMs). Fusion then follows the QP rules for sequential and incompatible measurements rather than classical averaging or voting. The model was evaluated on two video sentiment benchmarks against both decision-level and content-level fusion methods. According to the abstract, it substantially outperformed these baselines. It also handled every combination pattern of unimodal predictions, including the extreme case where all unimodal classifiers are wrong. Numerical margins are not quoted here. The approach is quantum-like in structure, since it uses the QP rules of cognition, but it is fitted to annotated benchmark labels, not to human decision data, and it runs on classical hardware. A limitation is that the incompatibility is assumed rather than tested. For autonomous agents, it is a direct template for QP-based fusion of conflicting sensor-level decisions.

Uprety et al. (2024) (E4; Quantum-like; abstract). Uprety et al. (2024) asked whether large language models (LLMs) reproduce the order effects that humans show in similarity judgements. The motivation is alignment: LLM-based agents increasingly make decisions and take actions in real-world settings, and human judgements are themselves affected by cognitive biases. The authors replicated a well-known human study showing order effects in similarity judgements, in which the rated similarity of two items depends on which is presented first. They administered the same judgements to several popular LLMs under different prompt settings. They report the specific settings in which LLMs show human-like order-effect bias and those in which they do not. They then discuss what these findings imply for designing LLM-based applications. The work is framed as an ongoing investigation. The abstract does not give the list of models, the sampling settings or quantitative effect sizes, so none are given here. The study comes from the quantum cognition and information retrieval group, and order effects are the standard QP signature of incompatible judgements. The paper, however, probes the effect empirically rather than fitting a QP model. Its limitations are its preliminary scope and its dependence on prompts. For autonomous agents, it shows that order-dependent judgement can surface in LLM decision components and should be audited.

Zhang et al. (2025) (E4; Quantum-inspired; abstract). Zhang et al. (2025) proposed a quantum-inspired neural network for semantic matching based on density matrices. Following earlier complex-valued and density-matrix matching models, sentences are represented as density matrices, and matching exploits a duality property of density matrices. Here the density matrix serves both as the state representation and as a measurement-like operator. Interactions between the two texts can therefore be computed within one QP formalism. The model was evaluated on semantic matching or question-answering benchmarks and compared with classical and earlier quantum-inspired baselines. The authors report improved or competitive effectiveness. The abstract and full text could not be read, so the exact definition of the duality, the datasets and the scores are not given here. They should be checked before this paragraph is finalised. The model is quantum-inspired and runs on classical hardware. Its likely limitations, shared with this family of models, are the cost of the density

matrices and interpretability claims that are not tested with users. For autonomous agents, the paper continues the line of work from (Li et al., 2019) and (Zhang et al., 2018). It shows how QP state and measurement operators can be learned jointly for relevance or matching decisions.

Maksimovic and Maksymov (2025) (E4; Quantum-inspired; abstract). Maksimovic and Maksymov (2025) looked at image-classification systems that fail with confusing or erroneous outputs, or that defer difficult cases to human operators. They asked whether a quantum-cognition-motivated network could emulate human-like perception, confidence and uncertainty. They used the recently proposed quantum-tunnelling neural network (QT-NN), whose activation function is inspired by the quantum-mechanical probability of a particle tunnelling through a potential barrier. Its outputs are interpreted in the light of quantum cognition theory as human-like judgements. The network was trained to classify image datasets, and its confidence and uncertainty patterns were compared with those expected in human decision making. The authors report that the QT-NN provides evidence of human-like decision-making. It could also be trained up to 50 times faster than a classical counterpart. These claims come from the abstract, and the datasets and the comparison protocol were not checked here. The model is quantum-inspired: it runs on classical hardware and uses a physics-derived activation rather than QP state updating. A limitation is that human-likeness is judged against simulated human decisions rather than behavioural data from participants. For human-AI teaming, the work points to classifiers whose uncertainty is closer to human uncertainty and therefore easier for operators to interpret.

Lo et al. (2025) (E4; Quantum-like; full text). Lo et al. (2025) gave the first large-scale test of quantum-like contextuality in natural language. They used a large language model as the source of probability distributions. The authors built a linguistic schema, 'PR-anaphora', modelled on the minimal contextual measurement scenario, the 3-cyclic scenario, whose strongly contextual model is PR-box-like. Each instance has two nouns, three adjectives and two pronouns, framed as a co-reference resolution problem. Noun pairs sharing adjectives were extracted from Simple English Wikipedia, giving 866,108 noun pairs and 51,966,480 instances. Empirical models were then built from bert-base-uncased predictions. Contextuality was assessed with two frameworks that tolerate signalling: a signalling-corrected sheaf-theoretic criterion and Contextuality-by-Default (CbD). Of all instances, 77,118 (0.148%) were sheaf-contextual and 36,938,948 (71.1%) were CbD-contextual. The authors derived a relation between the degree of contextuality and the Euclidean distance between BERT embedding vectors. Regression identified this distance as the best predictor of contextuality. Restricting the data to the top 1% most similar noun pairs raised the rates to 0.50% (sheaf) and 81.83% (CbD). The analysis is quantum-like and runs on classical hardware. Limitations are that it uses a single masked language model and a synthetic schema, and that it does not show that contextuality improves any task. For autonomous agents, it provides tools for detecting contextuality in the belief states of language-model components.

Aerts et al. (2026) (E4; Quantum-like; abstract). Aerts et al. (2026) asked whether the non-classical probability structures previously found in human conceptual combination also appear when large language models (LLMs) act as the 'participants'. They carried out two tests. In the first, ChatGPT and Google Gemini judged conceptual combinations within a Bell-type design. The abstract reports that Bell's inequalities were significantly violated, which indicates a probability model that does not satisfy Kolmogorov's axioms. In the second test, the same models generated text, and word distributions in large texts followed Bose-Einstein rather than Maxwell-Boltzmann statistics. Both results mirror the authors' earlier findings from human participants and corpus analyses. The authors interpret this as evidence of convergence between human and artificial cognition, driven by a meaning-bearing structure in semantic vector spaces rather than by network architecture. The analysis is quantum-like: QP structure is detected in LLM outputs on classical hardware. The CHSH values and sample sizes are in the full text, which was not read, so they are not reported here. Limitations include the dependence of LLM outputs on prompts and model versions, uncontrolled signalling in Bell tests, and the interpretive step from statistical signature to shared cognitive structure. For AI systems, the paper suggests that QP structure may emerge from language training without being built in.

Imannezhad et al. (2026) (E4; Quantum-like; abstract). Imannezhad et al. (2026) compared GPT-5's probability judgements with those of humans on a task known to elicit classical-probability violations. The violations considered were conjunction fallacies, disjunction fallacies and breaches of binary complementarity. The authors used an existing large human dataset, in which participants showed several types of fallacy, and ran GPT-5 on the same task. Each session was matched to a participant's profile, giving 553 persona sessions. Unlike the humans, GPT-5 produced only single conjunction or disjunction fallacies. It also complied almost perfectly with binary complementarity. The article reports conjunction-fallacy rates of about 18.8% for GPT-5 against 58.5% for humans, and double conjunction fallacies of 0.5% against 37.0%. The model's overall pattern matched the predictions of early quantum-probability models of judgement better than later variants that add noise. The authors conclude that GPT-5 reasons about probability more coherently and consistently than naïve

human participants. The study is quantum-like: QP models are used as the theoretical yardstick for both humans and the LLM. Limitations are that only one proprietary model was tested and that personas are an imperfect proxy for real participants. For human–AI teaming, the study shows that LLM advisers cannot be assumed to share human biases, which matters when an AI system models or supports human decisions.

Kang (2026) (E4; Quantum-like; abstract). Kang (2026) turned the QQ equality into an audit criterion for sequential yes/no judgements by autoregressive large language models (LLMs). The QQ equality is a parameter-free prediction of the projective quantum question-order model, which human survey data approximately satisfy. On the theoretical side, the author characterised the mechanism families that satisfy QQ robustly. He showed that classical repetition can reproduce the equality exactly, so QQ alone does not diagnose quantum-like structure. He then combined QQ with the rank-2 Contextuality-by-Default criterion through the bound $|q_QQ| \leq OSS$, which separates order sensitivity, QQ imbalance and residual contextuality. On the methodological side, a committed multi-turn forced-branch protocol reconstructs order-conditioned joint distributions from next-token log-probabilities, with counterbalanced label mappings and pre-specified health gates. A pilot on an open-weight instruction-tuned model, reported as Qwen3-4B-Instruct, passed all health gates. However, the distributions were near-deterministic for 17 of 18 item pairs under direct-evaluation framing and 7 of 8 under persona framing. Label assignment changed several QQ verdicts, and no item was certified as residually contextual. The main result is a caveat: next-token probabilities should not be treated as survey-response distributions without saturation screening and label counterbalancing. The limitations are a single small model and few items. For AI agents, the paper sets methodological standards for auditing QP signatures in LLMs.

Nebli et al. (2026) (E2; Quantum-inspired; abstract). Nebli et al. (2026) proposed a complex-valued sequence model in which hidden states evolve under unitary (Hamiltonian) dynamics and outputs are read by the Born rule, so that phase information between token paths is preserved and can interfere. Cayley discretisation and a low-rank interaction Hamiltonian keep the cost practical and decompose probability currents into interpretable coupling channels, and a separation theorem shows that Born-rule decoding is more expressive than standard decoding on a constructed family of tasks. The authors motivate interference by analogy with quantum cognition. The work is theoretical: no experiments on language data are reported, and whether interference matters for natural language is left as the central open question.

Reinforcement learning and agent learning

Dong et al. (2008) (E4; Quantum-inspired; abstract). Dong et al. (2008) propose quantum reinforcement learning (QRL), which recasts the states and actions of a reinforcement-learning problem as eigenstates and eigenactions of a quantum system. The state–action space is represented as a superposition; an action is selected by a simulated measurement that collapses the action superposition, with probabilities given by squared amplitudes, and rewards update the amplitudes through a Grover-type amplitude-amplification step, so that exploration and exploitation are balanced by the amplitudes rather than by an ε -greedy or softmax rule. The authors analyse convergence and optimality and argue that, on genuine quantum hardware, quantum parallelism could update many state values at once. The simulated experiments, run on a classical computer in a gridworld navigation task, compare QRL with TD(0) using ε -greedy exploration; the preprint reports that QRL learns faster and converges over a much wider range of learning rates. The limitations are that the ‘quantum’ algorithm is only simulated classically, the environments are small tabular problems, and no physical implementation or function approximation is provided. The paper founded the Dong–Chen line of quantum and quantum-inspired RL ((Dong et al., 2012), (Wei et al., 2022)) and is a common reference point for quantum decision rules in autonomous agents.

Dong et al. (2012) (E5; Quantum-inspired; full text). Dong et al. (2012) address the sensitivity of reinforcement learning (RL) to learning rates and initial settings in mobile-robot navigation. They propose a quantum-inspired RL (QiRL) algorithm in which each state carries a superposition-like vector of action amplitudes: action selection is probabilistic, mimicking measurement collapse, and reinforcement adjusts amplitudes by a rule inspired by Grover-type amplitude amplification; a convergence result is stated for the temporal-difference value update. Everything runs on classical hardware. The method is first tested on a 58-state Markovian transition task, where QiRL converged after about 2000 episodes from four different start states, whereas a TD baseline needed between 1000 and 4000 episodes depending on the start state, and QiRL performed well for learning rates from 0.01 to 0.12 while TD failed to converge for rates of 0.02 and above. It is then evaluated in a simulated 600×400 grid world with 16 range sensors (obstacle avoidance) and on a real two-wheeled robot, MT-R, which learned to follow an office corridor into a target room. Limitations are the tabular state representation, comparison against a single TD baseline, and a largely qualitative real-robot demonstration. The work is an early example of quantum-inspired exploration rules transferred to physical robot learning.

Briegel and De las Cuevas (2012) (E4; Quantum-inspired; full text). Briegel and De las Cuevas (2012) propose projective simulation (PS), a model of a learning agent that, before acting, 'projects itself' into possible future situations by running a random walk through a network of clips, i.e. elementary patches of episodic memory. This episodic and compositional memory is updated by adapting transition probabilities between clips on reward, by adding clips from new percepts, and by composing new clips from existing ones, which the authors present as an elementary form of creative behaviour. Emotion tags on transitions record remembered rewards and allow a 'reflection' mechanism in which the agent repeats its internal simulation when the proposed action is not tagged as rewarded. The model is illustrated with a simulated invasion game, in which a defender agent learns to block an attacker from symbolic cues; learning curves show that a forgetting (dissipation) parameter trades adaptation speed against maximum blocking efficiency, and that a longer reflection time speeds up learning. The simulations are small and classical, and no comparison with standard RL benchmarks is made. The paper matters for the survey because the random walk over clips is designed to generalise naturally to quantum walks, providing the basis for later quantum-enhanced agents ((Paparo et al., 2014)).

Melnikov et al. (2018) (E4; Quantum-inspired; abstract). Melnikov et al. (2018) benchmark the projective simulation (PS) agent of (Briegel and De las Cuevas, 2012) against standard tabular reinforcement learning. They apply PS, Q-learning and SARSA to two classic navigation problems, a grid world and the mountain car task, and compare learning performance and the effort needed to tune model parameters. According to the abstract, PS and the standard methods perform qualitatively and quantitatively similarly, but optimal PS parameters are much easier to find, with computational effort for parameter selection reduced by one to two orders of magnitude, because PS has few model parameters. The study is entirely classical and uses small tabular benchmarks, so it says nothing directly about quantum speed-up; its value is in showing that the classical PS model, which admits a quantum-walk generalisation ((Paparo et al., 2014)), is a competitive baseline learner. For the survey, it anchors claims about PS-based agents in robotics to a standard benchmark comparison rather than to toy games such as the invasion game alone.

Wei et al. (2022) (E4; Quantum-inspired; abstract). Wei et al. (2022) propose deep reinforcement learning with quantum-inspired experience replay (DRL-QER), a classical training scheme that changes how transitions are sampled from the replay buffer. Each stored transition is given a quantum-like representation, a qubit-style amplitude, which is modified by a 'preparation' operation, relating its temporal-difference error to its importance in a manner inspired by amplitude amplification, and by a 'depreciation' operation that lowers the priority of transitions already replayed many times to preserve diversity. Sampling probabilities follow from these amplitudes, so the method adaptively balances exploration and exploitation in the choice of experiences. The authors evaluate DRL-QER on Atari 2600 games against prioritised experience replay (DRL-PER) and DCRL, and report in the abstract that it outperforms these methods on most games with improved training efficiency, and that it can be combined with double and dueling network architectures. No quantum hardware is involved; the 'quantum' element is a sampling heuristic whose advantage depends on tuning and has been shown mainly on game benchmarks. For autonomous agents it is a practical example of quantum-inspired ideas improving the sample efficiency of deep RL, continuing the line of (Dong et al., 2008).

Robots and embodied agents

Lukac and Perkowski (2007) (E2; Quantum circuit (design or simulation); abstract). Lukac and Perkowski (2007) present an emotional model for Cynthea, a humanoid robot developed at Portland State University, organised in two levels. A cognitive level perceives the environment and executes behaviour scripts written in a robot command language, while an emotional level alters how those commands are executed, independently of their content, through string-rewriting operations such as repeating, modifying or deleting commands. The emotional level is modelled with quantum finite automata built from Hadamard and CNOT gates, over six emotional variables (anger, happiness, melancholy, depression, fear and contentment), so that superposition and entanglement generate less predictable, more lifelike behaviour; the quantum part runs on a software simulator, and dialogue relies on an existing chatbot system. As far as could be determined from the available text, no quantitative evaluation is reported and the authors describe the testing as partial. The quantum model is therefore a design proposal, and its behavioural benefit over a classical stochastic automaton is not demonstrated. The paper is nevertheless one of the earliest attempts to use quantum formalism, rather than quantum hardware, to drive robot affect, anticipating later quantum emotion models for robots ((Yan et al., 2019)).

Aerts et al. (2011) (E1; Quantum-like; full text). Aerts et al. (2011b) is a short position paper that reviews the Brussels 'quantum interaction' programme and draws out its implications for artificial intelligence and robotics. The authors summarise how Hilbert-space and quantum-probability models, building on the SCoP (state-context-property) formalism, reproduce Hampton's membership-weight data on the conjunction and

disjunction of concepts, including the Guppy (Pet-Fish) effect, and how related models address the disjunction effect and the Ellsberg paradox. From this they propose that human thought has two superposed layers: a classical logical layer and a non-Kolmogorovian quantum conceptual layer. For AI, they argue that tensor-product representations, as in Smolensky’s binding proposal, could supply the word order and syntax that bag-of-words latent semantic analysis lacks. For robotics, they contrast the quantum-robot programme of (Benioff, 1998) and (Dong et al., 2006), which relies on building real quantum computers, with the view that macroscopic, classically realised systems can carry quantum structure; they suggest that AI’s shortfalls may stem from its reliance on classical computation. No new model, data or experiment is presented, and the claims about robotics remain speculative. The paper is nonetheless an early explicit statement of the idea that quantum-like cognition, rather than quantum hardware, could shape robot design.

Hangl et al. (2016) (E5; Quantum-inspired; abstract). Hangl et al. (2016) tackle generalisation in complex manipulation, such as bimanual or in-hand tasks, without large supervised datasets or complex controllers. Their idea is to keep simple controllers that work in a few specific situations and to have the robot ‘play’ with objects, using previously trained basis skills to estimate the state of the environment or to transform a novel situation into one where the solution is known; skills are organised hierarchically. Projective simulation ((Briegel and De las Cuevas, 2012)) serves as the learning mechanism, both for choosing which haptic sensing action to use and for choosing the preparatory skill given the perceived state. The approach is evaluated on a dual-arm robot in pick-and-place scenarios, notably grasping a book that may lie in different orientations and placing objects into a closed box, where the robot learned to use a flipping skill to open the lid. According to the paper (read via a web summary), sensing by sliding was far more reliable than poking or pressing, and roughly 80 roll-outs sufficed to reach 90% success in simulation based on real success rates. The state and skill sets are small and hand-designed and the learning is classical. It is an early demonstration of a quantum-compatible learning model (projective simulation) controlling a physical robot, later extended in (Hangl et al., 2020).

Yan et al. (2019) (E2; Quantum circuit (design or simulation); abstract). Yan et al. (2019) propose a quantum emotion space (QES) for representing and manipulating a robot’s emotional state. Building on a pleasure–arousal style model extended with time, emotions are encoded as sequences of entangled qubits, so that an emotional trajectory appears as a curve inside a cuboid, and transitions between emotions are expressed as unitary transformation matrices. The authors argue that this encoding needs fewer resources than a classical representation and makes it easy to track and manipulate emotional transitions over time, for instance with an ‘emotion sensor’; as a proof of concept they show an inverse operation, implemented with quantum NOT gates, that maps an ‘anger’ state to a ‘relaxation’ state. The motivation is anticipatory: quantum robots, alone or in hybrid systems, may eventually need native emotional representations for human–robot interaction. The work is purely formal. According to the abstract and available summary there are no simulations on a robot, no human participants and no comparison with classical affect models, and the claimed resource savings would only materialise on quantum hardware. It is cited as the starting point of a quantum-computing approach to robot emotion ((Yan et al., 2021c), (Yan et al., 2021a)), distinct from quantum-probability models of human affect.

Moreira et al. (2020) (E1; Quantum-like; abstract). Moreira et al. (2020a) is an open peer commentary on Lieder and Griffiths’ target article proposing resource-rational analysis, which explains cognitive biases as optimal use of limited computational resources. The authors argue for an alternative, unifying framework based on quantum probability that yields more general cognitive and decision models able to represent more information than classical ones. They outline the ingredients of such a quantum-like cognitive architecture: indefinite (superposed) belief states represented as vectors in a Hilbert space, beliefs expressed through rotated bases, contextuality in the sense that no single joint distribution covers all decision contexts, and interference effects that account for violations of classical probability such as the disjunction effect. The commentary refers to their earlier quantum-like Bayesian network work ((Moreira and Wichert, 2016)) and to literature on contextuality in human judgement. It presents no new model, data or evaluation, and its argument that quantum models are more economical than resource-rational ones is asserted rather than demonstrated. For autonomous agents it is useful as a short, citable statement of what a quantum-like decision architecture would comprise, and of how it positions itself relative to bounded and resource-rational accounts that dominate classical agent design.

Lanza et al. (2020) (E2; Quantum-like; abstract). Lanza et al. (2020) explore whether quantum-probability formalisms, specifically state superposition and interference, can model robot perception in the way quantum cognition models human bistable perception such as binocular rivalry. In a deliberately minimal case study, a stationary circular robot with a front and a back presence sensor observes a randomly moving object. The relative frequency of back-sensor events within a time window is encoded into a single qubit by a

Y-axis rotation, so that the qubit state summarises the robot’s two mutually exclusive percepts and repeated measurements recover their probabilities. The model was run both as a classical simulation of the circuit and on real quantum processors: the designed behaviour was reproduced in simulation, whereas actual quantum hardware showed notable errors, particularly for unbalanced probability distributions. The authors therefore recommend simulation-based, quantum-like implementations over current quantum backends. The study is a feasibility check with binary percepts, a single object and no comparison with classical probabilistic perception, so it does not show any benefit of the quantum-like representation. It is nonetheless one of the first explicit transfers of quantum cognition to robot perception and motivates the multisensory extension in (Lanza et al., 2021).

Hangl et al. (2020) (E5; Quantum-inspired; abstract). Hangl et al. (2020) address how a robot can extend manipulation skills that initially work only in a narrow range of situations. The robot plays autonomously with objects, applying previously trained skills and behaviours to learn how to bring a situation into one for which a successful strategy is already known. Behaviour selection uses projective simulation ((Briegel and De las Cuevas, 2012)), a classical reinforcement-learning model based on random walks through an episodic and compositional memory; the experience collected during play also trains an environment model that is used for active learning and for composing novel preparatory behaviours. The approach is demonstrated on a real dual-arm robot across several manipulation tasks, including book grasping, grasping objects of different sizes by selecting grasp strategies, placement on shelves and tower disassembly; composition allows previously unsolvable tasks such as tower disassembly to be solved. Using success statistics from the real-world trials, simulations indicate that active learning improves learning speed by around 30%. The method remains classical, relies on hand-designed skill libraries, supervision for rewards and a fixed sense–prepare–execute structure, and its scalability to large state spaces is untested. It is the clearest example in the corpus of a quantum-compatible learning model running on a physical robot.

Lanza et al. (2021) (E2; Quantum-like; abstract). Lanza et al. (2021) generalise their single-qubit quantum-like perception model ((Lanza et al., 2020)) to integrate several sensory inputs into a multidimensional representation of the world. Each of n normalised sensor readings is encoded into its own qubit by a Y-rotation, so the joint register holds a superposition over 2^n basis states whose measurement probabilities express the robot’s confidence in each world configuration; ‘query operators’ are defined that re-assign the meaning of basis states so that the robot’s degree of belief in a particular target state can be read out. The demonstration uses a camera-like sensor whose average RGB values are treated as three scalar sensors, giving a three-qubit model simulated in IBM Qiskit; according to the article summary, the model was evaluated over a systematic sampling of the colour space with many simulated measurements per input, showing a definite but non-linear relation between encoded inputs and measurement outcomes. There is no physical robot, no temporal integration in this version, each query destroys the state, and no comparison with classical Bayesian or fuzzy sensor fusion is made. The contribution is conceptual and formal: it shows how quantum-like state representations and projective queries could express graded belief in robot perception, but not that they improve it.

Yan et al. (2021) (E2; Quantum circuit (design or simulation); abstract). Yan et al. (2021a) extend the quantum emotion space work of (Yan et al., 2019) to companion robots by building on Plutchik’s wheel of emotions, which arranges eight primary emotions with graded intensities. The authors define a quantum counterpart of the wheel model, in which emotional states are encoded in qubit registers arranged as a sequence of ‘discs’ forming an emotion space, and design quantum circuit schemes for generating an emotion and for transitions between emotions, so that changes of emotional state are realised as sequences of quantum gates. The motivation is that a quantum encoding can represent mixed and graded emotions compactly and may later run on quantum processors embedded in or connected to robots. We could not access the abstract or full text of this paper, only its title and figure captions, so the evaluation setting (simulation or hardware), any quantitative results and the exact circuit designs could not be verified. Like the earlier quantum emotion space, the approach appears to be a formal design without user studies or comparison with classical affect models. It is relevant to the survey as part of the quantum-computing strand of robot affect, which differs from quantum-probability models of human emotion and decision.

Ho and Hoorn (2022) (E2; Quantum circuit (design or simulation); abstract). Ho and Hoorn (2022) attempt a quantum-computational construction of robot affect intended to handle ambiguous emotional states, probabilistic decisions and parallel cognitive processes in human–robot interaction. Their Quantum Coppélia (Q-Coppélia) translates the fuzzy-logic Silicon Coppélia model, which simulates how a robot’s attitude towards its user develops, into quantum logic. Psychological scales are represented as qubits, so that superposition captures indefinite affective states and controlled operations and entanglement capture coexisting, ambivalent

appraisals; the framework covers feature encoding, relevance and valence appraisal, use intentions and the trade-off between involvement and distance. The paper presents the full circuitry of the framework and argues that it relates quantum algorithms to (neuro)psychological processes. As far as the abstract and article summary indicate, there is no empirical validation: no human data, no run on a robot, and no comparison with the original fuzzy system or a classical probabilistic model, and whether circuits were executed on a simulator or hardware is not clear. The contribution is therefore a formal design showing how an existing affective architecture for social robots could be mapped onto gate-based quantum computation. It is relevant to the survey as the most complete quantum rendering of an established robot-affect model.

Song et al. (2022) (E4; Quantum-like; abstract). Song et al. (2022) apply quantum cognition to a core autonomous-driving problem: estimating the intentions of pedestrians whose behaviour may be boundedly rational. They argue that classical models assume fully rational traffic participants, whereas real crossing decisions often violate classical probability. Using a quantum-like Bayesian (QLB) network, in which probabilities are obtained from complex amplitudes and interference terms modify the classical law of total probability, they model pedestrian crossing decisions. A case analysis is used to argue that the QLB model captures apparently unreasonable crossing behaviour better than a classical probability model, and a trajectory-prediction experiment on pedestrian trajectory data compares the QLB-based predictor with the data-driven Social-LSTM, reporting that QLB covers edge events in interactive scenes and is closer to real trajectories. All computation is quantum-like, on classical hardware. The exact datasets and error figures could not be verified here, and the interference parameters appear to be set rather than learned, which limits claims of generality; the scenarios are also limited to pedestrian crossing. The paper is one of the few applications of quantum-like Bayesian networks (cf. Moreira and Wichert (2016)) to a safety-critical perception-and-prediction problem for autonomous vehicles.

Widdows et al. (2023) (E5; Quantum circuit (QPU); full text). Widdows et al. (2023) show that some established quantum-probability models of human judgement can be written as small quantum circuits and executed on real quantum hardware. They build circuit components for question-order effects (non-commuting projections, illustrated with the Clinton–Gore poll, following question-order models such as (Wang et al., 2014)), for ‘subjective bias’ via entanglement with an extra qubit, and for interference in disjunction effects, combining them into a circuit that reproduces the Prisoner’s Dilemma pattern and, with other parameters, the two-stage gamble and Hawaii vacation problems of (Shafir and Tversky, 1992). All circuits use four qubits or fewer and were run with 10,000 shots on an 11-qubit IonQ trapped-ion machine. The Prisoner’s Dilemma circuits reproduced the target defection probabilities (82%, 73% and 64% for partner betrays, cooperates or unknown) with a largest discrepancy of 0.37%, while order-effect circuits were within about 2–3% of expected values, comparable to the reported two-qubit gate fidelity. The authors stress that this is not a quantum advantage and not a claim that brains use qubits; parameters are set by hand from published effects rather than fitted to new data. For agent design, it offers reusable circuit ‘components’ for embedding quantum-like decision biases directly on quantum hardware.

Mannone et al. (2024) (E2; Quantum circuit (design or simulation); abstract). Mannone et al. (2024) adopt the D’Ariano–Faggin view, in which thinking and consciousness are described by quantum-state dynamics and mental complexity by entanglement of ‘qualia’ states, and use it to model a robot’s inner processing as a quantum–classical dialogue. Their case study is improvised robot dance: classical, periodic multisensory stimuli (visual and auditory, each encoded as calm or activating) are fed into a small quantum circuit representing the dancer’s inner state, and measurement outcomes are mapped to a repertoire of poses. According to the article summary, the circuit uses three qubits with controlled-Hadamard and Toffoli gates and is run on the IBM Qiskit simulator; a second case study adds environmental feedback so that repeated cycles produce increasingly unpredictable sequences, which the authors interpret as creativity, and one run shows reduced activity when an excited state is stimulated further, read as sensory overload. The results are preliminary and illustrative: initial states are set by hand, stimuli are binary, and there is no evaluation of the resulting behaviour by observers or comparison with a classical stochastic controller. The paper is relevant as an example of quantum circuits used as a generative ‘inner state’ for expressive robot behaviour rather than as an optimiser.

Daglarli (2025) (E4; Quantum circuit (design or simulation); full text). Daglarli (2025) proposes a brain-inspired cognitive architecture for an autonomous game agent that combines a computational prefrontal-cortex model (meta-cognition and situation awareness), a transformer-based deep reinforcement learning module, and quantum algorithms. Deutsch–Jozsa and Bernstein–Vazirani circuits are used during training to classify environmental states and extract bit patterns, and Grover’s search is placed in the decision loop to amplify the probability of high-reward actions; all circuits are written in Qiskit and run on a classical simulator. The agent is evaluated in Minecraft (with the MineRL human-play dataset) on ‘explorer’, ‘survivor’ and mixed

scenarios, comparing configurations with and without the quantum components. The abstract reports that the quantum-enhanced agent reaches an 80% task success rate about twice as fast and obtains roughly 15% higher cumulative reward than a classical deep RL baseline, with smaller gains in highly uncertain, reactive scenarios; much of the results section is qualitative, and details on runs and statistical testing are limited. Because the quantum routines are classically simulated, any 'speed-up' cannot be physical, and the role of Deutsch–Jozsa and Bernstein–Vazirani in learning is weakly motivated. The paper illustrates how quantum algorithms are being embedded in agent architectures, but its evidence should be read cautiously.

De Carolis et al. (2025) (E1; Quantum circuit (design or simulation); full text). De Carolis et al. (2025) present a short state-of-the-art review that forms the first task of QUADRI, an Italian project on quantum-enhanced human–robot interaction. Drawing on more than 35 papers published between 2021 and 2025, they group work into reviews and theoretical foundations, quantum-based cognitive models for robots (including (Ho and Hoorn, 2022), (Mannone et al., 2024) and (Widdows et al., 2023)), quantum and quantum-inspired emotion recognition and decision-making, and quantum approaches to path finding and navigation (including (Hohenfeld et al., 2024)). The project aims to connect a social robot to a quantum simulator and to replace some reinforcement-learning and agent-based components with quantum search (e.g. Grover) and quantum machine learning, focusing on verbal interaction. The authors conclude that hardware limits on qubit numbers are the main common obstacle and that hybrid designs, perceiving the environment classically and processing selected information quantumly, are preferable for now. The paper offers no implementation or evaluation, mixes genuinely quantum, quantum-like and quantum-inspired methods without distinguishing them, and repeats some favourable claims uncritically. It is useful as a map of current social-robotics interest in quantum methods.

Chella et al. (2026) (E4; Quantum circuit (design or simulation); abstract). Chella et al. (2026) propose a hybrid classical–quantum architecture for a teleo-reactive robot, i.e. an agent that must classify its perceived situation and select an action under uncertainty arising from sensor noise, ambiguous perceptual configurations or poorly separable action regions. A classical sensor vector is mapped into a quantum feature space in which learned concepts or rule-defined perceptual regions are stored as reference states, and similarities between the current perception and these references support action selection. Two implementations are evaluated offline on a public wall-following robot dataset: a quantum-kernel classifier using ZZ feature maps with a support-vector classifier, compared with a classical RBF kernel, and an illustrative circuit that encodes sensor conditions directly into qubits and selects actions by measurement after amplitude amplification. According to the article summary, configurations with two sensors performed better than four, and the circuit reached an accuracy of about 0.91 on the two-sensor test set, with errors concentrated near boundaries between action regions. The authors state explicitly that this is an offline proxy for reactive control, not a closed-loop controller and not evidence of quantum advantage. The paper extends the group’s earlier quantum planner ((Chella et al., 2022)) from deliberative planning to reactive perception–action mapping.

Hoorn and Ho (2019) (E2; Quantum circuit (design or simulation); abstract). Hoorn and Ho (2019) addressed two obstacles to artificially sentient robots: conventional computers do not process information in parallel in the way the human affective system can produce concurrent emotions (for example, feeling happy and sad at once), and cognition and affect are usually modelled separately. They proposed a quantum computational account of mixed states of reflection and affect, mapping known psychological dimensions onto quantum dynamics, and illustrated possible robot behaviours with small Einstein–Podolsky–Rosen-type circuits as a precursor to simulation. No simulation results, robot implementation or user study are reported, and the claims about quantum processes in neurology are speculative. The work is the precursor of the authors’ later Q-Coppélia model (Ho and Hoorn, 2022), which gives the full circuit design.

Multi-agent systems and teams

Yukalov et al. (2018) (E2; Quantum-like; abstract). Yukalov et al. (2018) extend quantum decision theory (QDT; cf. Yukalov and Sornette (2011)) from single individuals to a society of interacting decision-makers taking decisions in multistep processes. Each agent evaluates prospects by two criteria: a utility factor (rational, classical part) and an attraction factor (quantum interference term reflecting subconscious attractiveness). Agents exchange information between steps, and this exchange progressively reduces the attraction (interference) term. The model yields three predictions about the dynamic disjunction effect: behaviour is initially probabilistic and deviates from utility-based choice; as information accumulates, prospect probabilities converge towards the utility factors; and repeated information sharing leads to a consensus. The authors report that these predictions agree closely with empirical observations on the dynamic disjunction effect, although the comparison relies on existing experimental results rather than a new study. The framework is quantum-like,

being operator-based QP on classical computation, and its agent-interaction rules are stylised. For multi-agent AI, it gives an analytically tractable account of how communication between agents can damp irrational (interference-driven) choice and produce collective convergence.

Ried et al. (2019) (E4; Quantum-inspired; abstract). Ried et al. (2019) propose modelling collective motion with learning agents rather than featureless self-propelled particles with pre-specified interaction rules. Each individual is a projective simulation (PS) agent ((Briegel and De las Cuevas, 2012)): percepts and actions are 'clips' in an episodic memory network, decisions are random walks over the network, and edge weights (h-values) are reinforced by reward with a forgetting (damping) term. In the marching-locust case study, agents in a one-dimensional ring world (80 cells, sensory range 5) perceive the net flow of nearby conspecifics, may either continue or turn around, and are rewarded for moving with the local majority. With $N = 10, 40$ and 70 agents the learned behaviour reproduces the known density-dependent regimes: disordered motion, alignment with periodic switching, and stable long-term alignment; individual response probabilities converge within roughly 10^2 time-steps, and a Fokker-Planck analysis gives drift and diffusion coefficients that qualitatively match experimental locust data. PS here is a classical reinforcement-learning model whose design is inspired by quantum-walk ideas; no quantum hardware or QP is used. The setting is minimal (1-D, two actions). For robot swarms, it shows how interaction rules can emerge from individual reward rather than being hand-coded.

Lopez-Incera et al. (2020) (E4; Quantum-inspired; abstract). López-Incera et al. (2020) use projective simulation (PS; (Briegel and De las Cuevas, 2012)), a transparent reinforcement-learning model with episodic and compositional memory, to study how swarm behaviour develops under different foraging pressures. Each of 60 agents in a one-dimensional circular world (500 positions) perceives neighbour density and orientation within a visual range of six positions (25 percepts), chooses to continue or turn around, and is rewarded for reaching a food source placed at varying distance. Training runs for 10,000 trials per scenario. When food is distant (e.g. 21 positions), strongly aligned swarms emerge (alignment order parameter around 0.8 or higher) after roughly 200 trials; when food is near, agents remain weakly aligned (about 0.2). Individual trajectories differ accordingly: distant-resource agents show composite correlated random walks resembling Lévy walks, whereas nearby-resource agents show mainly Brownian motion. As in (Ried et al., 2019), PS is a classical algorithm with quantum-walk motivation; quantum speed-ups are not exercised. Limitations are the 1-D world, binary actions and a single reward type. For robotics, it shows that collective strategies suited to the task (search versus exploitation) can be learned rather than prescribed, with an interpretable policy.

Lawless (2020) (E1; Quantum-like; abstract). Lawless (2020) argues that no adequate mathematics yet exists for designing and safely operating autonomous human-machine teams (A-HMTs), and that rational, individual-level models fail under the uncertainty in which such teams operate. He proposes interdependence theory with quantum-like properties: bistability (an analogue of wave-particle duality, e.g. two competing interpretations of reality), a measurement problem (measuring one aspect of a team increases uncertainty about its complement) and non-factorability (team performance cannot be decomposed into individual contributions, an analogue of entanglement and the no-cloning theorem). The argument draws on entropy production and Kullback-Leibler divergence and is contrasted with six field case studies, including the USS Greeneville and USS McCain collisions, the USS Vincennes incident and the Uber autonomous-vehicle fatality. A central claim is that a machine teammate must be able to explain itself to humans in causal language. The paper is conceptual; the quantum analogies are not formalised as a fitted QP model, and the case studies are interpreted qualitatively. For autonomous systems it frames why team-level, context-dependent models, rather than aggregation of individual models, may be needed for trustworthy human-machine teaming.

Yan et al. (2021) (E2; Quantum circuit (design or simulation); full text). Yan et al. (2021b) propose a quantum affective computing (QAC) framework in which a robot's emotion is a single-qubit state $|E(\xi)\rangle = \cos \xi|0\rangle + \sin \xi|1\rangle$, with the angle ξ placing the emotion on a bipolar spectrum. Standard gates acquire affective readings: X inverts an emotion, Z negates its angle, H 'neutralises' it, and Ry rotations increase or decrease emotional intensity; measurement statistics recover the emotion as a probability distribution. For groups, the authors adopt an image-style encoding (akin to flexible quantum image representations) in which robots on a $2^n \times 2^n$ lattice are indexed by $2n$ position qubits entangled with one emotion qubit, so that $2n + 1$ qubits hold the fused emotion of all robots, and controlled rotations change the emotions of chosen sub-regions in one operation. The demonstrations are small classical simulations, for example 64 robots on an 8×8 lattice split into quadrants, showing probability distributions before and after controlled rotations. The paper itself notes that state preparation costs $O(2^{4n})$ elementary operations and that read-out requires many repeated measurements, which offsets the claimed qubit economy; no hardware runs, human data or behavioural validation are reported. This is a quantum-circuit formalism rather than a quantum-like cognitive model. For multi-robot systems it is a representative example of mapping affective state onto circuits, useful mainly as a design sketch.

Yukalov et al. (2022) (E2; Quantum-like; abstract). Yukalov et al. (2022) study networks of agents operating under quantum decision theory, in which each agent evaluates alternatives by utility, attractiveness and the information available to it, so that choice is a multimodal process. Information reaches agents in two forms: direct agent-to-agent exchange and a collective information field generated by the whole society. For agents with short-term memory, decision probabilities often oscillate for a prolonged period before settling. Adding the collective information field smooths sharp variations in temporal behaviour, reduces oscillation amplitude, can remove oscillations entirely and forces convergence to fixed limits; the model also reproduces the dynamic disjunction effect. The work builds on (Yukalov et al., 2018) and is analytical and numerical; no new behavioural data are collected and the communication structure is idealised. The formalism is quantum-like (QP maths on classical computation). For multi-agent AI, it suggests that a shared broadcast signal, such as a common operating picture, can stabilise otherwise oscillating collective decisions.

Yukalov (2023) (E2; Quantum-like; abstract). Yukalov (2023) reviews the principles on which artificial intelligence should be based if it is to imitate how humans who experience emotions actually take decisions. Two approaches are compared, the quantum-theoretic description of decision-making (quantum decision theory) and a formulation of affective decision-making in the language of classical probability; both are principally probabilistic and share several similarities. The paper elucidates analogies between quantum measurements under intrinsic noise and affective decision-making, treating emotions as a noise-like component that coexists with rational (cognitive) evaluation. It argues that an affective artificial intelligence that accounts for this cognition–emotion duality avoids many behavioural paradoxes of traditional decision theory, such as those violating expected-utility axioms. The contribution is theoretical and synthetic: no implementation, benchmark or agent evaluation is reported, and the practical architecture of such an AI is left open. For autonomous agents it offers a principled route to building emotion-sensitive decision modules whose probabilities remain well defined, which is relevant to agents that must predict or emulate human choices.

Lawless et al. (2023) (E1; Quantum-like; abstract). Lawless et al. (2023) review the case for designing quantum-like algorithms that structure human–machine teams to reach maximum performance in open, real-world systems characterised by complexity, uncertainty and conflict. They contrast an individual approach, which aggregates the thoughts, behaviours and skills of team members and which, they argue, has produced few generalisable results and is affected by the replication crisis, with an interdependence approach. The latter is quantum-like: it allows bistability of mind and body, poses a measurement problem, and is non-factorable, so team performance cannot be decomposed into individual performances and the effective degrees of freedom are reduced. The paper surveys laboratory versus field team research, the relevant mathematics, and results from the authors’ model in the open field, notably a trade-off in which low structural entropy production (cohesion) supports maximum entropy production (productivity). It is largely a review and position piece; the quantum-like formalism is analogical and the field evidence is drawn from the authors’ earlier work rather than new controlled tests. For autonomous systems it motivates team-level, non-separable models of human–machine teaming.

Essalmi et al. (2025) (E4; Quantum-inspired; abstract). Essalmi et al. (2025) address interaction-aware decision-making in automated driving, where neglecting or oversimplifying interactions with other road users can make the ego vehicle overly conservative. They propose two quantum game models, QG-U1 (a single-unitary formulation) and QG-G4 (a gate-based formulation), which extend classical two-player, two-strategy games with superposition, interference and entanglement in the Eisert–Wilkens–Lewenstein tradition ((Eisert et al., 1999)). Both run in real time on a standard computer, so they are quantum-inspired classical simulations rather than quantum hardware. The models are evaluated in simulated merging and roundabout scenarios against classical game-theoretic methods and baselines (IDM, MOBIL and a utility-based technique). The authors report that QG-G4 achieves lower collision rates and higher success rates than the baselines, and that both quantum models yield higher expected payoffs than classical games under certain parameter settings. These findings need caution: an independent review reported that the collision results could not be reproduced, and the advantage is parameter-dependent and limited to two-agent, two-action games in simulation. For autonomous vehicles, the work is an early test of whether quantum game formalisms can encode interaction-aware negotiation.

Kong et al. (2026) (E4; Quantum-inspired; abstract). Kong et al. (2026) ask why swarms form elongated longitudinal queues during long-distance navigation, noting that deep reinforcement learning models of collective behaviour are black boxes that hide the microscopic rules. They propose an interpretable model based on projective simulation with episodic and compositional memory ((Briegel and De las Cuevas, 2012)), in which each individual learns a decision policy on a transparent state–action network. In simulation, swarms self-organise into stable, highly elongated queues: polarisation converges to about 0.99 within 100 steps and the peak

elongation ratio reaches 8.31 in steady-state navigation. Visualising the learned policies reveals a target-priority rule: when alignment with neighbours conflicts with the global target direction, individuals prioritise the target, and this drives queue formation. Action-space granularity has a nonlinear effect on stability, with moderate precision (15 actions) best; polarisation stays above 0.97 with sensory noise up to 30°, and the approach scales across groups of 30, 100 and 150. The model is classical and quantum-inspired only through the projective-simulation lineage, and it is tested in simulation only. For robot swarms it demonstrates an interpretable learned navigation policy whose rules can be inspected and verified.

Essalmi et al. (2026) (E4; Quantum-inspired; abstract). Essalmi et al. (2026) extended their earlier two-player quantum game models of driving (Essalmi et al., 2025) to a multi-player, multi-strategy quantum game for interaction-aware decision-making in automated driving. Each vehicle can hold superposed and entangled strategies, combining classical game theory with quantum-game payoffs, and the model is simulated classically. In simulated interactive scenarios the authors report higher success rates and fewer collisions than classical game-theoretic approaches, especially when interaction is dense. The study is simulation-only and has not been tested with human drivers or on a vehicle. It is the only decision-making study in the corpus set on a vehicle task with several interacting agents.

Human-AI trust

Ashtiani and Abdollahi Azgomi (2019) (E4; Quantum-like; abstract). Ashtiani and Abdollahi Azgomi (2019) propose a trust evolution algorithm for computational trust models used in decision support systems, building on a quantum-like representation of trust. Trust in candidate trustees is encoded as probability amplitudes, and trust evolves by amplitude amplification inspired by Grover’s quantum search algorithm, so that evidence from recommendations and interactions increases the amplitudes of trustworthy candidates and attenuates others. The algorithm is evaluated in simulated scenarios with ten trustee candidates and ten recommenders: recommendation-based trust, detection of malicious recommenders (three perfect, four normal, three malicious; the malicious recommenders’ amplitudes fall to about 0.0009 after ten iterations), adaptation to a trustee who turns malicious after ten honest transactions, and accuracy comparison. The authors report lower mean absolute error and root mean square error than classical approaches including BLADE, TidalTrust, max-min, max-mean and Bayesian methods. The work runs entirely on classical computers; the Grover analogy is an update heuristic rather than a quantum speed-up. Scenarios are small and synthetic, with no human trust data. For multi-agent systems and human-machine teaming, it gives an early example of quantum-like trust updating that can be embedded in agents that decide whom to rely on.

Pushparaj et al. (2021) (E3; Quantum-inspired; abstract). Pushparaj et al. (2021) model air traffic controller (ATCO) trust in automation by combining two schools of thought on trust through the quantum principles of superposition and complementarity, so that both views are held simultaneously until a decision is made. The model was refined with behavioural indicators and verified with functional magnetic resonance imaging (fMRI) while ATCOs performed conflict-detection tasks aided by ATS-CAP, software that generates credible flight plans with visible waypoints and airports. The sample was very small, five participants including student and professional ATCOs, and each accepted the automation’s advice in 40% of ambiguous cases. The authors argue that grounding the model in behavioural and neural data rather than self-report makes it more robust for ATCO trust under uncertainty than generalised questionnaire-based models. The approach is quantum-inspired, using superposition as a descriptive principle on classical data; the abstract gives no formal fit statistics, and the sample precludes generalisation. For human-automation teaming in safety-critical operations, it is one of the few studies linking a QP-style trust model to neuroimaging of professional operators.

Humr et al. (2023) (E2; Quantum-like; abstract). Humr et al. (2023) argue that as AI takes over parts of the decision space, human decision-makers must rationalise AI outputs and may struggle to form trust in them, and that existing analytical methods give only partial insight into trust. They propose a generative, open-system modelling approach in which trust is represented as a function of time by a single probability distribution, so that its temporal evolution can be captured. The paper is a short conference contribution in the Human Factors and Ergonomics Society proceedings; the abstract reports no participant numbers or fit statistics, and the approach is presented as having potential rather than as validated. The authors suggest that results could inform machine behaviours that steer human preferences towards Bayesian-optimal rationality in stressful decision settings. The model is quantum-like, drawing on open quantum system dynamics simulated classically. For human-AI teaming, it motivates dynamic trust models that could be embedded in adaptive AI decision aids.

Roeder et al. (2023) (E3; Quantum-like; abstract). Roeder et al. (2023) investigate how a person’s judgements of an AI system’s reliability evolve during interaction, with two aims: to compare quantum and

Markov random-walk models of these judgements and to identify a neural correlate of perturbed reliability expectations. In a Wizard-of-Oz style experiment, seven adults classified face images (294 real, 294 AI-generated) as authentic or synthetic across 560 trials in 20 blocks, receiving manipulated feedback in which the 'AI' agreed or disagreed with them, and gave reliability ratings. The quantum random-walk model predicted the evolving ratings better than the Markov model, which the authors attribute to its capacity to represent the pronounced within-subject variability of ratings. EEG showed a clear event-related potential when reliability expectations were perturbed. The study is exploratory with a small sample, and the AI's behaviour was scripted rather than a real classifier. The model is quantum-like (Hilbert-space dynamics simulated classically). For human-AI teaming, the EEG marker points towards adapting a quantum trust model's parameters in real time, a possible mechanism for trust-aware autonomous systems.

Humr et al. (2025) (E2; Quantum-like; abstract). Humr et al. (2025) argue that incongruities between how AI systems and humans rationalise information create uncertainties in human-in-the-loop decision-making that classical probability theory cannot capture adequately. They propose quantum probability theory (QPT) as a better framework because it represents interference effects, order effects, the interaction of categorisation and decision, and ontic uncertainty, meaning uncertainty that is resolved only when a judgement is made. The paper is a perspective rather than an empirical study. It supports the argument with prior findings, including a Gallup question-order effect of about 15 percentage points that QP models explain, and with illustrative simulations of oscillating decision confidence in an information-seeking scenario, a hypothetical user searching for medical information on Lyme disease. No new human data are collected and the proposed use of QPT in AI systems is not implemented. The approach is quantum-like (QP maths on classical computers). For human-AI teaming, it provides a structured argument for building QP-based models of the human into decision-support systems so that systems can anticipate order and interference effects in user judgements.

van der Meer et al. (2025) (E3; Quantum-like; abstract). van der Meer et al. (2025) model how a person's trust in an AI system evolves during human-AI interaction in high-risk settings, using quantum random-walk models of reliability ratings. Their contribution is to make the Hamiltonian, which governs the dynamics of the belief state, depend on the interaction: different Hamiltonians are selected according to empirically derived parameters describing what happens in the interaction, for example whether the AI agrees or disagrees with the human. The authors conclude that using empirical parameters to inform the choice of Hamiltonian is a promising way to model the evolution of trust. The work builds on the reliability-rating paradigm and data of (Roeder et al., 2023), although the abstract does not report sample sizes or fit statistics. It is an arXiv preprint and should be treated as preliminary. The modelling is quantum-like, run on classical computers. For trust-aware autonomous systems, interaction-dependent Hamiltonians offer a way for an agent's model of human trust to switch dynamics in response to its own actions.

Humr and Canan (2026) (E1; Quantum-like; abstract). Humr and Canan (2026) consider military decision-making supported by AI, where the probabilistic foundations of AI tools can violate the axioms of probability or return contradictory answers, inducing aporia and ontic uncertainty in the human and fostering over-reliance and inadvertent acceptance of the AI's choice. The chapter reviews challenges and attack surfaces for AI systems and examines three popular generative AI (GenAI) applications, showing that they fail to identify conjunction fallacies (cf. Tversky and Kahneman (1983)). It then summarises quantum cognition and proposes a system-level intervention, the quantum cognitive alignment layer (QCAL), which would use quantum models of cognition to detect and mitigate human-AI mismatches in rational deliberation, together with a research agenda. The GenAI probe is small and qualitative, and QCAL is a concept without implementation or evaluation. The approach is quantum-like. For AI decision support and autonomous systems, it proposes an architectural place for QP-based human models between the AI and the operator.

S9.2 Theory and evidence studies

Khrennikov (1999) (E2; full text). Khrennikov (1999) developed classical and quantum mechanical formalisms on 'information spaces' built from p-adic numbers, treating 'transformers of information' as basic objects and information processes as their dynamics. The motivation is that the p-adic (ultrametric) distance naturally expresses hierarchical association between ideas, so that cognitive, social and even 'anomalous' phenomena could be described as flows of information rather than matter. The paper is purely mathematical: it defines information states, Hamiltonian-type dynamics and p-adic probability, with qualitative illustrations such as hierarchical societies and systems of dogmas, but no fitting to data. Its limitation is that the link to measurable behaviour remains speculative, and later quantum-like work dropped the p-adic machinery in favour of standard Hilbert-space probability. It matters for the survey historically, as one of the first explicit attempts to apply quantum formalism to cognition, preceding the quantum-probability models later adopted for agents.

Aerts et al. (2000) (E2; abstract). Aerts et al. (2000) analysed why the disjunction ('or') of quantum logic differs from the classical logical 'or'. Taking a physical perspective, they argued that the difference stems from EPR-like correlation states of the entity under consideration, in which potentiality (outcomes not yet actualised) makes the quantum disjunction non-equivalent to the classical one. They showed that the same structure can arise in macroscopic situations where such correlation states exist, and then suggested applying the analysis to concepts to alleviate known problems in cognitive science, such as the non-compositional behaviour of concept combinations. The work is a formal, conceptual argument without experimental data. Its limitation is that the cognitive application is only sketched; the quantitative modelling of concept disjunctions came later with (Aerts, 2009). For the survey, it marks the early Brussels argument that non-classical logical connectives, not just probabilities, may be needed to represent how agents combine concepts.

Busemeyer et al. (2006) (E2; abstract). Busemeyer et al. (2006) proposed a quantum dynamic model of decision-making and compared it with an established Markov model. Both are formulated as random-walk evidence-accumulation processes, but the Markov model evolves real-valued probabilities whereas the quantum model evolves complex-valued probability amplitudes under Schrödinger-type dynamics. Because amplitudes can cancel, the quantum model generates interference effects, in which the probability of the union of two paths can be smaller than that of either path alone, something impossible for Markov processes. The authors derived choice probabilities and response-time distributions for the quantum model and contrasted its predictions with the Markov model. The paper is mainly a formal development; decisive empirical tests came in later work such as (Busemeyer et al., 2009) and (Kvam et al., 2015). For autonomous agents, it provides the template for replacing Markov evidence accumulation with a quantum walk while keeping a classical implementation.

Pothos and Busemeyer (2009) (E3; abstract). Pothos and Busemeyer (2009) addressed violations of the sure-thing principle, where people defect in a one-shot prisoner's dilemma when they know the opponent's move but cooperate more often when it is unknown, together with the analogous two-stage gambling results. They built a quantum probability model in which beliefs about the opponent and the player's own actions are represented in a four-dimensional Hilbert space, with a unitary evolution driven by payoffs and a cognitive-dissonance-type term that couples beliefs and actions. Interference between belief states in the unknown condition allows the probability of defection to fall outside the range permitted by the classical law of total probability. The model was fitted to published behavioural findings rather than new data. A limitation is the number of free parameters relative to the few data points. For agents, it shows how a quantum-like state can make an agent's action depend on whether uncertainty about another agent is resolved.

Busemeyer et al. (2009) (E3; abstract). Busemeyer et al. (2009) compared Markov and quantum theories as frameworks for probabilistic-dynamical models of decision-making. They noted that the two share many ideas but differ in key properties: Markov models obey the law of total probability but not the doubly stochastic law, whereas quantum models obey the doubly stochastic law but not the law of total probability. The article derived two parameter-free methods for testing which property holds empirically, presented a new experimental test, and reviewed findings from cognitive psychology that bear on these two properties. The strength of the approach is that the tests do not depend on fitting parameters. A limitation is that the conclusions depend on the specific paradigms examined, and later work refined the comparison, for example (Busemeyer et al., 2020). For agent design, it gives a principled, data-driven way to decide whether a Markov or a quantum dynamic best describes an operator's or agent's decision process.

Aerts (2009) (E3; full text). Aerts (2009) asked how concepts combine and carry meaning, and proposed modelling concept combinations with the Hilbert-space and Fock-space formalism of quantum mechanics. Using Hampton's (1988) membership-weight data for pairs of concepts and their conjunctions and disjunctions, he showed that over- and underextension effects (e.g. the guppy effect) cannot be reproduced by any classical probability model, and built an explicit quantum model in which superposition and interference account for these deviations. The Fock-space construction led to a proposed two-layer structure of thought: a classical logical layer and a quantum conceptual layer, whose superposition determines the measured weights (e.g. worked examples such as Poppyseeds for 'Spices or Herbs'). The paper also relates this to the disjunction effect, conjunction fallacy and Allais and Ellsberg paradoxes. A limitation is that the model reconstructs existing data with considerable freedom rather than testing new predictions. For AI, it suggests that concept representations in agents may need contextual, non-compositional structure.

Khrennikov (2010) (E2; abstract). Khrennikov (2010) is a monograph setting out the 'quantum-like' programme: applying the probabilistic formalism of quantum theory, without assuming quantum physics in the brain, to psychology, cognition, economics and finance. It builds on a contextual (Växjö) view of probability in which probabilities depend on experimental contexts, and presents a representation algorithm that maps

contextual probability data exhibiting interference into complex (and, in some cases, hyperbolic) probability amplitudes. Applications include interference of probabilities in cognitive experiments, the disjunction effect in decision-making, and quantum-like models of financial markets. The book is primarily theoretical, drawing illustrations from published data. Its limitation is that many applications are sketches rather than systematic model comparisons. For the survey, it is the main foundational statement of the quantum-like stance, the idea that QP mathematics can run on classical hardware to model agents operating under contextual uncertainty.

Trueblood and Busemeyer (2011) (E3; abstract). Trueblood and Busemeyer (2011) studied order effects in inference, where the final judgement about a hypothesis depends on the order in which evidence is presented. In a mock-jury style task, participants rated the likelihood of guilt after reading prosecution and defence evidence in different orders, producing recency effects in which later evidence had greater impact. The authors modelled this with a quantum probability model, in which evidence is processed by rotating a belief state vector across incompatible perspectives, and compared it with the heuristic belief-adjustment (anchoring-and-adjustment) model. According to the later review by (Bruza et al., 2015), the anchoring-and-adjustment model failed even to reproduce the ordering of order-effect sizes across conditions, whereas the quantum model reproduced all of them. A limitation is that the evidence comes from vignette judgements rather than consequential decisions. For autonomous systems, the work models how sequential evidence in human-agent teams may produce order-dependent beliefs that a classical Bayesian filter would not predict.

Busemeyer et al. (2011) (E3; full text). Busemeyer et al. (2011) introduced a quantum probability model of human probability judgement to explain conjunction, disjunction, inverse and conditional fallacies, as well as unpacking and partitioning effects. Events are subspaces of a Hilbert space; incompatible questions must be evaluated sequentially, so a conjunction is judged by projecting the belief state first onto the more likely event and then onto the less likely one, which can yield a conjunction probability higher than that of the less likely constituent (as in the Linda problem of (Tversky and Kahneman, 1983)). The model predicts order effects, which the authors link to published data showing larger conjunction errors when the conjunction is rated first. They compared the quantum account with the representativeness heuristic, an averaging model and a memory-retrieval model across 15 major empirical findings, concluding that the quantum model is more comprehensive for conjunction and disjunction errors, while not claiming general superiority. Limitations include qualitative rather than quantitative fitting. For AI, it offers a principled generator of human-like judgement 'errors'.

Aerts and Sozzo (2011) (E3; full text). Aerts and Sozzo (2011) argued that entanglement arises naturally when concepts combine and tested this experimentally. Treating 'The Animal Acts' as a combination of the concepts Animal and Acts, they defined four experiments in which participants chose which of two exemplars (e.g. Horse or Bear; Growls or Whinnies) was a good example of the concept, and combined them into coincidence experiments to construct a Clauser-Horne-Shimony-Holt (CHSH) version of Bell's inequality. With 81 participants, the CHSH expression reached 2.4197, exceeding the classical bound of 2, which the authors interpret as evidence of entanglement between the component concepts, grounded in the state context property (SCoP) formalism. A limitation is that Bell violations in cognition can also stem from signalling or context-dependent marginals, an issue later formalised by Contextuality-by-Default ((Dzhafarov and Kujala, 2016)). For AI, it suggests that meaning in combined concepts may not factorise into independent component representations.

Aerts et al. (2011) (E2; full text). Aerts et al. (2011c) presented the Brussels quantum-cognition programme to the quantum-technology community. They summarised experiments showing that concept combination and decision-making violate classical logic and probability (guppy effect, Allais and Ellsberg paradoxes, disjunction effect), the state context property (SCoP) formalism in which contexts change the state of a concept, and quantum models aligned with experimental data. The results are interpreted as a double-layered structure of thought: classical logical thought and quantum conceptual thought. The paper then links this to information retrieval, citing quantum-logic negation and Hilbert-space retrieval models, and web-search evidence of guppy-like effects and Bell-inequality violations, and suggests 'quantised' knowledge organisation systems. An appendix gives a simple interference model for concept disjunction. The limitation is that it is an overview without new data. It matters because it explicitly names artificial intelligence and robotics as application domains for quantum structure in cognition.

Aerts et al. (2011) (E3; full text). Aerts et al. (2011a) connected violations of Savage's sure-thing principle, and hence of expected utility, to the disjunction effect. They focused on the Hawaii problem, in which more than half of participants chose to buy a vacation package when they knew the exam outcome (54% after passing, 57% after failing) but only 32% did so when the outcome was unknown. Drawing on their earlier concept-theory work, they explained this through 'quantum conceptual thought' operating alongside 'classical logical thought', and built a quantum mechanical model of the Hawaii situation in which the decision dynamics

are generated by the conceptual landscape surrounding the choice. They also relate the analysis to overextension in concept disjunctions (e.g. Olive for 'Fruits or Vegetables') and to the Ellsberg paradox. A limitation is that the model is fitted to one classic data set with no out-of-sample test. For agents, it frames sure-thing violations as meaningful context effects rather than noise.

Yukalov and Sornette (2011) (E3; abstract). Yukalov and Sornette (2011) developed Quantum Decision Theory (QDT), in which prospects (lotteries or actions) are represented as vectors in a Hilbert space of mental states and decision-making is modelled as a quantum measurement. The probability of choosing a prospect is decomposed into a classical utility factor and a quantum interference (attraction) factor that reflects subjective attractiveness and uncertainty; composite prospects can be entangled. The authors used this framework to explain a range of classical decision paradoxes, including the disjunction effect, conjunction fallacy and Allais and Ellsberg paradoxes, and compared qualitative predictions with published experimental results. The approach is mathematically general and offers a priori estimates of interference terms, developed further in (Yukalov and Sornette, 2016). A limitation is that the interference factor is only partially constrained, leaving flexibility in fitting individual paradoxes. For autonomous agents, QDT provides an explicit utility-plus-attraction choice rule that can be implemented on classical hardware.

Bussemeyer and Bruza (2012) (E3; full text). Bussemeyer and Bruza (2012) is the first comprehensive textbook on quantum models of cognition and decision. After a non-physicist introduction to quantum probability (Chapter 2), it applies the theory to order effects in attitude judgements, probability judgement errors, concept combination, conjoint memory recognition and the mental lexicon; introduces quantum dynamics and contrasts them with Markov models; applies them to paradoxical decision findings; and covers quantum computing and quantum neural networks. Chapter 1 motivates the approach with examples including interference in categorisation-decision tasks (reporting an experiment with 26 undergraduates) and a simple two-dimensional model of the conjunction fallacy. The book stresses that quantum probability is used as a mathematical framework, not as a claim about quantum brain physics. Its limitation, like the field's, is reliance on data from laboratory paradigms. For the survey, it is the standard reference for implementing quantum-like models on classical computers for agent decision-making.

Pothos and Bussemeyer (2013) (E3; full text). Pothos and Bussemeyer (2013) argued in a target article that quantum probability (QP) theory offers a viable alternative to classical (Bayesian) probability for cognitive modelling. They introduced QP in cognitive terms, highlighting that judgements are context and order dependent, that belief states can be superpositions indefinite with respect to a question, and that composite systems can be entangled. They reviewed evidence including the conjunction fallacy, violations of the sure-thing principle in the prisoner's dilemma and two-stage gambles, and question order effects such as a Gallup poll in which 'yes' rates for 'Is Clinton honest?' and 'Is Gore honest?' were 50% and 68% in one order but 57% and 60% in the reverse order. They also discussed rationality and model complexity, remaining agnostic about quantum brain physics. As a target article it was published with open peer commentary, where model flexibility and testability were debated. For the survey, it is the reference statement that QP models are cognitive, not physical, models, which is what makes them portable to classical agents.

Wang et al. (2013) (E1; abstract). Wang et al. (2013) introduced a Topics in Cognitive Science special issue on quantum cognition and set out the case for using quantum theory to build models of cognition. They argued that the mathematical principles of quantum probability, such as incompatibility of questions, superposition of belief states and entanglement, provide a coherent formal basis for phenomena that are awkward for classical probability, including order effects, violations of the law of total probability and non-compositional concept combination. The paper emphasises that the approach is a probabilistic modelling framework, independent of claims about quantum physics in the brain, and outlines open issues such as model testability and the relation to classical Bayesian and heuristic accounts. It contains no new data. Its limitation is its programmatic nature: the evidential weight rests on the articles it introduces, such as (Wang and Bussemeyer, 2013) and (Aerts et al., 2013). For the survey, it documents the field's consolidation phase and its explicitly non-physical interpretation.

Wang and Bussemeyer (2013) (E3; abstract). Wang and Bussemeyer (2013) derived an a priori, parameter-free prediction of the quantum question-order model, the quantum question (QQ) equality: the order effect for 'yes-yes' plus 'no-no' answer sequences must cancel against the order effect for 'yes-no' plus 'no-yes' sequences, so that the summed order effects equal zero. The prediction follows from representing answers as projections in a Hilbert space and holds regardless of the initial belief state. They tested it against published survey data on question order effects, including well-known attitude questions, and found it supported, whereas many classical models that produce order effects do not generally imply it. A limitation is that the tests use existing survey

data, and the QQ equality does not by itself rule out all non-quantum explanations. A much larger test followed in (Wang et al., 2014). For agents that query humans sequentially, the QQ equality offers a concrete diagnostic of whether answers behave as quantum-like projections.

Aerts et al. (2013) (E3; abstract). Aerts et al. (2013) presented, for a cognitive-science audience, the Brussels quantum-theoretic approach to concepts and their dynamics. Concepts are treated as entities in states that change under context, so that exemplar typicality and membership weights are context dependent. Using Hilbert space and Fock space, the authors modelled the guppy effect, over- and underextension in conjunctions and disjunctions of concepts, and entanglement in concept combinations, and interpreted the results as a two-layer structure of human thought (quantum conceptual and classical logical) as in (Aerts, 2009) and (Aerts and Sozzo, 2011). They argued that these quantum effects reflect how meaning is carried by concepts rather than failures of reasoning. A limitation is that the models are largely reconstructions of existing experimental data, with high representational freedom. For AI and robotics, the paper motivates context-dependent concept representations for agents that must interpret combined linguistic commands or categories.

Wang et al. (2014) (E3; abstract). Wang et al. (2014) tested the quantum question (QQ) equality, the parameter-free prediction of the quantum question-order model introduced in (Wang and Busemeyer, 2013), on a large body of survey data. According to the later review by (Bruza et al., 2015), the equality was statistically supported across 70 national field experiments on question order covering a wide range of topics, plus two laboratory experiments, and anchoring-and-adjustment accounts failed to reproduce the observed patterns. The authors argued that this pattern is predicted a priori by quantum probability yet is not implied by many classical models that produce order effects, so that question order effects reveal a quantum-like structure of human judgement. A limitation is that the QQ equality is necessary rather than sufficient evidence, and later analyses (e.g. Khrennikov et al. (2014)) showed that simple projective models face difficulties once response replicability is also considered. For human-AI interaction, it is the strongest large-scale evidence that sequential queries change the responder’s state.

Khrennikov et al. (2014) (E2; abstract). Khrennikov et al. (2014) identified a basic difficulty for quantum models of psychological measurement. Question order effects require the corresponding projectors to be non-commuting, but people also show response replicability: when a question is asked again, including after an intervening second question, they tend to repeat their earlier answer. The authors showed that standard quantum models with projective measurements and Lüders state updating cannot reproduce order effects together with such replicability, and argued that simple generalisations of the measurement model do not resolve the conflict, hence ‘an unsolved problem’. The analysis is formal, drawing on the empirical properties of question-order data such as the QQ equality of (Wang et al., 2014). A limitation is that it assumes particular measurement classes; later work proposed resolutions using more general quantum instruments ((Ozawa and Khrennikov, 2021)). For agents, it warns that naive projective QP modules may be inconsistent with the stability of human answers in repeated interactions.

Bruza et al. (2015) (E3; full text). Bruza et al. (2015) reviewed quantum cognition as a third approach alongside heuristic and rational (Bayesian) accounts of judgement and decision. They contrasted the classical and quantum axioms (in boxed summaries), explaining that the two agree when all events are compatible and diverge only when incompatible events force sequential evaluation. Key evidence reviewed includes the Linda conjunction fallacy, question order effects and the quantum question (QQ) equality, reported as statistically supported across 70 national field experiments; order effects in inference, where the quantum model outperformed anchoring-and-adjustment; violations of the law of total probability in the prisoner’s dilemma; and categorisation-decision interference in which attack rates without categorisation fell outside the classical bounds. They also discussed rationality, suggesting quantum models may approximate optimal performance with lower representational demands than full joint distributions, and listed open questions on dynamics and contextuality. As a review, it presents no new data. For agents, its compatibility argument clarifies when a quantum-like module adds anything over a Bayesian one.

Busemeyer and Wang (2015) (E1; abstract). Busemeyer and Wang (2015) gave a short, accessible introduction to quantum cognition for a general psychology readership. They explained that quantum cognition uses the mathematical principles of quantum probability, not quantum physics of the brain, to model how people form judgements and decisions, emphasising incompatible questions, superposition states and the resulting order and context effects. They illustrated applications such as question order effects, conjunction and disjunction fallacies, and violations of the sure-thing principle, and indicated how quantum models can generate a priori testable predictions. The article is a review and contains no new data or model fits. Its limitation is brevity: it

cannot address competing classical explanations in depth. For the survey, it is a convenient entry-level citation for the quantum-like stance adopted when transferring these models to artificial agents.

Kvam et al. (2015) (E3; abstract). Kvam et al. (2015) tested whether evidence accumulation in perceptual decisions behaves like a Markov or a quantum random walk. In a random-dot motion task, participants gave confidence ratings at two time points, with or without making an explicit choice at the first point. A Markov (classical random walk) model implies that making an intermediate choice should not change the distribution of later confidence, apart from the information already accumulated, whereas a quantum walk predicts an interference effect, because the choice collapses an indefinite belief state. The authors found that the intermediate choice did affect later confidence, consistent with interference, and that a quantum random walk model accounted for the data better than Markov models. A limitation is that the paradigm is a simple two-alternative perceptual task. For robots and agents, it supports quantum walk models of evidence accumulation in which committing to a decision changes subsequent confidence, relevant to human-robot teaming where operators report decisions.

Yearsley and Busemeyer (2016) (E2; abstract). Yearsley and Busemeyer (2016) wrote a tutorial on building quantum models of cognition and decision, aimed at researchers familiar with linear algebra but not necessarily with quantum physics. Rather than surveying the field, they focused on the practical construction of models, using order effects in judgement as the running example. They showed how quantum theory naturally produces order effects through incompatible (non-commuting) projections and derived the quantum question (QQ) equality, a prediction that competing frameworks do not generally derive and that has received strong experimental support ((Wang et al., 2014)). The paper covers the essential mathematical tools, model specification and applications. As a tutorial, it contains no new data. Its limitation is its restriction to pure states and ideal measurements, extended in (Yearsley, 2017). For the survey, it is the natural methodological reference for engineers implementing a quantum-like judgement module on classical hardware.

Wang and Busemeyer (2016) (E3; abstract). Wang and Busemeyer (2016) examined how explicit categorisation influences subsequent decisions. Across three experiments with 721 participants, people viewed faces and either categorised them and then decided on an action (C-D trials), were told the category and then decided (X-D trials), or decided without categorising (D-alone trials). In some conditions an interference effect appeared: the probability of action on D-alone trials differed from the total probability computed from the categorisation trials, violating the classical law of total probability and challenging Markov and signal-detection models. To explain this, the authors proposed a belief-action entanglement (BAE) model that uses superposition and entanglement between belief and action states, and reported that it predicted the observed pattern across experiments. A limitation is that interference appeared only in some conditions, and the stimuli are laboratory faces with artificial reward structures. For autonomous agents and human-robot teams, it models how forcing an explicit classification step (e.g. friend or foe) can change the downstream action probabilities.

Yearsley and Pothos (2016) (E3; abstract). Yearsley and Pothos (2016) tested a distinctive prediction of quantum cognition: the quantum Zeno effect, whereby frequent measurement of a system slows or freezes its evolution. Applied to cognition, the hypothesis is that repeatedly asking a person to report their belief while evidence accumulates will slow the change in that belief compared with fewer judgements. The authors built a quantum model of belief change under sequential evidence and conducted experiments in which participants evaluated a scenario while the number of intermediate judgements was manipulated, reporting results consistent with a Zeno-type slowing of belief change that is difficult to reconcile with standard classical accounts. A limitation is that alternative explanations, such as commitment or consistency effects from repeated reporting, must be carefully excluded, and the effect was studied in a specific judgement paradigm. For agents that frequently query human operators, it suggests that the querying rate itself may alter the operator’s belief trajectory.

Yukalov and Sornette (2016) (E2; full text). Yukalov and Sornette (2016) gave a general definition of quantum probability that covers elementary and composite events, operationally testable and inconclusive measurements, and commuting and non-commuting observables, so that quantum measurement and quantum decision-making share one mathematical footing. Within their Quantum Decision Theory ((Yukalov and Sornette, 2011)), a prospect probability splits into a utility factor and an attraction (interference) factor; a ‘quarter law’ motivates a non-informative prior of about ± 0.25 for the attraction factor. In the prisoner’s dilemma example, published data give cooperation and defection probabilities of 0.37 and 0.63 when the partner’s move is unknown, and QDT with the ± 0.25 prior predicts 0.35 and 0.65. They also argued that the attraction factor decays roughly exponentially with added information, explaining preference reversal, and stated a quantum-classical correspondence principle specifying when classical decision theory is recovered. A limitation is that

empirical support rests on a single illustrative example here. For agents, QDT offers a parameter-light choice rule usable on classical hardware.

Dzhafarov and Kujala (2016) (E2; abstract). Dzhafarov and Kujala (2016) gave a systematic presentation of Contextuality-by-Default (CbD), restricted to finite systems of categorical random variables. In CbD, each random variable is identified by its content (what is measured) and its context (the conditions of measurement), and two variables have a joint distribution only if they share a context. Variables sharing a context form 'bunches', and variables with the same content in different contexts form 'connections'. A system is contextual if no probabilistic coupling of the bunches can make all connections satisfy the required (maximal-agreement) properties simultaneously, meaning that context affects variables beyond direct influences. The framework is classical probability theory and is neutral about quantum mechanics, yet it provides the tool for deciding whether behavioural data are genuinely contextual or merely show context-dependent marginals. A limitation is that it tests, rather than models, cognition. For the survey, it is the reference method for auditing contextuality claims about human or agent decisions.

Yearsley (2017) (E2; abstract). Yearsley (2017) extended the tutorial of (Yearsley and Busemeyer, 2016) to the more advanced tools needed for realistic quantum models of cognition and decision. The tutorial introduces density matrices to represent mixed states and imperfect knowledge, positive-operator-valued measures (POVMs) for noisy or imprecise measurements, and open-system dynamics in which the cognitive system interacts with an environment. The common aim is to incorporate noise and measurement uncertainty into quantum cognitive models rather than assuming idealised pure states and projective measurements. The paper is methodological and contains no new data. Its limitation is that these richer tools add parameters, increasing model flexibility and the burden of constraint by data. For autonomous agents, it is the key methodological bridge to open-system (e.g. Lindblad-type) quantum-like models of noisy decision processes, which are implementable on classical hardware and relevant to decoherence-style accounts of settling on a decision.

Pothos et al. (2017) (E2; abstract). Pothos et al. (2017) examined whether behaviour described by quantum probability can be regarded as rational. Classical probability underpins normative accounts of rationality, for example through Dutch book arguments, so the quantum account of apparent fallacies such as the conjunction fallacy raises the question of whether such judgements are simply errors. The authors analysed the conditions under which quantum-probabilistic reasoning can be vulnerable to, or protected from, Dutch book-style exploitation, and argued that quantum probability can be seen as a principled framework for reasoning under incompatible perspectives and limited resources, rather than a catalogue of mistakes. A limitation is that the argument concerns rationality in a restricted sense and depends on assumptions about how judgements are elicited. For the survey, the question matters directly: designing agents with quantum-like decision modules requires a position on whether their deviations from Bayesian norms are defensible or should be corrected.

Trueblood et al. (2017) (E3; abstract). Trueblood et al. (2017) asked how inference can be Bayesian in some situations and not in others, and proposed a unified quantum probability account. They postulate a hierarchy of mental representations running from fully quantum to classical, in which moving between levels changes the assumptions about compatibility, that is, how joint events are represented. Using data from three experiments, the authors show that the framework accounts for five inference phenomena: order effects, reciprocity (the inverse fallacy), memorylessness, violations of the Markov condition and anti-discounting; they state that no existing theory explains all five. They also examine transitions within the hierarchy, finding that representations become more classical as individuals gain familiarity with a task, and that individual representations relate to the Cognitive Reflection Test. A limitation is the added flexibility of choosing a hierarchy level per person and task. For autonomous agents, the idea of a graded, learnable shift from incompatible to compatible representations offers a principled way to model human teammates whose inference becomes more normative with experience.

Broekaert et al. (2017) (E2; abstract). Broekaert et al. (2017) examined the dynamical side of quantum probability theory (QPT) in cognitive modelling, asking how the mind's driving potentials, encoded as Hamiltonian and Lindbladian operators, shape the time evolution of a mental state. Noting that existing QPT models usually adopt strong simplifying assumptions about dynamics, they construct a series of nested models designed for a simple hypothetical decision-making paradigm. They consider Hamiltonians more complex than those traditionally used, vary the initial state (a mixed state with a specific orthogonal decomposition versus a general mixed state), and extend the dynamics with Lindbladian operators that destroy separability structure, giving an open-system extension. Relations between the models are illustrated mathematically and numerically, but no human data are fitted. The main limitation is therefore the absence of empirical constraint on the added

flexibility. For autonomous agents, the paper is a useful map of the modelling choices, Hamiltonian versus Lindbladian, pure versus mixed, available when designing open-system decision dynamics.

Aerts et al. (2017) (E4; abstract). Aerts et al. (2017) contrasted two quantum accounts of the conjunction fallacy: one in which respondents reason sequentially, so that the fallacy arises from question order effects, and one in which the conjunction is evaluated as a whole, so that the fallacy arises from the emergence of new meaning as an overextension effect. To discriminate between them, the authors use the World Wide Web as an information space that can be interrogated both sequentially and non-sequentially, treating web-based counts as a proxy for collective conceptual structure rather than running a new laboratory study. They conclude that emergence effects, not order effects, should be considered the main mechanism producing the observed conjunction fallacies. The limitation is the indirect link between web statistics and individual judgement processes. For AI systems that draw on web-scale text, the result suggests that conjunction-type biases may be inherited from corpus-level meaning structure rather than from sequential inference alone.

Aerts and Sassoli de Bianchi (2017) (E3; full text). Aerts and Sassoli de Bianchi (2017) ask whether the Born rule suffices for question order effects. Their general tension-reduction (GTR) model is a hidden-measurement framework in which Born probabilities emerge as an average over non-uniform contextual fluctuations, so departures are expected when averaging is incomplete or measurements are combined. Using an exact GTR solution on the extended Bloch sphere, they reproduce the Gallup data reported by Moore (2002) for the Clinton/Gore and Rose/Jackson question pairs. They argue that both data sets are irreducibly non-Hilbertian, that the QQ-equality of (Wang et al., 2014) can also hold for non-Hilbertian models and is therefore not a sufficient test, and that the joint occurrence of order effects and response replicability, which projective quantum models cannot handle, is naturally captured by the GTR model. The work re-describes published aggregate data without new experiments or out-of-sample tests, and the added fluctuation structure makes the model very flexible. For autonomous agents, it suggests Hilbert-space belief models may be only a first approximation when sequential queries interact.

Busemeyer and Wang (2018) (E3; abstract). Busemeyer and Wang (2018) presented Hilbert space multidimensional (HSM) theory, a general account of measurement context effects. A context is the subset of psychological variables a person evaluates on a given occasion; context effects occur when judgements across different, possibly overlapping, contexts cannot be derived from a single joint probability distribution over all variables. HSM represents all variables within a low-dimensional vector space and uses quantum probability to generate the pattern of judgements across contexts, with parameter estimates that the authors argue are simple to interpret. The theory is tested on a new large experiment, with HSM fits compared against Bayesian network fits, supporting its viability. The authors conclude that context effects are broadly applicable in the social and behavioural sciences. A limitation is that model comparison is against one classical family, and dimensionality must be chosen by the modeller. For human-AI teaming, HSM offers a compact way to represent operator judgements collected under varying question subsets.

Cervantes and Dzhamfarov (2018) (E3; abstract). Cervantes and Dzhamfarov (2018) reported what they describe as unambiguous experimental evidence for quantum-like probabilistic contextuality in psychology. Earlier attempts had failed because behavioural data grossly violate marginal selectivity, making traditional quantum-mechanical tests inapplicable. In a crowdsourcing experiment, respondents made two simple choices, one of characters from Andersen’s *The Snow Queen* and one of characteristics such as Kind and Evil, with the design imitating the Bohm version of the Einstein–Podolsky–Rosen/Bell paradigm. Marginal selectivity was violated, showing that the choices directly influenced one another, but the Contextuality-by-Default test, which extends the traditional test to such data, indicated a strong presence of true contextuality not reducible to direct influences. The limitation is that the task is deliberately artificial and the design was engineered to produce contextuality. For autonomous agents, the paper matters because it supplies a rigorous test that separates genuine contextuality from ordinary context-dependent signalling in human responses.

Busemeyer et al. (2019) (E3; full text). Busemeyer et al. (2019) compare Markov and quantum dynamic models of belief change during evidence monitoring. In the Markov (random-walk) model the belief occupies one point on an evidence scale; in the quantum model it is a superposition over evidence states that evolves unitarily and is collapsed by each rating request, predicting interference between successive ratings. Eleven participants completed about 1000 dot-motion trials each at four coherence levels, giving two probability ratings at pairs of time points across three conditions. Interference was small, significant only at low coherence and for 3 of 11 participants. In a generalisation test (parameters fitted to two conditions, predictions for the third), summed G^2 differences favoured the quantum model at all coherence levels (350, 306, 260, 40), but against a Markov model with drift-rate variability the quantum model won only at low coherence; the authors conclude that

features of both are needed. The sample is small. For autonomous agents, the design is a template for testing whether querying an agent’s belief perturbs later updating.

Basieva et al. (2019) (E3; abstract). Basieva et al. (2019) addressed whether human judgement systems exhibit true contextuality, as opposed to context-dependence caused by direct causal influences of context on responses. Using Contextuality-by-Default theory, which separates these two sources, they first reanalyse numerous earlier claimed demonstrations of contextuality in human judgement and show that all can be accounted for by direct influences, with no true contextuality present. They then present a series of crowdsourcing experiments on simple decision making, elaborating the design of the Snow Queen study (Cervantes and Dzhafarov, 2018), and show that these experiments do exhibit true contextuality. The limitation is that the demonstrating tasks are specially constructed, so the prevalence of true contextuality in everyday cognition remains open. For autonomous agents, the work sets a methodological bar: context effects in human data should be tested for direct influences before quantum-like models are claimed to be necessary.

Busemeyer et al. (2020) (E2; abstract). Busemeyer et al. (2020) reviewed how open quantum system models can combine Markov and quantum dynamics in cognition. Markov processes such as random walks have long been used to model choice and decision time, while quantum walk models have recently been introduced for the dynamics of choice and confidence; the authors argue that empirical evidence calls for both. They formulate a quantum–Markov master equation (Lindblad form) in which a weight parameter between 0 and 1 sets the relative contribution of the quantum and Markov components, illustrate it with a 101-dimensional state space for evidence or preference levels, and discuss the challenges of formulating such models, including how to derive choice and response-time predictions. The paper is primarily methodological, with illustrations rather than new experiments. For autonomous agents and human–machine systems, it offers a practical template for decision dynamics that interpolate between classical diffusion and quantum-like oscillation.

Ozawa and Khrennikov (2020) (E2; abstract). Ozawa and Khrennikov (2020) responded to a no-go result against quantum models of psychology. The question order effect had been modelled by representing questions as Hermitian observables with orthogonal projectors for the back-action of answering, but this representation cannot also reproduce the response replicability effect, whereby a person repeats an earlier answer to the same question; the no-go result had been extended to atomic quantum instruments, prompting doubts about quantum formalism in psychology. The authors show that both effects can be modelled together within the theory of non-atomic quantum instruments, which generalise projective measurement. The contribution is purely mathematical, with no data fitting. The limitation is that the more general instruments add representational freedom whose psychological interpretation remains to be constrained. For agents interacting with humans, the paper indicates that sequential questioning (order effects plus stable repeated answers) can still be captured with a quantum-instrument measurement model.

Yukalov (2020) (E2; abstract). Yukalov (2020) reviewed the basics of quantum decision theory (QDT) with an emphasis on temporal processes in decision making. The review explains how an operationally testable rational choice between alternatives differs from a choice influenced by irrational feelings, the latter entering through an attraction or quality factor that supplements a utility factor. Quantum–classical correspondence is emphasised, so that QDT reduces to classical decision theory when the quantum term vanishes. The paper also describes a model of a quantum intelligence network, in which decision makers exchange information and their probabilities evolve over time, and argues that dynamic inconsistencies in preferences are resolved within QDT. As a review of the author’s own framework, it offers illustrations rather than new empirical tests, and its parameters are largely set by general arguments. For multi-agent systems, the network model is relevant as a formal description of how interacting agents’ decision probabilities can converge through information exchange.

Kvam et al. (2021) (E3; abstract). Kvam et al. (2021) investigated how preferences are constructed over time by tracking dynamic changes in preference strength. Across two experiments, they observed that mean preference strength oscillated systematically over time, and that eliciting a choice early strongly affected the later pattern of oscillation: preferences following a choice alternated between being stronger and weaker than those without a prior choice. To account for this, they developed an open system model that merges the dynamics of a Markov random walk with those of quantum processes. The model represents two sources of uncertainty: epistemic uncertainty about the decision maker’s state at a given time, and ontic uncertainty about what judgement will be produced when a person in some state is measured. This allows it to capture both the oscillations and the effect of choice on preference formation. A limitation is that the oscillation evidence rests on laboratory preference tasks with aggregated means. For autonomous agents, the work shows how intermediate queries can alter a human’s later preference, relevant to interactive elicitation.

Ozawa and Khrennikov (2021) (E3; full text). Ozawa and Khrennikov (2021) address a known obstacle for quantum-like models: projective (von Neumann–Lüders) measurement cannot jointly produce the question order effect (QOE) and the response replicability effect (RRE), in which a repeated question receives the same answer even after an intervening question. Working in the general theory of quantum instruments, they construct instruments that simultaneously satisfy QOE, RRE and the QQ-equality of (Wang et al., 2014), and they state postulates that generalise the Wang–Busemeyer postulates for quantum-like decision modelling. Applied to the Clinton–Gore Gallup poll data, the model reproduces the statistics closely with a prior belief state that does not depend on question order, and the authors use it to ‘correct’ the order effect and estimate the underlying opinion distribution. Its empirical content is limited to a single, already published poll, and the instrument construction is a proof of possibility rather than a model selected against alternatives. For autonomous agents, instrument-based updating offers a principled way to represent queries that disturb internal state while still allowing stable repeated answers.

Epping et al. (2023) (E3; abstract). Epping et al. (2023) extended the open system model of decision making, a hybrid of Markov and quantum random walk dynamics, from preference strength to pairwise choice and response time. Sequential sampling models conventionally follow Kolmogorov axioms (random walks, diffusion), while quantum random walk models have been proposed as an alternative; earlier work found that the hybrid open system model accounted for preference strength better than either pure model (Kvam et al., 2021). The authors report a new experiment on preferential choice between gift cards and compare the open system model with pure Markov and pure quantum random walk models using AIC, BIC and chi-squared tests. The pure Markov model was favoured for most participants, but a substantial number required the more complex open system model. The limitation is that the added complexity is justified only for a subset of individuals, which the authors report candidly. For autonomous agents, the result supports hybrid, individually selected decision dynamics rather than assuming quantum-like behaviour across all users.

Khrennikov (2023) (E2; abstract). Khrennikov (2023) reviewed the application of quantum formalism to model the behaviour of complex biosystems, from genomes and proteins to animals, humans and social systems, stressing that such quantum-like models must be distinguished from genuine quantum physical modelling of biology. Because any isolated biosystem is dead, he argues that biological and mental processes should be modelled with the general theory of open quantum systems, and explains its application through quantum instruments and the quantum master equation. Possible interpretations are discussed, with QBism favoured as the most useful. The paper is conceptual and mathematical, drawing on prior work rather than new data, and its broad scope limits detailed validation. For autonomous agents, it provides the theoretical rationale for treating an agent’s decision state as an open system continuously coupled to its environment.

Khrennikov (2024) (E2; full text). Khrennikov (2024) develops the contextual measurement model (CMM), a general probabilistic framework based on contextual probability spaces that is intended to clarify quantum foundations in line with Bohr’s emphasis on experimental contexts. CMM treats probability update as the core operation and is designed to cover classical, quantum and semi-classical measurements within one scheme. The paper shows how Kolmogorov probability, von Neumann projective measurement and quantum-instrument theory can each be framed as special cases of CMM, and gives abstract contextual definitions of the order effect, replicability and response replicability, interference of probabilities (a modified formula of total probability), entanglement and Bell-inequality violation. It also links CMM to the linear space representation used in generalised probability theory. The contribution is purely theoretical, illustrated only with simple examples and not tested against behavioural data. Its relevance to autonomous agents is that it offers a formal vocabulary in which classical, quantum-like and hybrid belief-update rules can be compared, which matters when choosing an update rule for an artificial agent.

Waddup et al. (2023) (E3; abstract). Waddup et al. (2023) imported the temporal Bell (Leggett–Garg-type) inequality from physics to test whether recollection alters the information later retrieved from memory. A temporal Bell inequality tests sensitivity to the context of previous measurements in sequential measurement scenarios, as quantum theory predicts. The authors designed a memory paradigm using a novel change judgement, in which participants decide whether there has been a change in a question’s answer across different time points of a scenario. Across two experiments, they observed a violation of a temporal Bell inequality in one case, which they interpret as evidence for quantum-like processes in memory. The limitation is that the violation was found in only one case, so the evidence is suggestive rather than conclusive. For autonomous agents, the paradigm illustrates that querying a human’s memory may itself change subsequent reports, relevant to repeated situation-awareness probes.

Velož and Sobetska (2024) (E3; full text). Velož and Sobetska (2024) examine the conjunction fallacy extensionally, as a set of possible observed facts, rather than asking what causes it. Any conjunction experiment yields a point $(P(A), P(B), P(A \wedge B))$ in the unit cube, and they ask which regions of this space the empirical literature has actually sampled. Reviewing 4080 articles citing (Tversky and Kahneman, 1983) up to 2016, they found 272 with probability-estimate experiments and only 37 that reported all three estimates, yielding 416 data points. These points were distributed very unevenly: single-fallacy cases were well explored, whereas the no-fallacy region and especially the 'double fallacy' $(P(A \wedge B) \text{ above both constituents})$ were rarely probed. The authors argue that explanations of the fallacy, classical and quantum alike, rest on a narrow and biased empirical base, and call for experiments designed to reach unexplored regions. The analysis is descriptive, uses a coarse partition of the cube, and proposes no model. For the survey it cautions that fits of quantum-like conjunction models to agent or human data may reflect limited coverage of possible judgements.

Huang et al. (2025) (E3; abstract). Huang et al. (2025a) tackled the reconciliation of Bayesian normative expectations with fallacies in probability judgement. Recent Bayesian accounts attribute fallacies to sampling noise, whereas quantum models invoke different probability rules. The authors argue a framework must incorporate both Bayesian and quantum influences, and must also explain findings beyond any current model. They develop the Quantum Sequential Sampler, which integrates Bayesian and quantum reasoning with a sequential sampling process that maps subjective probability estimates onto observable responses. It is compared with the leading Bayesian Sampler using a new experiment which they describe as producing one of the largest data sets in probabilistic reasoning to date. The empirical tests also revealed a new, systematic overestimation of probabilities. A limitation is the complexity of the combined model and its comparison with a single main classical rival. For AI systems estimating human judgements, it shows how quantum and sampling noise accounts can be combined in one generative model.

Huang et al. (2025) (E2; full text). Huang et al. (2025c) present, as a conference poster paper, a tutorial example of a 'quantum heuristics toolbox' that encodes fast-and-frugal heuristics as quantum-controlled unitary operations. They focus on the recognition heuristic: a gate controlled by two recognition qubits rotates or flips the target decision state, reproducing the classical deterministic rule when the cues are measured but producing probabilistic choice when the cues remain in superposition; graded judgements are obtained by coupling the operator to a quantum sequential sampler (Huang et al., 2025a). The key claimed advantage over classical probabilistic heuristics is that projective measurement naturally yields hindsight bias in an ABA paradigm, where recognition of a city is judged again after an intervening comparison question, without ad hoc memory assumptions. They outline extensions to interference, order and CHSH-type super-correlation effects in multi-cue reasoning. The paper is short, uses the quantum-circuit formalism only as a modelling language on classical computers, and reports no fitted data. For autonomous agents it suggests a route to bounded-rational, rule-based decision policies expressed in a circuit form compatible with quantum AI.

Fuyama et al. (2025) (E2; full text). Fuyama et al. (2025) ask which class of quantum measurements best matches quantum-like models of cognition. Projective measurements, the canonical class in physics, cannot jointly reproduce the question order and response replicability effects, so the authors argue for sharp repeatable non-projective ($SR\bar{P}$) measurements, a class practically unused in physics. They show that $SR\bar{P}$ instruments can both satisfy and violate the QQ-equality, which accommodates opinion-poll data that obey it as well as aesthetic-perception data reported to violate it. A central conceptual point is the distinction between two sources of non-classicality: non-commutativity of observables, as in physics, and non-commutativity of the state-update maps induced by measurement back-action. In $SR\bar{P}$ models the observables can commute while the update maps do not, so order effects need not imply incompatible questions. The work is theoretical and does not fit new data. For autonomous agents it clarifies that 'quantum-like' belief updating can reside in how a query changes the internal state rather than in incompatible representations, which affects how such agents might be designed and tested.

Asano and Khrennikov (2026) (E2; full text). Asano and Khrennikov (2026) review and extend open-system models of cognition based on the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation, in which the mental state is a density operator that evolves dissipatively through interaction with an informational environment and relaxes to a steady state interpreted as a decision ('decision making through decoherence'). They distinguish 'passive' and 'active' Hamiltonians, treating non-commutation of the Hamiltonian with projections onto the decision basis as a mathematical signature of cognitive agency and of escape from classical equilibria, and they show how such dynamics can stabilise non-Nash outcomes in Prisoner's Dilemma-type games. They further introduce 'cognitive beats', slow modulations produced when competing Liouvillian channels operate at similar frequencies, as a proposed marker of internal deliberation. The paper is theoretical: it cites earlier eye-tracking data but notes that these lacked the temporal resolution to detect beats, and it

only sketches possible experiments. For autonomous agents it supplies a continuous-time, dissipative model of deliberation that could inform agents whose commitment to an action should emerge gradually.

Bruza et al. (2009) (E2; abstract). Bruza et al. (2009) took up Nelson and McEvoy’s claim that word associations can display ‘spooky activation at a distance’ and proposed quantum theory as a framework for modelling the human mental lexicon. They focus on results from intralist and extralist cued-recall word association experiments, in which the associative structure around a target word affects its recall. Initial quantum models of these association effects are discussed, and the authors propose experiments capable of testing these models. The contribution is programmatic: the models are exploratory and new tests are proposed rather than conducted. A limitation is therefore the lack of direct empirical discrimination between quantum and classical associative accounts. For language-based AI agents and LLMs, the paper is an early source for the idea that lexical semantics may be non-separable, motivating later quantum-inspired information retrieval and NLP models.

Blutner et al. (2013) (E2; abstract). Blutner et al. (2013) studied vagueness, the property of natural concepts with fuzzy boundaries, borderline cases and susceptibility to the sorites paradox. They review experiments on judgements of borderline cases and borderline contradictions, where people accept statements such as ‘x is tall and not tall’. They first propose a classical probabilistic model, based on a Hopfield network for predicting truth values, that extends Alxatib and Pelletier’s analysis, and show that it falls short of covering the relevant empirical results. They then replace standard probabilities with quantum probabilities, so that borderline contradictions are explained as an interference phenomenon. The work is largely theoretical, applied to existing experimental findings rather than new data. For language-grounded agents and LLMs, it offers a formal account of why vague predicates yield contradictory yet acceptable descriptions near category boundaries.

Pothos et al. (2013) (E2; abstract). Pothos et al. (2013) revisited Tversky’s (1977) demonstrations that similarity judgements violate symmetry and the triangle inequality and show context effects, which undermine geometric models of similarity. They propose a quantum geometric model in which concepts are subspaces and similarity is a quantum probability, computed from sequential projections. Asymmetries then emerge as order effects, while violations of the triangle equality and the diagnosticity effect relate to context-dependent properties of quantum probability. The model thus retains a geometric representation while addressing several of Tversky’s main findings, and the authors discuss implications for similarity theory in general. The work is primarily theoretical, reproducing established empirical patterns rather than fitting new data. For AI and embedding-based agents, it shows how vector representations can yield asymmetric, context-sensitive similarity, relevant to retrieval and categorisation.

Trueblood and Hemmer (2017) (E3; abstract). Trueblood and Hemmer (2017) addressed two memory phenomena: over-distribution, where acceptance probabilities across mutually exclusive conditions violate the disjunction rule, and subadditivity, where probabilities across exclusive and exhaustive instruction conditions sum to more than one. Signal-detection theory struggles with both, and the dual-trace over-distribution (OD) model explains only over-distribution. They propose the Generalized Quantum Episodic Memory (GQEM) model, extending the Quantum Episodic Memory model of Brainerd, Wang and Reyna (2013). GQEM is compared with OD using data from a novel item-memory experiment and a previously published source-memory experiment (Kellen, Singmann and Klauer, 2014). Using best-fit parameters, the authors show that GQEM also accounts for subadditivity. A limitation is the restricted set of paradigms and a single main classical comparison. For agents relying on human testimony, the work formalises why people endorse mutually exclusive recollections.

Busemeyer et al. (2025) (E2; full text). Busemeyer et al. (2025) consider how quantum models of judgement can incorporate episodic memory of one’s own recent responses, which a Bayesian model treats as uninformative but which empirically influences later judgements. They review system-plus-environment (indirect measurement) representations and quantum instruments, and then compare three models against a puzzling combination of findings: repeated answers in AA paradigms, question order effects that obey the QQ-equality in AB/BA designs, and the response replicability effect in ABA designs. The models are the Ozawa–Khrennikov instrument account ((Ozawa and Khrennikov, 2021)), Busemeyer’s system-plus-environment account with an explicit memory-recall operation, and a new reformulation of the latter in instrument terms. The comparison is analytical: the accounts differ in whether order effects arise from non-commuting projections or from belief updates by person ‘types’, and in whether participants must always repeat their earlier answer. No new data are fitted, and the authors note that the empirical ABA evidence is itself contested. For autonomous agents the work is relevant to how an agent’s record of its own past outputs should enter subsequent belief updates.

Costello and Watts (2014) (E3; abstract). Costello and Watts (2014) challenged the view that systematic biases in probability judgement show that people use heuristics rather than probability theory. They propose that people reason according to probability theory but subject to random noise in the reasoning process, and show that the effect of noise cancels for certain probabilistic expressions. Analysing data from two experiments, they find that for these expressions people’s judgements are strikingly close to those required by probability theory, whereas for other expressions noise produces systematic biases. The account explains four reliable biases: conservatism, subadditivity, and the conjunction and disjunction fallacies. The limitation, from the survey’s perspective, is that the model does not address all phenomena targeted by quantum models, such as order effects in sequential questions. It matters here as the principal classical rival to quantum-like accounts of judgement, providing a benchmark that any agent-side model of human probability errors should be compared against.

Boyer-Kassem et al. (2016) (E3; abstract). Boyer-Kassem et al. (2016) investigated the empirical adequacy of the major quantum-like models of the conjunction fallacy, in which people judge a conjunction more probable than one of its conjuncts. They focus on models that represent beliefs with quantum states and argue that these can be tested in three different ways, including in a question order effect configuration that differs from the traditional conjunction fallacy experiment. They then run the proposed experiment using methodologies from experimental economics. The results are at odds with the predictions of the quantum-like models, which the authors take as strong evidence that this account of the conjunction fallacy fails; future research paths are discussed. A limitation is that the critique targets a specific class of projective quantum-like models, leaving other quantum accounts, such as emergence-based models (Aerts et al., 2017), less directly tested. For the survey, it is a key empirical critique that tempers claims of quantum-like explanations for judgement fallacies.

Dzhafarov et al. (2016) (E3; abstract). Dzhafarov et al. (2016) asked whether behavioural and social systems genuinely exhibit contextuality. Most relevant experiments involve cyclic systems with binary outcomes, analogous to Klyachko–Can–Binicioglu–Shumovsky, EPR–Bell and Leggett–Garg systems in physics. Using Contextuality-by-Default theory, which defines and measures contextuality even when measurements are context-dependent, they derive necessary and sufficient non-contextuality conditions for cyclic binary systems. Reviewing datasets on opinion polls, visual illusions, conjoint choices, word combinations and psychophysical matching, they found no evidence of contextuality. They hypothesise that behavioural and social systems are non-contextual, with apparent contextual effects arising because response distributions depend on context elements beyond those targeted. The limitation is that the conclusion rests on the datasets reviewed; later experiments (Cervantes and Dzhafarov, 2018) reported true contextuality. For the survey, this is a key critique warning against over-interpreting context effects in human data as quantum.

Moreira and Wichert (2017) (E3; full text). Moreira and Wichert (2017) ask whether quantum projection models of question order effects genuinely require quantum principles. Using the Clinton–Gore Gallup poll reported by Moore (2002), they compare a quantum projection model, in which order effects arise from rotating the basis between non-commuting questions, with a classical projection model built in a real-valued vector space. They show that the quantum model reproduces the order effects by varying a rotation parameter, but that quantum interference plays no role in the computed probabilities, so the classical projection model achieves the same results; on grounds of parsimony, the classical model should therefore be preferred unless interference is needed. They also propose a ‘relativistic’ interpretation of the rotation parameters as reflecting uncertainty about a personal basis state, the world and the question context, rather than similarity between questions. The analysis is confined to a two-dimensional model and a single published data set. For autonomous agents, it is a reminder that geometric, order-sensitive belief representations need not involve complex amplitudes, which simplifies implementation.

Busemeyer and Wang (2017) (E3; abstract). Busemeyer and Wang (2017) responded to a published claim that quantum models of psychological measurement face a problem: if questions are incompatible, an intermediate measurement B should disturb an initial measurement A, and critics argued that people would instead remain consistent when A is repeated. The authors tested the quantum prediction with an ‘ABA’ design, in which an opinion question A is asked, followed by B, then A again. The study recruited 325 participants from Amazon Mechanical Turk and used 12 pairs of opinion questions, with responses on a 0–100 scale. Participants showed moderate opinion changes between repeated A measurements, and the pattern of correlations across the question pairs was consistent with quantum model predictions, contrary to the earlier sceptical claim. A limitation is reliance on online opinion ratings rather than consequential decisions. For human–AI teaming, it shows that intermediate queries can measurably shift a person’s later responses.

Kellen et al. (2018) (E2; abstract). Kellen et al. (2018) examined whether the question order effects that have been presented as signatures of quantum probability, in particular the mirrored pattern captured by the QQ equality (Wang et al., 2014), genuinely require a quantum account. The authors show that classical-probability models, in which order affects the process that generates responses, can also produce mirrored order effects and satisfy the QQ constraint. They thereby argue that such patterns do not uniquely discriminate quantum from classical explanations. The work is a modelling critique drawing on existing order-effect data rather than a new large experiment. Its limitation is that it establishes the possibility of classical accounts rather than showing they are preferable on all criteria. For the survey, it is an important caution that quantum-like signatures in human data must be compared against suitably flexible classical baselines before being used to justify quantum-like agent models.

Costello and Watts (2018) (E3; abstract). Costello and Watts (2018) discussed three invariants or identities reported in people’s probabilistic reasoning: the QQ identity, the addition law identity and the Bayes rule identity. Each represents agreement with normative probability theory, and strikingly these identities appear to hold in the same judgements that show strong, systematic biases against probability theory. The QQ identity had been proposed as evidence for quantum probability models of order effects (Wang et al., 2014). The authors examine these invariants within their probability-theory-plus-noise account (Costello and Watts, 2014), arguing that noisy application of classical probability explains why the identities hold despite the biases. The limitation is that the argument concerns aggregate identities and does not settle every phenomenon addressed by quantum models. For the survey, it shows that a key quantum signature can also be predicted by a classical noisy-reasoning model.

Martinez-Martinez and Sanchez-Burillo (2016) (E2; abstract). Martínez-Martínez and Sánchez-Burillo (2016) model a decision-maker as an open quantum system whose choice probabilities arise as the stationary distribution of a quantum stochastic walk (QSW) on a ‘cognitive network’. The network extends a classical decision tree: nodes are action–state combinations, connected components correspond to states of the world, and edge weights combine a profitability matrix built from Luce’s choice axiom (with a discrimination parameter) and a belief-formation matrix. Dynamics follow a Lindblad–Kossakowski master equation, and a single parameter interpolates between purely unitary (quantum walk) and purely dissipative (classical random walk) evolution; uniqueness of the stationary state is proved via Spohn’s theorem. A Prisoner’s Dilemma case study shows that the model can reproduce violations of the sure-thing principle and of the law of total probability. The work is formal and illustrative, not fitted to new behavioural data, and it adds free parameters. For autonomous agents it offers a concrete open-system (Lindblad) decision mechanism that converges to a well-defined choice distribution, a template for agents that must settle on actions over time.

Waddup et al. (2021) (E3; abstract). Waddup et al. (2021) define sensitivity to context for two agents each answering two binary questions: responses to one agent’s question depend on which question the other agent faces, so that the joint probabilities cannot be represented by a single four-way distribution. They report two experiments with a Prisoner’s-Dilemma-like task constructed without a Nash equilibrium; Experiment 1 recruited 100 participants. Behaviour showed sensitivity to context, adding to evidence that such games can be built so that choices are inconsistent with baseline classical probability and with decisions revealing pre-existing values. Two closely matched models, one classical and one quantum, were fitted, and the quantum model fitted better; in it, context sensitivity coincides with (epiphenomenal) entanglement. The effect is shown in one task family, and the classical comparison model is a baseline rather than an exhaustive alternative. For multi-agent AI and human–robot interaction, the result indicates that a partner’s choice may depend on the joint question context, which matters when agents predict human responses in interactive settings.

Busemeyer et al. (2017) (E2; abstract). Busemeyer et al. (2017) address how the brain could implement the operations used in quantum cognition models, such as state vectors, projections, incompatible bases and unitary evolution, without appealing to quantum physics in neurons. They discuss how these mathematical operations could be realised by classical neural mechanisms, drawing on work on neural oscillations and synchronisation, so that quantum-like probability arises from classical neural dynamics. The paper is theoretical and does not test the proposal against neural data. It is quantum-like in the sense of the survey: QP structure without quantum hardware in the brain. For AI it supports the view that QP models can be implemented in classical neural networks, a precondition for building quantum-like cognitive modules into artificial agents.

Khrennikov et al. (2018) (E2; abstract). Khrennikov et al. (2018) ask how neurons could generate the quantum-like statistics seen in cognitive psychology. They represent uncertainty in generating action potentials, arising from complex electrochemical processes, with quantum information states that extend classical neural codes such as quiescent/firing, so that information processing involves superpositions of these states. A neuronal

group performing a psychological function is treated as an open quantum system interacting with its electrochemical environment, described by the Gorini–Kossakowski–Sudarshan–Lindblad equation; decision-making is decoherence in the eigenbasis of that function, and the decision is the resulting steady state. This linear representation of nonlinear electrochemical dynamics guarantees exponentially fast convergence to the decision state. The model is formal and not fitted to neural data. It is quantum-like: it does not claim quantum physics in the brain. For AI, it gives an open-system recipe for decision units that settle quickly to a choice.

Khrennikov et al. (2025) (E2; abstract). Khrennikov et al. (2025b) aim to couple quantum-like models of cognition with the functioning of neuronal networks within the framework of generalised probability theory, which contains classical and quantum probability as special cases. The paper continues the programme of (Khrennikov et al., 2018) in seeking a neural grounding for quantum-like statistics without assuming quantum physics in the brain, and it connects network-level neuronal descriptions to the state-and-measurement structure used in quantum cognition. It is a theoretical contribution in the *Journal of Mathematical Psychology* and does not report new behavioural or neural experiments. For AI, it is relevant as a candidate mathematical bridge between neural-network computation and quantum-like decision models on classical hardware.

Khrennikov et al. (2025) (E2; abstract). Khrennikov et al. (2025a) seek to connect the functioning of oscillatory neuronal networks with quantum-like models of cognition, and to draw implications for quantum-inspired computation and artificial intelligence. The central idea, consistent with (Busemeyer et al., 2017) and (Khrennikov et al., 2018), is that classical oscillatory neural dynamics can carry the complex-amplitude, phase-dependent structure that quantum-like probability models require, so that interference-type effects in cognition need no quantum physics in the brain. The paper is theoretical and does not test the proposal against neural or behavioural data. It is quantum-like and quantum-inspired rather than quantum hardware. For AI, it suggests that oscillator-based or phase-based classical networks could serve as substrates for quantum-like decision modules in artificial agents.

Khrennikov and Yamada (2025) (E2; full text). Khrennikov and Yamada (2025) tackle the hardest step in grounding quantum-like (QL) cognition in classical neural dynamics: how classical oscillatory networks could generate ‘mental entanglement’. Building on prequantum classical statistical field theory, in which QL states arise as covariance operators of random neuronal fields (Khrennikov et al., 2025b), they use two notions of entanglement: observation entanglement, where a tensor-product structure is induced by cross-commuting local algebras of observables, and state entanglement generated by spatially separated but interacting networks, attributed to classical electromagnetic (including ephaptic) coupling. They stress that this ‘nonlocality’ is classical signalling within the brain, not quantum nonlocality, and is distinct from quantum-brain hypotheses such as Orch OR. The paper is explicitly conceptual: it contains no data, but proposes EEG/MEG tests, maps QL constructs onto functional-connectivity measures, discusses in vitro networks and Bell-type tests, and speculates on feature binding and consciousness. For artificial agents, it suggests how QL representations might be realised by classical, networked substrates, which is the relevant model for quantum-like reasoning on conventional hardware.

References

- Alberto Acuto, Paola Barillà, Ludovico Bozzolo, Matteo Conterno, Mattia Pavese, and Antonio Policicchio. Variational quantum soft actor-critic for robotic arm control, 2022.
- Diederik Aerts. Quantum structure in cognition. *Journal of Mathematical Psychology*, 53(5):314–348, 2009. doi: 10.1016/j.jmp.2009.04.005.
- Diederik Aerts and Massimiliano Sassoli de Bianchi. Beyond-quantum modeling of question order effects and response replicability in psychological measurements. *Journal of Mathematical Psychology*, 79:104–120, 2017.
- Diederik Aerts and Sandro Sozzo. Quantum structure in cognition: Why and how concepts are entangled. In Dawei Song, Massimo Melucci, Ingo Frommholz, Peng Zhang, Lei Wang, and Sachi Arafat, editors, *Quantum Interaction: 5th International Symposium, QI 2011, Aberdeen, UK, June 26–29, 2011, Revised Selected Papers*, volume 7052 of *Lecture Notes in Computer Science*, pages 116–127, Berlin, Heidelberg, 2011. Springer. doi: 10.1007/978-3-642-24971-6_12.
- Diederik Aerts, Ellie D’Hondt, and Liane Gabora. Why the disjunction in quantum logic is not classical. *Foundations of Physics*, 30(9):1473–1480, 2000. doi: 10.1023/A:1026457918361.

- Diederik Aerts, Jan Broekaert, Marek Czachor, and Bart D’Hooghe. A quantum-conceptual explanation of violations of expected utility in economics. In Dawei Song, Massimo Melucci, Ingo Frommholz, Peng Zhang, Lei Wang, and Sachi Arafat, editors, *Quantum Interaction: 5th International Symposium, QI 2011, Aberdeen, UK, June 26–29, 2011, Revised Selected Papers*, volume 7052 of *Lecture Notes in Computer Science*, pages 192–198, Berlin, Heidelberg, 2011a. Springer. doi: 10.1007/978-3-642-24971-6_19.
- Diederik Aerts, Marek Czachor, and Sandro Sozzo. Quantum interaction approach in cognition, artificial intelligence and robotics. In Vladimir Privman and Victor Ovchinnikov, editors, *Proceedings of the Fifth International Conference on Quantum, Nano and Micro Technologies (ICQNM 2011)*, pages 35–40. IARIA, 2011b.
- Diederik Aerts, Liane Gabora, Sandro Sozzo, and Tomas Veloz. Quantum structure in cognition: Fundamentals and applications. In *Proceedings of the Fifth International Conference on Quantum, Nano and Micro Technologies (ICQNM 2011)*. IARIA, 2011c.
- Diederik Aerts, Liane Gabora, and Sandro Sozzo. Concepts and their dynamics: A quantum-theoretic modeling of human thought. *Topics in Cognitive Science*, 5(4):737–772, 2013. doi: 10.1111/tops.12042.
- Diederik Aerts, Jonito Aerts Arguëlles, Lester Beltran, Lyneth Beltran, Massimiliano Sassoli de Bianchi, Sandro Sozzo, and Tomas Veloz. Testing quantum models of conjunction fallacy on the World Wide Web. *International Journal of Theoretical Physics*, 56(12):3744–3756, 2017. doi: 10.1007/s10773-017-3288-8.
- Diederik Aerts, Jonito Aerts Arguëlles, Lester Beltran, Suzette Geriente, Roberto Leporini, Massimiliano Sassoli de Bianchi, and Sandro Sozzo. Identifying quantum structure in AI language: Evidence for evolutionary convergence of human and artificial cognition. *Entropy*, 28(6):622, 2026. doi: 10.3390/e28060622.
- Masanari Asano and Andrei Khrennikov. Quantum-like models of cognition and decision making: open-systems and Gorini–Kossakowski–Sudarshan–Lindblad dynamics, 2026.
- Mehrdad Ashtiani and Mohammad Abdollahi Azgomi. A novel trust evolution algorithm based on a quantum-like model of computational trust. *Cognition, Technology & Work*, 21(2):201–224, 2019. doi: 10.1007/s10111-018-0496-9.
- Irina Basieva, Víctor H. Cervantes, Ehtibar N. Dzhafarov, and Andrei Khrennikov. True contextuality beats direct influences in human decision making. *Journal of Experimental Psychology: General*, 148(11):1925–1937, 2019. doi: 10.1037/xge0000585.
- Paul Benioff. Quantum robots and environments. *Physical Review A*, 58(2):893–904, 1998. doi: 10.1103/PhysRevA.58.893.
- Reinhard Blutner, Emmanuel M. Pothos, and Peter Bruza. A quantum probability perspective on borderline vagueness. *Topics in Cognitive Science*, 5(4):711–736, 2013. doi: 10.1111/tops.12041.
- Thomas Boyer-Kassem, Sébastien Duchêne, and Eric Guerci. Quantum-like models cannot account for the conjunction fallacy. *Theory and Decision*, 81(4):479–510, 2016. doi: 10.1007/s11238-016-9549-9.
- Hans J. Briegel and Gemma De las Cuevas. Projective simulation for artificial intelligence. *Scientific Reports*, 2:400, 2012. doi: 10.1038/srep00400.
- Jan Broekaert, Irina Basieva, Pawel Blasiak, and Emmanuel M. Pothos. Quantum-like dynamics applied to cognition: A consideration of available options. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 375(2106):20160387, 2017. doi: 10.1098/rsta.2016.0387.
- Peter Bruza, Kirsty Kitto, Douglas Nelson, and Cathy McEvoy. Is there something quantum-like about the human mental lexicon? *Journal of Mathematical Psychology*, 53(5):363–377, 2009. doi: 10.1016/j.jmp.2009.04.004.
- Peter D. Bruza, Zheng Wang, and Jerome R. Busemeyer. Quantum cognition: A new theoretical approach to psychology. *Trends in Cognitive Sciences*, 19(7):383–393, 2015. doi: 10.1016/j.tics.2015.05.001.
- Jerome R. Busemeyer and Peter D. Bruza. *Quantum Models of Cognition and Decision*. Cambridge University Press, Cambridge, 2012. doi: 10.1017/CBO9780511997716. Second edition: Cambridge University Press, 2024.
- Jerome R. Busemeyer and Zheng Wang. What is quantum cognition, and how is it applied to psychology? *Current Directions in Psychological Science*, 24(3):163–169, 2015. doi: 10.1177/0963721414568663.

- Jerome R. Busemeyer and Zheng Wang. Is there a problem with quantum models of psychological measurements? *PLoS ONE*, 12(11):e0187733, 2017. doi: 10.1371/journal.pone.0187733.
- Jerome R. Busemeyer and Zheng Wang. Hilbert space multidimensional theory. *Psychological Review*, 125(4): 572–591, 2018. doi: 10.1037/rev0000106.
- Jerome R. Busemeyer, Zheng Wang, and James T. Townsend. Quantum dynamics of human decision-making. *Journal of Mathematical Psychology*, 50(3):220–241, 2006. doi: 10.1016/j.jmp.2006.01.003.
- Jerome R. Busemeyer, Zheng Wang, and Ariane Lambert-Mogiliansky. Empirical comparison of Markov and quantum models of decision making. *Journal of Mathematical Psychology*, 53(5):423–433, 2009. doi: 10.1016/j.jmp.2009.03.002.
- Jerome R. Busemeyer, Emmanuel M. Pothos, Riccardo Franco, and Jennifer S. Trueblood. A quantum theoretical explanation for probability judgment errors. *Psychological Review*, 118(2):193–218, 2011. doi: 10.1037/a0022542.
- Jerome R. Busemeyer, Pegah Fakhari, and Peter Kvam. Neural implementation of operations used in quantum cognition. *Progress in Biophysics and Molecular Biology*, 130(Part A):53–60, 2017. doi: 10.1016/j.pbiomolbio.2017.04.007.
- Jerome R. Busemeyer, Peter D. Kvam, and Timothy J. Pleskac. Markov versus quantum dynamic models of belief change during evidence monitoring. *Scientific Reports*, 9:18025, 2019. doi: 10.1038/s41598-019-54383-9.
- Jerome R. Busemeyer, Qizi Zhang, S. N. Balakrishnan, and Zheng Wang. Application of quantum–Markov open system models to human cognition and decision. *Entropy*, 22(9):990, 2020. doi: 10.3390/e22090990.
- Jerome R. Busemeyer, Masanao Ozawa, Emmanuel M. Pothos, and Naotsugu Tsuchiya. Incorporating episodic memory into quantum models of judgment and decision. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 383(2309):20240387, 2025. doi: 10.1098/rsta.2024.0387.
- Víctor H. Cervantes and Ehtibar N. Dzhafarov. Snow Queen is evil and beautiful: Experimental evidence for probabilistic contextuality in human choices. *Decision*, 5(3):193–204, 2018. doi: 10.1037/dec0000095.
- Antonio Chella, Salvatore Gaglio, Giovanni Pilato, Filippo Vella, and Salvatore Zammuto. A quantum planner for robot motion. *Mathematics*, 10(14):2475, 2022. doi: 10.3390/math10142475.
- Antonio Chella, Salvatore Gaglio, Giovanni Pilato, and Filippo Vella. An architecture for a quantum teleo-reactive robot. *Entropy*, 28(7):731, 2026. doi: 10.3390/e28070731.
- Samuel Yen-Chi Chen, Chao-Han Huck Yang, Jun Qi, Pin-Yu Chen, Xiaoli Ma, and Hsi-Sheng Goan. Variational quantum circuits for deep reinforcement learning. *IEEE Access*, 8:141007–141024, 2020. doi: 10.1109/ACCESS.2020.3010470.
- Fintan Costello and Paul Watts. Surprisingly rational: probability theory plus noise explains biases in judgment. *Psychological Review*, 121(3):463–480, 2014. doi: 10.1037/a0037010.
- Fintan Costello and Paul Watts. Invariants in probabilistic reasoning. *Cognitive Psychology*, 100:1–16, 2018. doi: 10.1016/j.cogpsych.2017.11.003.
- Evren Daglarli. A generative neuro-cognitive architecture using quantum algorithms for the autonomous behavior of a smart agent in a simulation environment. *Computers, Materials & Continua*, 84(3), 2025. doi: 10.32604/cmc.2025.065572.
- Berardina Nadja De Carolis, C. Loglisci, G. Palestra, and S. Violante. Quantum-enhanced social robotics: The QUADRI project. In *Adjunct Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP Adjunct '25)*, New York, NY, USA, 2025. ACM. doi: 10.1145/3708319.3735545.
- Daoyi Dong, Chunlin Chen, Chenbin Zhang, and Zonghai Chen. Quantum robot: Structure, algorithms and applications. *Robotica*, 24(4):513–521, 2006. doi: 10.1017/S0263574705002596.
- Daoyi Dong, Chunlin Chen, Hanxiong Li, and Tzyh-Jong Tarn. Quantum reinforcement learning. *IEEE Transactions on Systems, Man, and Cybernetics, Part B: Cybernetics*, 38(5):1207–1220, 2008. doi: 10.1109/TSMCB.2008.925743.
- Daoyi Dong, Chunlin Chen, Jian Chu, and Tzyh-Jong Tarn. Robust quantum-inspired reinforcement learning for robot navigation. *IEEE/ASME Transactions on Mechatronics*, 17(1):86–97, 2012. doi: 10.1109/TMECH.2010.2090896.

- Ehtibar N. Dzhafarov and Janne V. Kujala. Context–content systems of random variables: The Contextuality-by-Default theory. *Journal of Mathematical Psychology*, 74:11–33, 2016. doi: 10.1016/j.jmp.2016.04.010.
- Ehtibar N. Dzhafarov, Ru Zhang, and Janne Kujala. Is there contextuality in behavioural and social systems? *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2058):20150099, 2016. doi: 10.1098/rsta.2015.0099.
- Jens Eisert, Martin Wilkens, and Maciej Lewenstein. Quantum games and quantum strategies. *Physical Review Letters*, 83(15):3077–3080, 1999. doi: 10.1103/PhysRevLett.83.3077.
- Gunnar P. Epping, Peter D. Kvam, Timothy J. Pleskac, and Jerome R. Busemeyer. Open system model of choice and response time. *Journal of Choice Modelling*, 49:100453, 2023. doi: 10.1016/j.jocm.2023.100453.
- Karim Essalmi, Fernando Garrido, and Fawzi Nashashibi. Quantum game models for interaction-aware decision-making in automated driving, 2025. arXiv preprint.
- Karim Essalmi, Fernando Garrido, and Fawzi Nashashibi. Multi-player, multi-strategy quantum game model for interaction-aware decision-making in automated driving, 2026. arXiv preprint.
- Miho Fuyama, Andrei Khrennikov, and Masanao Ozawa. Quantum-like cognition and decision-making in the light of quantum measurement theory. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 383(2309):20240372, 2025. doi: 10.1098/rsta.2024.0372.
- X. Gao, H. Yang, L. Pan, and D. Pelusi. Quantum-like evidence networks decision-making model. *Engineering Applications of Artificial Intelligence*, 157:111368, 2025. doi: 10.1016/j.engappai.2025.111368.
- Dimitrios Gkoumas, Qiuchi Li, Shahram Dehdashti, Massimo Melucci, Yijun Yu, and Dawei Song. Quantum cognitively motivated decision fusion for video sentiment analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 827–835, 2021. doi: 10.1609/aaai.v35i1.16165.
- Yeshwanth Guru and Dev Kunwar Singh Chauhan. Quantum technologies for autonomous robotics: A systematic survey of methods, evidence, and deployment constraints, 2026. Companion manuscript, prepared for ACM Computing Surveys.
- Simon Hangl, Emre Ugur, Sandor Szedmak, and Justus Piater. Robotic playing for hierarchical complex skill learning. In *Proceedings of the 2016 IEEE/RSSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2799–2804. IEEE, 2016.
- Simon Hangl, Vedran Dunjko, Hans J. Briegel, and Justus Piater. Skill learning by autonomous robotic playing using active learning and exploratory behavior composition. *Frontiers in Robotics and AI*, 7:42, 2020. doi: 10.3389/frobt.2020.00042.
- Johnny K. W. Ho and Johan F. Hoorn. Quantum affective processes for multidimensional decision-making. *Scientific Reports*, 12:20468, 2022. doi: 10.1038/s41598-022-22855-0.
- Hans Hohenfeld, Dirk Heimann, Felix Wiebe, and Frank Kirchner. Quantum deep reinforcement learning for robot navigation tasks. *IEEE Access*, 12:87217–87236, 2024. doi: 10.1109/ACCESS.2024.3417808.
- Johan F. Hoorn and Johnny K. W. Ho. Robot affect: the amygdala as Bloch sphere, 2019. arXiv preprint.
- Jiaqi Huang, Jerome R. Busemeyer, Zoe Ebelt, and Emmanuel M. Pothos. Bridging the gap between subjective probability and probability judgments: the Quantum Sequential Sampler. *Psychological Review*, 132(4): 916–955, 2025a. doi: 10.1037/rev0000489.
- Jiaqi Huang, Gunnar P. Epping, Jennifer S. Trueblood, James M. Yearsley, Jerome R. Busemeyer, and Emmanuel M. Pothos. An overview of the quantum cognition research program. *Psychonomic Bulletin & Review*, 32:2507–2556, 2025b. doi: 10.3758/s13423-025-02675-9.
- Jiaqi Huang, Timothy J. Pleskac, and Jerome R. Busemeyer. Quantum recognition heuristics: A tutorial example of quantum heuristics. In *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*. IEEE, 2025c. doi: 10.1109/QCE65121.2025.10486.
- Scott Humr and Mustafa Canan. System level intervention for AI-supported decision-making: A quantum cognition perspective. In *Quantum AI and NLP (QNLPAI 2025)*, volume 2659 of *Communications in Computer and Information Science*, pages 250–273. Springer, Cham, 2026. doi: 10.1007/978-3-032-13883-5_14.
- Scott Humr, Mustafa Canan, and Mustafa Demir. A quantum probability approach to improving human–AI decision making. *Entropy*, 27(2):152, 2025. doi: 10.3390/e27020152.

- Scott A. Humr, Mustafa Canan, and Mustafa Demir. Temporal evolution of trust in artificial intelligence-supported decision-making. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 67(1):1936–1941, 2023. doi: 10.1177/21695067231193672.
- Pegah Imannezhad, Emmanuel M. Pothos, and Andy J. Wills. Divergent patterns of probabilistic reasoning in humans and GPT-5. *Frontiers in Psychology*, 17:1782184, 2026. doi: 10.3389/fpsyg.2026.1782184.
- Sofiene Jerbi, Casper Gyurik, Simon C. Marshall, Hans J. Briegel, and Vedran Dunjko. Parametrized quantum policies for reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- Subhash Kak. Order effects for queries in intelligent systems, 2018. arXiv preprint.
- Pilsung Kang. Auditing question-order effects in large language models with the QQ equality: Mechanism characterization and a saturation caveat, 2026.
- David Kellen, Henrik Singmann, and William H. Batchelder. Classic-probability accounts of mirrored (quantum-like) order effects in human judgments. *Decision*, 5(4):323–338, 2018. doi: 10.1037/dec0000080.
- Farbod Khoshnoud, Marco B. Quadrelli, Ibrahim I. Esat, and Dario Robinson. Quantum cooperative robotics and autonomy, 2020.
- Andrei Khrennikov. Classical and quantum mechanics on information spaces with applications to cognitive, psychological, social, and anomalous phenomena. *Foundations of Physics*, 29(7):1065–1098, 1999. doi: 10.1023/A:1018885632116.
- Andrei Khrennikov. *Ubiquitous Quantum Structure: From Psychology to Finance*. Springer, Berlin, Heidelberg, 2010. doi: 10.1007/978-3-642-05101-2.
- Andrei Khrennikov. Open systems, quantum probability, and logic for quantum-like modeling in biology, cognition, and decision-making. *Entropy*, 25(6):886, 2023. doi: 10.3390/e25060886.
- Andrei Khrennikov. Contextual measurement model and quantum theory. *Royal Society Open Science*, 11(3):231953, 2024. doi: 10.1098/rsos.231953.
- Andrei Khrennikov and Makiko Yamada. Quantum-like representation of neuronal networks’ activity: modeling “mental entanglement”. *Frontiers in Human Neuroscience*, 19:1685339, 2025. doi: 10.3389/fnhum.2025.1685339.
- Andrei Khrennikov, Irina Basieva, Ehtibar N. Dzhafarov, and Jerome R. Busemeyer. Quantum models for psychological measurements: An unsolved problem. *PLoS ONE*, 9(10):e110909, 2014. doi: 10.1371/journal.pone.0110909.
- Andrei Khrennikov, Irina Basieva, Emmanuel M. Pothos, and Ichiro Yamato. Quantum probability in decision making from quantum information representation of neuronal states. *Scientific Reports*, 8:16225, 2018. doi: 10.1038/s41598-018-34531-3.
- Andrei Khrennikov, Atsushi Iriki, and Irina Basieva. Constructing a bridge between functioning of oscillatory neuronal networks and quantum-like cognition along with quantum-inspired computation and AI. *BioSystems*, 257:105573, 2025a. doi: 10.1016/j.biosystems.2025.105573.
- Andrei Khrennikov, Masanao Ozawa, Felix Benninger, and Oded Shor. Coupling quantum-like cognition with the neuronal networks within generalized probability theory. *Journal of Mathematical Psychology*, 125:102923, 2025b. doi: 10.1016/j.jmp.2025.102923.
- Decheng Kong, Kai Xue, Ping Wang, and Zeyu Xu. Emergence of longitudinal queues in group navigation: An interpretable approach via projective simulation. *Biomimetics*, 11(3):201, 2026. doi: 10.3390/biomimetics11030201.
- Peter D. Kvam, Timothy J. Pleskac, Shuli Yu, and Jerome R. Busemeyer. Interference effects of choice on confidence: Quantum characteristics of evidence accumulation. *Proceedings of the National Academy of Sciences*, 112(34):10645–10650, 2015. doi: 10.1073/pnas.1500688112.
- Peter D. Kvam, Jerome R. Busemeyer, and Timothy J. Pleskac. Temporal oscillations in preference strength provide evidence for an open system model of constructed preference. *Scientific Reports*, 11:8169, 2021. doi: 10.1038/s41598-021-87659-0.
- Davide Lanza, Paolo Solinas, and Fulvio Mastrogiovanni. A preliminary study for a quantum-like robot perception model, 2020.

- Davide Lanza, Paolo Solinas, and Fulvio Mastrogiovanni. Multi-sensory integration in a quantum-like robot perception model. In Bruno Siciliano, Cecilia Laschi, and Oussama Khatib, editors, *Experimental Robotics: The 17th International Symposium (ISER 2020)*, volume 19 of *Springer Proceedings in Advanced Robotics*, pages 502–509. Springer, Cham, 2021. doi: 10.1007/978-3-030-71151-1_44.
- William F. Lawless. Quantum-like interdependence theory advances autonomous human–machine teams (A-HMTs). *Entropy*, 22(11):1227, 2020. doi: 10.3390/e22111227.
- William F. Lawless, Ira S. Moskowitz, and Katarina Z. Doctor. A quantum-like model of interdependence for embodied human–machine teams: Reviewing the path to autonomy facing complexity and uncertainty. *Entropy*, 25(9):1323, 2023. doi: 10.3390/e25091323.
- Qiuchi Li, Benyou Wang, and Massimo Melucci. CNM: An interpretable complex-valued network for matching. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4139–4148, Minneapolis, Minnesota, 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1420.
- Qiuchi Li, Dimitrios Gkoumas, Christina Lioma, and Massimo Melucci. Quantum-inspired multimodal fusion for video sentiment analysis. *Information Fusion*, 65:58–71, 2021a. doi: 10.1016/j.inffus.2020.08.006.
- Qiuchi Li, Dimitrios Gkoumas, Alessandro Sordani, Jian-Yun Nie, and Massimo Melucci. Quantum-inspired neural network for conversational emotion recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 13270–13278, 2021b. doi: 10.1609/aaai.v35i15.17567.
- Kin Ian Lo, Mehrnoosh Sadrzadeh, and Shane Mansfield. Quantum-like contextuality in large language models. *Proceedings of the Royal Society A*, 2025.
- Andrea López-Incera, Katja Ried, Thomas Müller, and Hans J. Briegel. Development of swarm behavior in artificial learning agents that adapt to different foraging environments. *PLoS ONE*, 15(12):e0243628, 2020. doi: 10.1371/journal.pone.0243628.
- Robin Lorenz, Anna Pearson, Konstantinos Meichanetzidis, Dimitri Kartsaklis, and Bob Coecke. QNLP in practice: Running compositional models of meaning on a quantum computer. *Journal of Artificial Intelligence Research*, 76:1305–1342, 2023. doi: 10.1613/jair.1.14329.
- Martin Lukac and Marek Perkowski. Quantum mechanical model of emotional robot behaviors. In *Proceedings of the 37th International Symposium on Multiple-Valued Logic (ISMVL 2007)*, Oulu, Finland, 2007. IEEE.
- Milan Maksimovic and Ivan S. Maksymov. Quantum-cognitive neural networks: Assessing confidence and uncertainty with human decision-making simulations. *Big Data and Cognitive Computing*, 9(1):12, 2025. doi: 10.3390/bdcc9010012.
- Maria Mannone, Valeria Seidita, and Antonio Chella. Categories, quantum computing, and swarm robotics: A case study. *Mathematics*, 10(3):372, 2022. doi: 10.3390/math10030372.
- Maria Mannone, Antonio Chella, Giovanni Pilato, Valeria Seidita, Filippo Vella, and Salvatore Gaglio. Modeling robotic thinking and creativity: A classic–quantum dialogue. *Mathematics*, 12(5):642, 2024. doi: 10.3390/math12050642.
- Ismael Martínez-Martínez and Eduardo Sánchez-Burillo. Quantum stochastic walks on networks for decision-making. *Scientific Reports*, 6:23812, 2016. doi: 10.1038/srep23812.
- A. Meghdadi, M.-R. Akbarzadeh-T., and K. Javidan. A quantum-like cognitive approach to modeling human biased selection behavior. *Scientific Reports*, 12:22545, 2022. doi: 10.1038/s41598-022-13757-2.
- Alexey A. Melnikov, Adi Makmal, and Hans J. Briegel. Benchmarking projective simulation in navigation problems. *IEEE Access*, 6:64639–64648, 2018. doi: 10.1109/ACCESS.2018.2876494.
- Nico Meyer, Christian Ufrecht, Maniraman Periyasamy, Daniel D. Scherer, Axel Plinge, and Christopher Mutschler. A survey on quantum reinforcement learning, 2022.
- Catarina Moreira. *Quantum Probabilistic Graphical Models for Cognition and Decision*. PhD thesis, Instituto Superior Técnico, Universidade de Lisboa, Lisbon, Portugal, 2017.
- Catarina Moreira and Andreas Wichert. Interference effects in quantum belief networks. *Applied Soft Computing*, 25:64–85, 2014. doi: 10.1016/j.asoc.2014.09.008.

- Catarina Moreira and Andreas Wichert. Quantum-like Bayesian networks for modeling decision making. *Frontiers in Psychology*, 7:11, 2016. doi: 10.3389/fpsyg.2016.00011.
- Catarina Moreira and Andreas Wichert. Are quantum models for order effects quantum? *International Journal of Theoretical Physics*, 56:4029–4046, 2017. doi: 10.1007/s10773-017-3424-5.
- Catarina Moreira, Lauren Fell, Shahram Dehdashti, Peter Bruza, and Andreas Wichert. Towards a quantum-like cognitive architecture for decision-making. *Behavioral and Brain Sciences*, 43:e17, 2020a. doi: 10.1017/S0140525X19001687.
- Catarina Moreira, Prayag Tiwari, Hari Mohan Pandey, Peter Bruza, and Andreas Wichert. Quantum-like influence diagrams for decision-making. *Neural Networks*, 132:190–210, 2020b. doi: 10.1016/j.neunet.2020.07.009.
- Ahmed Nebli, Hadi Saadatdoorabi, and Kevin Yam. Deep sequence modeling with quantum dynamics: Language as a wave function, 2026. arXiv preprint.
- Masanao Ozawa and Andrei Khrennikov. Application of theory of quantum instruments to psychology: Combination of question order effect with response replicability effect. *Entropy*, 22(1):37, 2020. doi: 10.3390/e22010037.
- Masanao Ozawa and Andrei Khrennikov. Modeling combination of question order effect, response replicability effect, and QQ-equality with quantum instruments. *Journal of Mathematical Psychology*, 100:102491, 2021. doi: 10.1016/j.jmp.2020.102491.
- Giuseppe Davide Paparo, Vedran Dunjko, Adi Makmal, Miguel Angel Martin-Delgado, and Hans J. Briegel. Quantum speedup for active learning agents. *Physical Review X*, 4(3):031002, 2014. doi: 10.1103/PhysRevX.4.031002.
- Emmanuel M. Pothos and Jerome R. Busemeyer. A quantum probability explanation for violations of ‘rational’ decision theory. *Proceedings of the Royal Society B: Biological Sciences*, 276(1665):2171–2178, 2009. doi: 10.1098/rspb.2009.0121.
- Emmanuel M. Pothos and Jerome R. Busemeyer. Can quantum probability provide a new direction for cognitive modeling? *Behavioral and Brain Sciences*, 36(3):255–274, 2013. doi: 10.1017/S0140525X12001525.
- Emmanuel M. Pothos and Jerome R. Busemeyer. Quantum cognition. *Annual Review of Psychology*, 73(1):749–778, 2022. doi: 10.1146/annurev-psych-033020-123501.
- Emmanuel M. Pothos, Jerome R. Busemeyer, and Jennifer S. Trueblood. A quantum geometric model of similarity. *Psychological Review*, 120(3):679–696, 2013. doi: 10.1037/a0033142.
- Emmanuel M. Pothos, James M. Yearsley, Richard M. Shiffrin, and Jerome R. Busemeyer. The rational status of quantum cognition. *Journal of Experimental Psychology: General*, 146(7):968–987, 2017. doi: 10.1037/xge0000312.
- Kiranraj Pushparaj, Gregoire Ky, Alvin J. Ayeni, Sameer Alam, and Vu N. Duong. A quantum-inspired model for human-automation trust in air traffic controllers derived from functional magnetic resonance imaging and correlated with behavioural indicators. *Journal of Air Transport Management*, 97:102143, 2021. doi: 10.1016/j.jairtraman.2021.102143.
- Katja Ried, Thomas Müller, and Hans J. Briegel. Modelling collective motion based on the principle of agency: General framework and the case of marching locusts. *PLoS ONE*, 14(2):e0212044, 2019. doi: 10.1371/journal.pone.0212044.
- Luisa Roeder, Pamela Hoyte, Johan van der Meer, L. Fell, P. Johnston, G. Kerr, and Peter Bruza. A quantum model of trust calibration in human–AI interactions. *Entropy*, 25(9):1362, 2023. doi: 10.3390/e25091362.
- Eldar Shafir and Amos Tversky. Thinking through uncertainty: Nonconsequential reasoning and choice. *Cognitive Psychology*, 24(4):449–474, 1992. doi: 10.1016/0010-0285(92)90015-T.
- Lulu She, Shilian Han, and Xinwang Liu. Application of quantum-like Bayesian network and belief entropy for interference effect in multi-attribute decision making problem. *Computers & Industrial Engineering*, 157:107307, 2021. doi: 10.1016/j.cie.2021.107307.
- Akash Sinha, Antonio Macaluso, and Matthias Klusch. Nav-Q: Quantum deep reinforcement learning for collision-free navigation of self-driving cars. *Quantum Machine Intelligence*, 7(1):19, 2025. doi: 10.1007/s42484-024-00226-4.

- Andrea Skolik, Sofiene Jerbi, and Vedran Dunjko. Quantum agents in the Gym: A variational quantum algorithm for deep Q-learning. *Quantum*, 6:720, 2022. doi: 10.22331/q-2022-05-24-720.
- Qingyuan Song, Wen Wang, Weiping Fu, Yuan Sun, Denggui Wang, and Zhiwei Gao. Research on quantum cognition in autonomous driving. *Scientific Reports*, 12:300, 2022. doi: 10.1038/s41598-021-04239-y.
- Alessandro Sordoni, Jian-Yun Nie, and Yoshua Bengio. Modeling term dependencies with quantum language models for IR. In *Proceedings of the 36th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '13)*, pages 653–662, Dublin, Ireland, 2013. ACM. doi: 10.1145/2484028.2484098.
- Edwin Stoudenmire and David J. Schwab. Supervised learning with tensor networks. In D. D. Lee, M. Sugiyama, U. V. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems 29 (NIPS 2016)*, pages 4799–4807. Curran Associates, Inc., 2016. URL <https://proceedings.neurips.cc/paper/2016/hash/5314b9674c86e3f9d1ba25ef9bb32895-Abstract.html>.
- Karl Svozil. Quantum contextuality for contextual word embeddings, 2025.
- Jennifer S. Trueblood and Jerome R. Busemeyer. A quantum probability account of order effects in inference. *Cognitive Science*, 35(8):1518–1552, 2011. doi: 10.1111/j.1551-6709.2011.01197.x.
- Jennifer S. Trueblood and Pernille Hemmer. The generalized quantum episodic memory model. *Cognitive Science*, 41(8):2089–2125, 2017. doi: 10.1111/cogs.12460.
- Jennifer S. Trueblood, James M. Yearsley, and Emmanuel M. Pothos. A quantum probability framework for human probabilistic inference. *Journal of Experimental Psychology: General*, 146(9):1307–1341, 2017. doi: 10.1037/xge0000326.
- Amos Tversky and Daniel Kahneman. Extensional versus intuitive reasoning: The conjunction fallacy in probability judgment. *Psychological Review*, 90(4):293–315, 1983. doi: 10.1037/0033-295X.90.4.293.
- Sagar Uprety, Haiming Liu, Amit Kumar Jaiswal, and Dawei Song. Investigating context effects in similarity judgements in large language models, 2024.
- Johan van der Meer, Pamela Hoyte, Luisa Roeder, and Peter Bruza. Modeling the quantum-like dynamics of human reliability ratings in human–AI interactions by interaction dependent Hamiltonians, 2025. arXiv preprint.
- Tomas Veloz and Olha Sobetska. Analyzing the conjunction fallacy as a fact, 2024.
- O. Waddup, James M. Yearsley, P. Blasiak, and Emmanuel M. Pothos. Temporal Bell inequalities in cognition. *Psychonomic Bulletin & Review*, 30(5):1946–1953, 2023. doi: 10.3758/s13423-023-02275-5.
- Oliver J. Waddup, Pawel Blasiak, James M. Yearsley, Bartosz W. Wojciechowski, and Emmanuel M. Pothos. Sensitivity to context in human interactions. *Mathematics*, 9(21):2784, 2021. doi: 10.3390/math9212784.
- Zheng Wang and Jerome R. Busemeyer. A quantum question order model supported by empirical tests of an a priori and precise prediction. *Topics in Cognitive Science*, 5(4):689–710, 2013. doi: 10.1111/tops.12040.
- Zheng Wang and Jerome R. Busemeyer. Interference effects of categorization on decision making. *Cognition*, 150:133–149, 2016. doi: 10.1016/j.cognition.2016.01.019.
- Zheng Wang, Jerome R. Busemeyer, Harald Atmanspacher, and Emmanuel M. Pothos. The potential of using quantum theory to build models of cognition. *Topics in Cognitive Science*, 5(4):672–688, 2013. doi: 10.1111/tops.12043.
- Zheng Wang, Tyler Solloway, Richard M. Shiffrin, and Jerome R. Busemeyer. Context effects produced by question orders reveal quantum nature of human judgments. *Proceedings of the National Academy of Sciences*, 111(26):9431–9436, 2014. doi: 10.1073/pnas.1407756111.
- Qing Wei, Hailan Ma, Chunlin Chen, and Daoyi Dong. Deep reinforcement learning with quantum-inspired experience replay. *IEEE Transactions on Cybernetics*, 52(9):9326–9338, 2022. doi: 10.1109/TCYB.2021.3053414.
- Andreas Wichert, Catarina Moreira, and Peter Bruza. Balanced quantum-like Bayesian networks. *Entropy*, 22(2):170, 2020. doi: 10.3390/e22020170.

- Dominic Widdows and Stanley Peters. Word vectors and quantum logic: Experiments with negation and disjunction. In *Proceedings of the 8th Mathematics of Language Conference (MOL 8)*, pages 141–154, Bloomington, IN, 2003.
- Dominic Widdows, Kirsty Kitto, and Trevor Cohen. Quantum mathematics in artificial intelligence. *Journal of Artificial Intelligence Research*, 72:1307–1341, 2021. doi: 10.1613/jair.1.12702.
- Dominic Widdows, Jahan Rani, and Emmanuel M. Pothos. Quantum circuit components for cognitive decision-making. *Entropy*, 25(4):548, 2023. doi: 10.3390/e25040548.
- F. Yan, X. Yang, N. Li, X. Yu, and H. Zhai. Emotion generation and transition of companion robots based on Plutchik’s model and quantum circuit schemes. *Security and Communication Networks*, 2021:6802521, 2021a. doi: 10.1155/2021/6802521.
- Fei Yan, Abdullah M. Ilyasu, Sihao Jiao, and Huamin Yang. Quantum structure for modelling emotion space of robots. *Applied Sciences*, 9(16):3351, 2019. doi: 10.3390/app9163351.
- Fei Yan, Abdullah M. Ilyasu, and Kaoru Hirota. Conceptual framework for quantum affective computing and its use in fusion of multi-robot emotions. *Electronics*, 10(2):100, 2021b. doi: 10.3390/electronics10020100.
- Fei Yan, Abdullah M. Ilyasu, and Kaoru Hirota. Emotion space modelling for social robots. *Engineering Applications of Artificial Intelligence*, 100:104178, 2021c. doi: 10.1016/j.engappai.2021.104178.
- James M. Yearsley. Advanced tools and concepts for quantum cognition: A tutorial. *Journal of Mathematical Psychology*, 78:24–39, 2017. doi: 10.1016/j.jmp.2016.07.005.
- James M. Yearsley and Jerome R. Busemeyer. Quantum cognition and decision theories: A tutorial. *Journal of Mathematical Psychology*, 74:99–116, 2016. doi: 10.1016/j.jmp.2015.11.005.
- James M. Yearsley and Emmanuel M. Pothos. Zeno’s paradox in decision-making. *Proceedings of the Royal Society B: Biological Sciences*, 283(1828):20160291, 2016. doi: 10.1098/rspb.2016.0291.
- Vyacheslav I. Yukalov. Evolutionary processes in quantum decision theory. *Entropy*, 22(6):681, 2020. doi: 10.3390/e22060681.
- Vyacheslav I. Yukalov. Quantum operation of affective artificial intelligence. *Laser Physics*, 33(6):065204, 2023. doi: 10.1088/1555-6611/accf7a.
- Vyacheslav I. Yukalov and Didier Sornette. Decision theory with prospect interference and entanglement. *Theory and Decision*, 70(3):283–328, 2011. doi: 10.1007/s11238-010-9202-y.
- Vyacheslav I. Yukalov and Didier Sornette. Quantum probability and quantum decision-making. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2058):20150100, 2016. doi: 10.1098/rsta.2015.0100.
- Vyacheslav I. Yukalov, Elizaveta P. Yukalova, and Didier Sornette. Information processing by networks of quantum decision makers. *Physica A: Statistical Mechanics and its Applications*, 492:747–766, 2018. doi: 10.1016/j.physa.2017.11.004.
- Vyacheslav I. Yukalov, Elizaveta P. Yukalova, and Didier Sornette. Role of collective information in networks of quantum operating agents. *Physica A: Statistical Mechanics and its Applications*, 598:127365, 2022. doi: 10.1016/j.physa.2022.127365.
- Won Joon Yun, Yunseok Kwak, Jae Pyoung Kim, Hyunhee Cho, Soyi Jung, Jihong Park, and Joongheon Kim. Quantum multi-agent reinforcement learning via variational quantum circuit design. In *Proceedings of the 2022 IEEE 42nd International Conference on Distributed Computing Systems (ICDCS)*, pages 1332–1335. IEEE, 2022. doi: 10.1109/ICDCS54860.2022.00151.
- Chenchen Zhang, Qiuchi Li, Dawei Song, and Prayag Tiwari. Quantum-inspired semantic matching based on neural networks with the duality of density matrices. *Engineering Applications of Artificial Intelligence*, 140:109667, 2025. doi: 10.1016/j.engappai.2024.109667.
- Peng Zhang, Zhan Su, Lipeng Zhang, Benyou Wang, and Dawei Song. A quantum many-body wave function inspired language modeling approach. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management (CIKM 2018)*, pages 1303–1312. ACM, 2018. doi: 10.1145/3269206.3271723.